

Introducing Spring

Code

Listing 1.1

```
ApplicationContext ctx =  
    new ClassPathXmlApplicationContext("/WEB-INF/  
applicationContext.xml");  
UserDAO dao = (UserDAO) ctx.getBean("userDAO");
```

Listing 1.2

```
<bean id="userDAO" class="org.appfuse.dao.hibernate.UserDAOHibernate">  
    <property name="sessionFactory">  
        <ref local="sessionFactory"/>  
    </property>  
</bean>
```

Listing 1.3

```
public List getUsers() throws DAOException {  
    List list = null;  
  
    try {  
        list = ses.find("from User u order by upper(u.username)");  
    } catch (HibernateException he) {  
        he.printStackTrace();  
        throw new DAOException(he);  
    }  
  
    return list;  
}
```

Listing 1.4

```
public List getUsers() {  
    return getHibernateTemplate()  
        .find("from User u order by upper(u.username)");  
}
```

Spring Quick Start Tutorial

Code

Listing 2.1

```
[echo] Available targets are:

[echo] compile    --> Compile all Java files
[echo] war         --> Package as WAR file
[echo] deploy      --> Deploy application as directory
[echo] deploywar    --> Deploy application as a WAR file

[echo] install     --> Install application in Tomcat
[echo] remove      --> Remove application from Tomcat
[echo] reload       --> Reload application in Tomcat
[echo] start        --> Start Tomcat application
[echo] stop         --> Stop Tomcat application
[echo] list         --> List Tomcat applications

[echo] clean --> Deletes compiled classes and WAR
[echo] new  --> Creates a new project
```

Listing 2.2

```
deploy=org.apache.catalina.ant.DeployTask
undeploy=org.apache.catalina.ant.UndeployTask
remove=org.apache.catalina.ant.RemoveTask
reload=org.apache.catalina.ant.ReloadTask
start=org.apache.catalina.ant.StartTask
stop=org.apache.catalina.ant.StopTask
list=org.apache.catalina.ant.ListTask
```

Listing 2.3

```
<!-- Tomcat Ant Tasks -->
<taskdef file="tomcatTasks.properties">
    <classpath>
        <pathelement
            path="${tomcat.home}/server/lib/catalina-ant.jar"/>
        </classpath>
    </taskdef>

<target name="install" description="Install application in Tomcat"
    depends="war">
    <deploy url="${tomcat.manager.url}"
        username="${tomcat.manager.username}"
        password="${tomcat.manager.password}"
        path="/${webapp.name}"
        war="file:${dist.dir}/${webapp.name}.war"/>
    </target>

<target name="remove" description="Remove application from Tomcat">
    <undeploy url="${tomcat.manager.url}"
        username="${tomcat.manager.username}"
        password="${tomcat.manager.password}"
        path="/${webapp.name}"/>
    </target>

<target name="reload" description="Reload application in Tomcat">
    <reload url="${tomcat.manager.url}"
        username="${tomcat.manager.username}"
        password="${tomcat.manager.password}"
        path="/${webapp.name}"/>
    </target>

<target name="start" description="Start Tomcat application">
    <start url="${tomcat.manager.url}"
        username="${tomcat.manager.username}"
        password="${tomcat.manager.password}"
        path="/${webapp.name}"/>
    </target>

<target name="stop" description="Stop Tomcat application">
    <stop url="${tomcat.manager.url}"
        username="${tomcat.manager.username}"
        password="${tomcat.manager.password}"
        path="/${webapp.name}"/>
    </target>

<target name="list" description="List Tomcat applications">
```

```
<list url="${tomcat.manager.url}"
      username="${tomcat.manager.username}"
      password="${tomcat.manager.password}"/>
</target>
```

Listing 2.4

```
# Properties for Tomcat Server
tomcat.manager.url=http://localhost:8080/manager
tomcat.manager.username=admin
tomcat.manager.password=admin
```

Listing 2.5

```
<user username="admin" password="admin" roles="manager"/>
```

Listing 2.6

```
package org.appfuse.dao;

// use your IDE to handle imports

public class UserDAOTest extends BaseDAOTestCase {
    private User user = null;
    private UserDao dao = null;

    protected void setUp() throws Exception {
        super.setUp();
        dao = (UserDao) ctx.getBean("userDAO");
    }

    protected void tearDown() throws Exception {
        super.tearDown();
        dao = null;
    }
}
```

Listing 2.7

```
public void testSaveUser() throws Exception {
    user = new User();
    user.setFirstName("Rod");
    user.setLastName("Johnson");

    dao.saveUser(user);
    assertNotNull("primary key assigned", user.getId());
}
```

```

        log.info(user);
        assertNotNull(user.getFirstName());
    }

    public void testAddAndRemoveUser() throws Exception {
        user = new User();
        user.setFirstName("Bill");
        user.setLastName("Joy");

        dao.saveUser(user);

        assertNotNull(user.getId());
        assertEquals(user.getFirstName(), "Bill");

        if (log.isDebugEnabled()) {
            log.debug("removing user...");
        }

        dao.removeUser(user.getId());

        assertNull(dao.getUser(user.getId()));
    }
}

```

Listing 2.8

```

package org.appfuse.dao;

// use your IDE to handle imports

public interface UserDao extends DAO {
    public List getUsers();
    public User getUser(Long userId);
    public void saveUser(User user);
    public void removeUser(Long userId);
}

```

Listing 2.9

```

package org.appfuse.model;

public class User extends BaseObject {
    private Long id;
    private String firstName;
    private String lastName;

    /*
     * Generate your getters and setters using your favorite IDE:
     */
}

```

```
In Eclipse:  
Right-click -> Source -> Generate Getters and Setters  
*/  
}
```

Listing 2.10

```
<?xml version="1.0" encoding="UTF-8"?>  
<!DOCTYPE hibernate-mapping PUBLIC  
    "-//Hibernate/Hibernate Mapping DTD 2.0//EN"  
    "http://hibernate.sourceforge.net/hibernate-mapping-2.0.dtd">  
  
<hibernate-mapping>  
    <class name="org.appfuse.model.User" table="app_user">  
  
        <id name="id" column="id" unsaved-value="0">  
            <generator class="increment" />  
        </id>  
        <property name="firstName" column="first_name"  
            not-null="true"/>  
        <property name="lastName" column="last_name" not-null="true"/>  
  
    </class>  
</hibernate-mapping>
```

Listing 2.11

```
<property name="mappingResources">  
    <list>  
        <value>org/appfuse/model/User.hbm.xml</value>  
    </list>  
</property>
```

Listing 2.12

```
INFO - SchemaExport.execute(98) | Running hbm2ddl schema export  
INFO - SchemaExport.execute(117) | exporting generated schema to  
database  
INFO - ConnectionProviderFactory.newConnectionProvider(53) |  
Initializing connection provider:  
org.springframework.orm.hibernate.LocalDataSourceConnectionProvider  
INFO - DriverManagerDataSource.getConnectionFromDriverManager(140) |  
Creating new JDBC connection to [jdbc:hsqldb:db/appfuse]  
INFO - SchemaExport.execute(160) | schema export complete
```

Listing 2.13

```
<listener>
  <listener-class>
    org.springframework.web.context.ContextLoaderListener
  </listener-class>
</listener>
```

Listing 2.14

```
<context-param>
  <param-name>contextConfigLocation</param-name>
  <param-value>/WEB-INF/sampleContext.xml</param-value>
</context-param>
```

Listing 2.15

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE beans PUBLIC "-//SPRING//DTD BEAN//EN"
  "http://www.springframework.org/dtd/spring-beans.dtd">

<beans>
  <bean id="dataSource"

class="org.springframework.jdbc.datasource.DriverManagerDataSource">
    <property name="driverClassName">
      <value>org.hsqldb.jdbcDriver</value>
    </property>
    <property name="url">
      <value>jdbc:hsqldb:db/appfuse</value>
    </property>
    <property name="username"><value>sa</value></property>
    <!-- Make sure <value> tags are on same line - if they're not,
      authentication will fail -->
    <property name="password"><value></value></property>
  </bean>

  <!-- Hibernate SessionFactory -->
  <bean id="sessionFactory"

class="org.springframework.orm.hibernate.LocalSessionFactoryBean">
    <property name="dataSource">
      <ref local="dataSource"/>
    </property>
    <property name="mappingResources">
      <list>
        <value>org/appfuse/model/User.hbm.xml</value>
      </list>
```



```

        </property>
        <property name="hibernateProperties">
            <props>
                <prop key="hibernate.dialect">
                    net.sf.hibernate.dialect.HSQLDialect
                </prop>
                <prop key="hibernate.hbm2ddl.auto">create</prop>
            </props>
        </property>
    </bean>

    <!-- Transaction manager for a single Hibernate SessionFactory -->
    <bean id="transactionManager"

class="org.springframework.orm.hibernate.HibernateTransactionManager">
        <property name="sessionFactory">
            <ref local="sessionFactory"/>
        </property>
    </bean>
</beans>

```

Listing 2.16

```

<bean id="dataSource"
    class="org.springframework.jndi.JndiObjectFactoryBean">
    <property name="jndiName">
        <value>java:comp/env/jdbc/appfuse</value>
    </property>
</bean>

```

Listing 2.17

```

package org.appfuse.dao.hibernate;

// organize imports using your IDE

public class UserDAOHibernate extends HibernateDaoSupport implements
UserDAO {
    private Log log = LogFactory.getLog(UserDAOHibernate.class);

    public List getUsers() {
        return getHibernateTemplate().find("from User");
    }

    public User getUser(Long id) {
        return (User) getHibernateTemplate().get(User.class, id);
    }

    public void saveUser(User user) {

```

```

        getHibernateTemplate().saveOrUpdate(user);

        if (log.isDebugEnabled()) {
            log.debug("userId set to: " + user.getId());
        }
    }

    public void removeUser(Long id) {
        Object user = getHibernateTemplate().load(User.class, id);
        getHibernateTemplate().delete(user);
    }
}

```

Listing 2.18

```

<bean id="userDAO" class="org.appfuse.dao.hibernate.UserDAOHibernate">
    <property name="sessionFactory">
        <ref local="sessionFactory"/>
    </property>
</bean>

```

Listing 2.19

```

package org.appfuse.service;

// use your IDE to organize imports

public class UserManagerTest extends TestCase {
    private static Log log = LogFactory.getLog(UserManagerTest.class);
    private ApplicationContext ctx;
    private User user;
    private UserManager mgr;

    protected void setUp() throws Exception {
        String[] paths = { "/WEB-INF/applicationContext.xml" };
        ctx = new ClassPathXmlApplicationContext(paths);
        mgr = (UserManager) ctx.getBean("userManager");
    }

    protected void tearDown() throws Exception {
        user = null;
        mgr = null;
    }

    // add testXXX methods here
}

```

Listing 2.20

```
public void testAddAndRemoveUser() throws Exception {
    user = new User();
    user.setFirstName("Easter");
    user.setLastName("Bunny");

    user = mgr.saveUser(user);

    assertNotNull(user.getId());

    if (log.isDebugEnabled()) {
        log.debug("removing user...");
    }

    String userId = user.getId().toString();
    mgr.removeUser(userId);

    user = mgr.getUser(userId);
    assertNull("User object found in database", user);
}
```

Listing 2.21

```
package org.appfuse.service;

// use your IDE to handle imports

public interface UserManager {
    public List getUsers();
    public User getUser(String userId);
    public User saveUser(User user);
    public void removeUser(String userId);
}
```

Listing 2.22

```
package org.appfuse.service.impl;

// use your IDE to handle imports

public class UserManagerImpl implements UserManager {
    private static Log log = LogFactory.getLog(UserManagerImpl.class);
    private UserDao dao;

    public void setUserDAO(UserDao dao) {
        this.dao = dao;
    }
}
```

```

public List getUsers() {
    return dao.getUsers();
}

public User getUser(String userId) {
    User user = dao.getUser(Long.valueOf(userId));

    if (user == null) {
        log.warn("UserId '" + userId + "' not found in database.");
    }

    return user;
}

public User saveUser(User user) {
    dao.saveUser(user);

    return user;
}

public void removeUser(String userId) {
    dao.removeUser(Long.valueOf(userId));
}
}

```

Listing 2.23

```

<bean id="userManager"
    class="org.appfuse.service.impl.UserManagerImpl">
    <property name="userDAO"><ref local="userDAO"/></property>
</bean>

```

Listing 2.24

```

<bean id="userManager"
    class="org.springframework.transaction.interceptor.TransactionProxy
FactoryBean">
    <property name="transactionManager">
        <ref local="transactionManager"/>
    </property>
    <property name="target">
        <ref local="userManagerTarget"/>
    </property>
    <property name="transactionAttributes">
        <props>
            <prop key="save*">PROPAGATION_REQUIRED</prop>
            <prop key="remove*">PROPAGATION_REQUIRED</prop>

```

```
        <prop key="*">PROPAGATION_REQUIRED,readOnly</prop>
    </props>
</property>
</bean>
```

Listing 2.25

```
<bean id="userManagerTarget"
    class="org.appfuse.service.impl.UserManagerImpl">
    <property name="userDAO"><ref local="userDAO"/></property>
</bean>
```

Listing 2.26

```
<logger name="org.springframework.transaction">
    <level value="DEBUG"/> <!-- INFO does nothing -->
</logger>
```

Listing 2.27

```
package org.appfuse.web;

// use your IDE to handle imports

public class UserActionTest extends MockStrutsTestCase {

    public UserActionTest(String testName) {
        super(testName);
    }

    public void testExecute() {
        setRequestPathInfo("/user");
        addRequestParameter("id", "1");
        actionPerform();
        verifyForward("success");
        verifyNoActionErrors();
    }
}
```

Listing 2.28

```
package org.appfuse.web;

// use your IDE to handle imports

public class UserAction extends DispatchAction {
    private static Log log = LogFactory.getLog(UserAction.class);
```

```

public ActionForward execute(ActionMapping mapping, ActionForm form,
                             HttpServletRequest request,
                             HttpServletResponse response)
    throws Exception {
    request.getSession().setAttribute("test", "succeeded!");

    log.debug("looking up userId: " + request.getParameter("id"));

    return mapping.findForward("success");
}
}

```

Listing 2.29

```

<action path="/user" type="org.appfuse.web.UserAction">
    <forward name="success" path="/index.jsp"/>
</action>

```

Listing 2.30

```

<form-bean name="userForm"
    type="org.apache.struts.action.DynaActionForm">
    <form-property name="user" type="org.appfuse.model.User"/>
</form-bean>

```

Listing 2.31

```

<action path="/user" type="org.appfuse.web.UserAction"
    name="userForm" scope="request">
    <forward name="success" path="/index.jsp"/>
</action>

```

Listing 2.32

```

public class UserActionTest extends MockStrutsTestCase {

    public UserActionTest(String testName) {
        super(testName);
    }

    // Adding a new user is required between tests because HSQL creates
    // an in-memory database that goes away during tests.
    public void addUser() {
        setRequestPathInfo("/user");
        addRequestParameter("method", "save");
        addRequestParameter("user.firstName", "Juergen");
    }
}

```

```

        addRequestParam("user.lastName", "Hoeller");
        actionPerform();
        verifyForward("list");
        verifyNoActionErrors();
    }

    public void testAddAndEdit() {
        addUser();

        // edit newly added user
        addRequestParam("method", "edit");
        addRequestParam("id", "1");
        actionPerform();
        verifyForward("edit");
        verifyNoActionErrors();
    }

    public void testAddAndDelete() {
        addUser();

        // delete new user
        setRequestPathInfo("/user");
        addRequestParam("method", "delete");
        addRequestParam("user.id", "1");
        actionPerform();
        verifyForward("list");
        verifyNoActionErrors();
    }

    public void testList() {
        addUser();
        setRequestPathInfo("/user");
        addRequestParam("method", "list");
        actionPerform();
        verifyForward("list");
        verifyNoActionErrors();

        List users = (List) getRequest().getAttribute("users");
        assertNotNull(users);
        assertTrue(users.size() == 1);
    }
}

```

Listing 2.33

```

public class UserAction extends DispatchAction {
    private static Log log = LogFactory.getLog(UserAction.class);
    private UserManager mgr = null;

    public void setUserManager(UserManager userManager) {

```

```

        this.mgr = userManager;
    }

    public ActionForward delete(ActionMapping mapping, ActionForm form,
                               HttpServletRequest request,
                               HttpServletResponse response)
        throws Exception {
        if (log.isDebugEnabled()) {
            log.debug("entering 'delete' method...");
        }

        mgr.removeUser(request.getParameter("user.id"));

        ActionMessages messages = new ActionMessages();
        messages.add(ActionMessages.GLOBAL_MESSAGE,
                     new ActionMessage("user.deleted"));

        saveMessages(request, messages);

        return list(mapping, form, request, response);
    }

    public ActionForward edit(ActionMapping mapping, ActionForm form,
                              HttpServletRequest request,
                              HttpServletResponse response)
        throws Exception {
        if (log.isDebugEnabled()) {
            log.debug("entering 'edit' method...");
        }

        DynaActionForm userForm = (DynaActionForm) form;
        String userId = request.getParameter("id");

        // null userId indicates an add
        if (userId != null) {
            User user = mgr.getUser(userId);

            if (user == null) {
                ActionMessages errors = new ActionMessages();
                errors.add(ActionMessages.GLOBAL_MESSAGE,
                          new ActionMessage("user.missing"));
                saveErrors(request, errors);

                return mapping.findForward("list");
            }

            userForm.set("user", user);
        }

        return mapping.findForward("edit");
    }
}

```



```

    public ActionForward list(ActionMapping mapping, ActionForm form,
                             HttpServletRequest request,
                             HttpServletResponse response)
        throws Exception {
        if (log.isDebugEnabled()) {
            log.debug("entering 'list' method...");
        }

        request.setAttribute("users", mgr.getUsers());

        return mapping.findForward("list");
    }

    public ActionForward save(ActionMapping mapping, ActionForm form,
                              HttpServletRequest request,
                              HttpServletResponse response)
        throws Exception {
        if (log.isDebugEnabled()) {
            log.debug("entering 'save' method...");
        }

        DynaActionForm userForm = (DynaActionForm) form;
        mgr.saveUser((User)userForm.get("user"));

        ActionMessages messages = new ActionMessages();
        messages.add(ActionMessages.GLOBAL_MESSAGE,
                     new ActionMessage("user.saved"));
        saveMessages(request, messages);

        return list(mapping, form, request, response);
    }
}

```

Listing 2.34

```

<plug-in
    className="org.springframework.web.struts.ContextLoaderPlugIn">
    <set-property property="contextConfigLocation"
        value="/WEB-INF/applicationContext.xml,
            /WEB-INF/action-servlet.xml"/>
</plug-in>

```

Listing 2.35

```

<action path="/user"
    type="org.springframework.web.struts.DelegatingActionProxy"

```

```
name="userForm" scope="request" parameter="method">
  <forward name="list" path="/userList.jsp"/>
  <forward name="edit" path="/userForm.jsp"/>
</action>
```

Listing 2.36

```
<bean name="/user" class="org.appfuse.web.UserAction"
  singleton="false">
  <property name="userManager">
    <ref bean="userManager"/>
  </property>
</bean>
```

Listing 2.37

```
user.saved=User has been saved successfully.
user.missing=No user found with this id.
user.deleted=User successfully deleted.
```

Listing 2.38

```
<message-resources parameter="messages"/>
```

Listing 2.39

```
<%@ include file="/taglibs.jsp"%>

<title>MyUsers ~ User List</title>

<button onclick="location.href='user.do?method=edit'">Add User</button>

<table class="list">
<thead>
<tr>
  <th>User Id</th>
  <th>First Name</th>
  <th>Last Name</th>
</tr>
</thead>
<tbody>
<c:forEach var="user" items="${users}" varStatus="status">
<c:choose>
  <c:when test="${status.count % 2 == 0}"><tr class="even"></c:when>
  <c:otherwise><tr class="odd"></c:otherwise>
</c:choose>
```

```

    <td><a href="user.do?method=edit&id=${user.id}">${user.id}</a></td>
    <td>${user.firstName}</td>
    <td>${user.lastName}</td>
</tr>
</c:forEach>
</tbody>
</table>

```

Listing 2.40

```

<target name="populate">
    <echo message="Loading sample data..." />
    <sql driver="org.hsqldb.jdbcDriver"
        url="jdbc:hsqldb:db/appfuse"
        userid="sa" password="">
        <classpath refid="classpath" />

        INSERT INTO app_user (id, first_name, last_name)
            values (5, 'Julie', 'Raible');
        INSERT INTO app_user (id, first_name, last_name)
            values (6, 'Abbie', 'Raible');

    </sql>
</target>

```

Listing 2.41

```

user.id=User Id
user.firstName=First Name
user.lastName=Last Name

```

Listing 2.42

```

<thead>
<tr>
    <th><bean:message key="user.id" /></th>
    <th><bean:message key="user.firstName" /></th>
    <th><bean:message key="user.lastName" /></th>
</tr>
</thead>

```

Listing 2.43

```
<display:table name="users" pagesize="10" styleClass="list"
    requestURI="user.do?method=list">
    <display:column property="id" paramId="id" paramProperty="id"
        href="user.do?method=edit" sort="true"/>
    <display:column property="firstName" sort="true"/>
    <display:column property="lastName" sort="true"/>
</display:table>
```

Listing 2.44

```
<%@ include file="/taglibs.jsp"%>

<title>MyUsers ~ User Details</title>

<p>Please fill in user's information below:</p>

<html:form action="/user" focus="user.firstName">
<input type="hidden" name="method" value="save"/>
<html:hidden property="user.id"/>
<table>
<tr>
    <th><bean:message key="user.firstName"/></th>
    <td><html:text property="user.firstName"/></td>
</tr>
<tr>
    <th><bean:message key="user.lastName"/></th>
    <td><html:text property="user.lastName"/></td>
</tr>
<tr>
    <td></td>
    <td>
        <html:submit styleClass="button">Save</html:submit>
        <c:if test="${not empty param.id}">
            <html:submit styleClass="button"
                onclick="this.form.method.value='delete'">
                Delete</html:submit>
        </c:if>
    </td>
</tr>
</table>
</html:form>
```

Listing 2.45

```
<plug-in className="org.apache.struts.validator.ValidatorPlugIn">
  <set-property
    property="pathnames" value="/WEB-INF/validator-rules.xml,
                               /WEB-INF/validation.xml"/>
</plug-in>
```

Listing 2.46

```
<formset>
  <form name="userForm">
    <field property="user.lastName" depends="required">
      <arg0 key="user.lastName"/>
    </field>
  </form>
</formset>
```

Listing 2.47

```
<form-bean name="userForm"
  type="org.apache.struts.validator.DynaValidatorForm">
  ...
```

Listing 2.48

```
if (log.isDebugEnabled()) {
    log.debug("entering 'save' method...");
}

// run validation rules on this form
ActionMessages errors = form.validate(mapping, request);
if (!errors.isEmpty()) {
    saveErrors(request, errors);
    return mapping.findForward("edit");
}

DynaActionForm userForm = (DynaActionForm) form;
```

Listing 2.49

```
<html:form action="/user" focus="user.firstName"
  onsubmit="return validateUserForm(this)">
  ...
</html:form>

<html:javascript formName="userForm"/>
```



The BeanFactory and How It Works

Code

Listing 3.1

```
public class ListUsers extends Action {

    public ActionForward execute(ActionMapping mapping,
                                ActionForm form,
                                HttpServletRequest request,
                                HttpServletResponse response)

    throws Exception {
        // get a connection from the database
        Connection conn = DatabaseUtils.getConnection();
        UserDAO dao = DAOFactory.createUserDAO("hibernate", conn);

        List users = dao getUsers();

        DatabaseUtils.closeConnection(conn);
        return mapping.findForward("success");
    }
}
```

Listing 3.2

```
public class ListUsers extends Action {
    private UserDAO dao;
    public void setUserDAO(UserDAO userDAO) {
        this.dao = userDAO;
    }

    public ActionForward execute(ActionMapping mapping,
                                ActionForm form,
```

```

        HttpServletRequest request,
        HttpServletResponse response)

    throws Exception {
        List users = dao getUsers();

        return mapping.findForward("success");
    }
}

```

Listing 3.3

```

<bean id="example" class="org.appfuse.util.Converter"/>

```

Listing 3.4

```

private int maxSize;

public void setMaxSize(int maxSize) {
    this.maxSize = maxSize;
}

```

Listing 3.5

```

<property name="maxSize"><value>1000</value></property>

```

Listing 3.6

```

<property name="maxSize" value="1000"/

```

Listing 3.7

```

<bean id="sessionFactory" class="...">
    <property name="dataSource">
        <ref local="dataSource"/>
    </property>
    <property name="mappingResources">
        <list>
            <value>org/appfuse/model/User.hbm.xml</value>
        </list>
    </property>
    ...
</bean>

```

Listing 3.8

```

<property name="dataSource" ref="dataSource"/>

```


Listing 3.9

```
<bean id="propertyConfigurer"  
  
class="org.springframework.beans.factory.config.PropertyPlaceholderConfigurer">  
    <property name="location">  
        <value>classpath:database.properties</value>  
    </property>  
</bean>  
  
<bean id="dataSource"  
    class="org.springframework.jdbc.datasource.DriverManagerDataSource">  
    <property name="driverClassName">  
        <value>${db.driverName}</value>  
    </property>  
    <property name="url"><value>${db.url}</value></property>  
    <property name="username"><value>${db.username}</value></property>  
    <property name="password"><value>${db.password}</value></property>  
</bean>
```

Listing 3.10

```
<bean id="userDAO" autowire="byName"  
    class="org.appfuse.dao.hibernate.UserDAOHibernate"/>
```

Listing 3.11

```
package org.appfuse.model;  
  
// organize imports with your IDE  
  
public class Smorgasbord extends BaseObject {  
    private Log log = LoggerFactory.getLog(Smorgasbord.class);  
    private int daysToJavaOne;  
    private boolean attendingJavaOne;  
    private Integer streetsInDenver;  
    private Long peopleInDenver;  
    private DataSource dataSource;  
  
    public void setDaysToJavaOne(int daysToJavaOne) {  
        this.daysToJavaOne = daysToJavaOne;  
    }  
  
    public void setAttendingJavaOne(boolean attendingJavaOne) {  
        this.attendingJavaOne = attendingJavaOne;  
    }  
}
```

```

    public void setStreetsInDenver(Integer streetsInDenver) {
        this.streetsInDenver = streetsInDenver;
    }

    public void setPeopleInDenver(Long peopleInDenver) {
        this.peopleInDenver = peopleInDenver;
    }

    public void setDataSource(DataSource dataSource) {
        this.dataSource = dataSource;
    }

    public String toString() {
        log.debug(super.toString());
        return super.toString();
    }
}

```

Listing 3.12

```

<bean id="smorgasbord" class="org.appfuse.model.Smorgasbord"
    dependency-check="all" init-method="toString">
    <property name="daysToJavaOne"><value>14</value></property>
    <property name="attendingJavaOne"><value>true</value></property>
    <property name="streetsInDenver"><value>334</value></property>
    <property name="peopleInDenver"><value>3000000</value></property>
    <property name="dataSource"><ref local="dataSource"/></property>
</bean>

```

Listing 3.13

```

<logger name="org.springframework.beans">
    <level value="INFO"/>
</logger>

```

Listing 3.14

```

public void setBeanFactory(BeanFactory beanFactory)
    throws BeansException;

```

Listing 3.15

```

public void setBeanName(java.lang.String name);

```

Listing 3.16

```
<!-- results in a setPeople(java.util.Properties) call -->
<property name="people">
  <props>
    <prop key="HarryPotter">The magic property</prop>
    <prop key="JerrySeinfeld">The funny property</prop>
  </props>
</property>
<!-- results in a setSomeList(java.util.List) call -->
<property name="someList">
  <list>
    <value>a list element followed by a reference</value>
    <ref bean="dataSource"/>
  </list>
</property>
<!-- results in a setSomeMap(java.util.Map) call -->
<property name="someMap">
  <map>
    <entry key="aString">
      <value>just some string</value>
    </entry>
    <entry key="aBean">
      <ref bean="dataSource"/>
    </entry>
  </map>
</property>
```

Listing 3.17

```
<entry key="ds" key-ref="dataSource"/>
```

Listing 3.18

```
String[] paths = {"/WEB-INF/applicationContext*.xml"};
ApplicationContext ctx = new
    ClassPathXmlApplicationContext(paths);
```

Listing 3.19

```
String[] paths =
    {"c:/source/myusers/web/WEB-INF/applicationContext*.xml"};
ApplicationContext ctx = new
    FileSystemXmlApplicationContext(paths);
```

Listing 3.20

```
StaticApplicationContext ctx = new StaticApplicationContext();
Map properties = new HashMap();
properties.put("basename", "messages");
ctx.registerSingleton("messageSource",
    ResourceBundleMessageSource.class,
    new MutablePropertyValues(properties));
ctx.refresh();
```

Listing 3.21

```
XmlWebApplicationContext ctx =
    new XmlWebApplicationContext();
context.setServletContext(ctx);
context.refresh();
```

Listing 3.22

```
public class StartupListener extends ContextLoaderListener
    implements ServletContextListener {

    public void contextInitialized(ServletContextEvent event) {

        // call Spring's context ContextLoaderListener to initialize
        // all the context files specified in web.xml
        super.contextInitialized(event);

        // get beans from WebApplicationContext in servletContext
    }
}
```

Listing 3.23

```
<beans>
    <import resource="applicationContext-hibernate.xml"/>
    <import resource="conf/applicationContext-security.xml"/>

    <bean id="firstBean" class="..."/>
</beans>
```

Listing 3.24

```
protected static ApplicationContext ctx = null;

static {
    ctx = new ClassPathXmlApplicationContext("/appContext.xml");
}
```

Listing 3.25

```
<bean id="messageSource"
class="org.springframework.context.support.ResourceBundleMessageSource"
>
  <property name="basename"><value>messages</value></property>
</bean>
```

Listing 3.26

```
<property name="basenames">
  <list>
    <value>messages</value>
    <value>errors</value>
  </list>
</property>
```



Spring's MVC Framework

Code

Listing 4.1

```
package org.springframework.web.servlet.mvc;

import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

import org.springframework.web.servlet.ModelAndView;

public interface Controller {

    /**
     * Process the request and return a ModelAndView object which the
     * DispatcherServlet will render. A null return is not an error:
     * It indicates that this object completed request processing
     * itself, thus there is no ModelAndView to render.
     */
    ModelAndView handleRequest(HttpServletRequest request,
                               HttpServletResponse response)
        throws Exception;
}
```

Listing 4.2

```
<servlet>
  <servlet-name>action</servlet-name>
  <servlet-class>
    org.apache.struts.action.ActionServlet
  </servlet-class>
  <load-on-startup>1</load-on-startup>
</servlet>
```

Listing 4.3

```
<servlet>
  <servlet-name>action</servlet-name>
  <servlet-class>
    org.springframework.web.servlet.DispatcherServlet
  </servlet-class>
  <load-on-startup>1</load-on-startup>
</servlet>
```

Listing 4.4

```
<servlet-mapping>
  <servlet-name>action</servlet-name>
  <url-pattern>*.html</url-pattern>
</servlet-mapping>
```

Listing 4.5

```
<context-param>
  <param-name>contextConfigLocation</param-name>
  <param-value>
    /WEB-INF/applicationContext1.xml
    /WEB-INF/applicationContext2.xml
  </param-value>
</context-param>
```

Listing 4.6

```
rm src/org/appfuse/web/UserAction.java
rm test/org/appfuse/web/UserActionTest.java
rm web/WEB-INF/lib/struts*
rm web/WEB-INF/struts-config.xml
```

Listing 4.7

```
<bean name="user" class="org.appfuse.web.UserAction"
  singleton="false">
  <property name="userManager">
    <ref bean="userManager"/>
  </property>
</bean>
```


Listing 4.8

```
public void setUp() {
    String[] paths = {"/WEB-INF/applicationContext.xml",
                      "/WEB-INF/action-servlet.xml"};
    ctx = new XmlWebApplicationContext();
    ctx.setConfigLocations(paths);
    ctx.setServletContext(new MockServletContext(""));
    ctx.refresh();
}
```

Listing 4.9

```
package org.appfuse.web;

// use your IDE (Eclipse and IDEA rock!) to add imports

public class UserControllerTest extends TestCase {

    private XmlWebApplicationContext ctx;

    public void setUp() {
        String[] paths = {"/WEB-INF/applicationContext.xml",
                          "/WEB-INF/action-servlet.xml"};
        ctx = new XmlWebApplicationContext();
        ctx.setConfigLocations(paths);
        ctx.setServletContext(new MockServletContext(""));
        ctx.refresh();
    }

    public void testGetUsers() throws Exception {
        UserController c = (UserController)
            ctx.getBean("userController");
        ModelAndView mav =
            c.handleRequest((HttpServletRequest) null,
                           (HttpServletResponse) null);
        Map m = mav.getModel();
        assertNotNull(m.get("users"));
        assertEquals(mav.getViewName(), "userList");
    }
}
```

Listing 4.10

```
package org.appfuse.web;

// Modern IDEs support easy importing
```

```

public class UserController implements Controller {
    private static Log log = LogFactory.getLog(UserController.class);
    private UserManager mgr = null;

    public void setUserManager(UserManager userManager) {
        this.mgr = userManager;
    }

    // put handleRequest() method here
}

```

Listing 4.11

```

public ModelAndView handleRequest(HttpServletRequest request,
                                HttpServletResponse response)
    throws Exception {
    if (log.isDebugEnabled()) {
        log.debug("entering 'handleRequest' method...");
    }

    return new ModelAndView("userList", "users", mgr.getUsers());
}

```

Listing 4.12

```

[junit] No bean named 'userController' is defined:
org.springframework.beans.factory.support.DefaultListableBeanFactory
defining beans
[dataSource, sessionFactory, transactionManager, userDao, userManagerTarget,
userManager]; root of BeanFactory hierarchy

```

Listing 4.13

```

<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE beans PUBLIC "-//SPRING//DTD BEAN//EN"
    "http://www.springframework.org/dtd/spring-beans.dtd">

<beans>
    <bean id="userController" class="org.appfuse.web.UserController">
        <property name="userManager" ref="userManager"/>
    </bean>
</beans>

```

Listing 4.14

```
<bean id="viewResolver"
class="org.springframework.web.servlet.view.InternalResourceViewResolver"
">
<property name="viewClass"
    value="org.springframework.web.servlet.view.JstlView"/>
    <property name="prefix" value="/" />
    <property name="suffix" value=".jsp" />
</bean>
```

Listing 4.15

```
<bean id="urlMapping"

class="org.springframework.web.servlet.handler.SimpleUrlHandlerMapping">
    <property name="mappings">
        <props>
            <prop key="/users.html">userController</prop>
        </props>
    </property>
</bean>
```

Listing 4.16

```
<bean name="/users.html" class="org.appfuse.web.UserController">
    <property name="userManager" ref="userManager"/>
</bean>
```

Listing 4.17

```
<%@ page language="java" errorPage="/error.jsp" %>
<%@ taglib uri="http://java.sun.com/jstl/core_rt" prefix="c" %>
<%@ taglib uri="http://java.sun.com/jstl/fmt" prefix="fmt" %>
<%@ taglib uri="http://www.springframework.org/tags/commons-validator"
prefix="html" %>
<%@ taglib uri="http://www.springframework.org/tags" prefix="spring" %>
<%@ taglib uri="http://www.opensymphony.com/sitemesh/decorator"
prefix="decorator"%>
```

Listing 4.18

```
<%-- Success Messages --%>
```

```

<c:if test="${not empty message}">
    <div class="message">${message}</div>
    <c:remove var="message" scope="session"/>
</c:if>

```

Listing 4.19

```

<%@ include file="/taglibs.jsp"%>

<title>MyUsers ~ User List</title>

<button onclick="location.href='editUser.html'">Add User</button>

<table class="list">
<thead>
<tr>
    <th><fmt:message key="user.id"/></th>
    <th><fmt:message key="user.firstName"/></th>
    <th><fmt:message key="user.lastName"/></th>
</tr>
</thead>
<tbody>
<c:forEach var="user" items="${users}" varStatus="status">
<c:choose>
    <c:when test="${status.count % 2 == 0}"><tr class="even"></c:when>
    <c:otherwise><tr class="odd"></c:otherwise>
</c:choose>
    <td><a href="editUser.html?id=${user.id}">${user.id}</a></td>
    <td>${user.firstName}</td>
    <td>${user.lastName}</td>
</tr>
</c:forEach>
</tbody>
</table>

```

Listing 4.20

```

<bean id="messageSource"
class="org.springframework.context.support.ResourceBundleMessageSource">
<property name="basename" value="messages"/>
<property name="useCodeAsDefaultMessage" value="true"/>
</bean>

```

Listing 4.21

```
<property name="basenames">
    <list>
        <value>messages</value>
        <value>errors</value>
    </list>
</property>
```

Listing 4.22

```
<bean id="messageSource"
class="org.springframework.context.support.ReloadableResourceBundleMessageSource">
    <property name="basename" value="/WEB-INF/messages "/>
    <property name="cacheSeconds" value="1"/>
</bean>
```

Listing 4.23

```
package org.appfuse.web;

// resolve imports using your IDE

public class UserFormControllerTest extends TestCase {
private static Log log =
    LogFactory.getLog(UserFormControllerTest.class);
private XmlWebApplicationContext ctx;
private UserFormController c;
private MockHttpServletRequest request;
private ModelAndView mv;
private User user;

public void setUp() throws Exception {
    String[] paths =
        { "/WEB-INF/applicationContext.xml",
          "/WEB-INF/action-servlet.xml" };
    ctx = new XmlWebApplicationContext();
    ctx.setConfigLocations(paths);
    ctx.setServletContext(new MockServletContext(""));
    ctx.refresh();
    c = (UserFormController) ctx.getBean("userFormController");

    // add a test user to the database
    UserManager mgr = (UserManager) ctx.getBean("userManager");
    user = new User();
    user.setFirstName("Matt");
    user.setLastName("Raible");
}
```

```

        user = mgr.saveUser(user);
    }

    public void tearDown() {
        ctx = null;
        c = null;
        user = null;
    }

    // put testXXX methods here
}

```

Listing 4.24

```

public void testEdit() throws Exception {
    log.debug("testing edit...");
    request = new MockHttpServletRequest("GET", "/editUser.html");
    request.addParameter("id", user.getId().toString());
    mv = c.handleRequest(request, new MockHttpServletResponse());
    assertEquals("userForm", mv.getViewName());
}

```

Listing 4.25

```

public MockHttpServletRequest(ServletContext servletContext)
public MockHttpServletRequest(ServletContext servletContext,
                               String method, String URI)
public MockHttpServletRequest()
public MockHttpServletRequest(String method, String URI)

```

Listing 4.26

```

public MyFormController() {
    super();
    setSessionForm(true);
}

```

Listing 4.27

```

package org.appfuse.web;

// resolve imports using your IDE

public class UserFormController extends SimpleFormController {
    private static Log log =
        LogFactory.getLog(UserFormController.class);
    private UserManager userManager = null;
}

```

```

    public void setUserManager(UserManager userManager) {
        this.userManager = userManager;
    }

    public UserFormController() {
        super();
        setCommandClass(User.class);
    }

    protected Object formBackingObject(HttpServletRequest request)
    throws ServletException {
        String id = RequestUtils.getStringParameter(request, "id", "");

        if (!"".equals(id)) {
            return userManager.getUser(id);
        } else {
            return new User();
        }
    }
}

```

Listing 4.28

```

<bean id="userFormController"
    class="org.appfuse.web.UserFormController">
    <property name="commandName" value="user"/>
    <property name="formView" value="userForm"/>
    <property name="successView" value="redirect:users.html"/>
    <property name="userManager" ref="userManager"/>
</bean>

```

Listing 4.29

```

<property name="commandClass" value="org.appfuse.model.User"/>

```

Listing 4.30

```

    return new ModelAndView(new RedirectView(getSuccessView()));

```

Listing 4.31

```

<property name="mappings">
    <props>

```

```
        <prop key="/users.html">userController</prop>
        <prop key="/editUser.html">userFormController</prop>
    </props>
</property>
```

Listing 4.32

```
<property name="mappings">
    <props>
        <prop key="/user/index.html">userController</prop>
        <prop key="/user/edit.html">userFormController</prop>
    </props>
</property>
```

Listing 4.33

```
<html:form action="/user">
```

Listing 4.34

```
<form action="<c:url value="/user.html"/>">
```

Listing 4.35

```
<form action="user.html">
```

Listing 4.36

```
<html:text property="user.firstName"/>
```

Listing 4.37

```
<spring:bind path="user.firstName">
    <input type="text" name="${status.expression}"
        value="${status.value}"/>
    <span class="fieldError">${status.errorMessage}</span>
</spring:bind>
```

Listing 4.38

```
<%@ include file="/taglibs.jsp"%>

<title>MyUsers ~ User Details</title>

<p>Please fill in user's information below:</p>
```



```

<form method="post" action="<c:url value="/editUser.html"/>">
<spring:bind path="user.id">
<input type="hidden" name="id" value="${status.value}"/>
</spring:bind>
<table>
<tr>
<th><fmt:message key="user.firstName"/>:</th>
<td>
<spring:bind path="user.firstName">
<input type="text" name="${status.expression}"
value="${status.value}"/>
<span class="fieldError">${status.errorMessage}</span>
</spring:bind>
</td>
</tr>
<tr>
<th><fmt:message key="user.lastName"/>:</th>
<td>
<spring:bind path="user.lastName">
<input type="text" name="${status.expression}"
value="${status.value}"/>
<span class="fieldError">${status.errorMessage}</span>
</spring:bind>
</td>
</tr>
<tr>
<td></td>
<td>
<input type="submit" class="button" name="cancel" value="Cancel"/>
</td>
</tr>
</table>
</form>

```

Listing 4.39

```

public void testSave() throws Exception {
    request = new MockHttpServletRequest("POST", "/editUser.html");
    request.addParameter("id", user.getId().toString());
    request.addParameter("firstName", user.getFirstName());
    request.addParameter("lastName", "Updated Last Name");

    mv = c.handleRequest(request, new MockHttpServletResponse());

    String errorsKey = BindException.ERROR_KEY_PREFIX +
        c.getCommandName();
}

```

```

        Errors errors = (Errors) mv.getModel().get(errorsKey);
        assertNull(errors);
        assertNotNull(request.getSession().getAttribute("message"));
    }

```

Listing 4.40

```

protected void initBinder(HttpServletRequest request,
                          ServletRequestDataBinder binder) {
    NumberFormat nf = NumberFormat.getNumberInstance();
    binder.registerCustomEditor(Long.class,
                                new CustomNumberEditor(Long.class, nf, true));
}

public ModelAndView onSubmit(HttpServletRequest request,
                             HttpServletResponse response,
                             Object command,
                             BindException errors)
    throws Exception {
    if (log.isDebugEnabled()) {
        log.debug("entering 'onSubmit' method...");
    }

    User user = (User) command;

    userManager.saveUser(user);
    // put success message into session so it survives redirects
    request.getSession().setAttribute("message",
                                       getMessageSourceAccessor().getMessage("user.saved",
                                       new Object[]
                                       { user.getFirstName() + ' ' + user.getLastName() }
                                       ));

    return new ModelAndView(getSuccessView());
}

```

Listing 4.41

```

<c:if test="${not empty message}">
    <div class="message">${message}</div>
    <c:remove var="message" scope="session"/>
</c:if>

```

Listing 4.42

```
userManager.saveUser(user);
String message = getMessageSourceAccessor()
    .getMessage("user.saved",
        new Object[] { user.getFirstName() + ' ' +
            user.getLastName() });

return new ModelAndView(getSuccessView(), "message", message);
```

Listing 4.43

```
<c:if test="${not empty param.message}">
    <div class="message">${param.message}</div>
</c:if>
```

Listing 4.44

```
<td>
    <input type="submit" class="button" name="save" value="Save"/>
    <input type="submit" class="button" name="cancel"
        value="Cancel"/>
</td>
```

Listing 4.45

```
/**
 * Redirect to the successView when cancel button has been pressed.
 */
public ModelAndView processFormSubmission(HttpServletRequest request,
                                           HttpServletResponse response,
                                           Object command,
                                           BindException errors)
    throws Exception {
    if (request.getParameter("cancel") != null) {
        return new ModelAndView(getSuccessView());
    }

    return super.processFormSubmission(request, response, command,
        errors);
}
```

Listing 4.46

```
public void testDelete() throws Exception {
    request = new MockHttpServletRequest("POST", "/editUser.html");
    request.addParameter("delete", "");
    request.addParameter("id", user.getId().toString());
    mv = c.handleRequest(request, new MockHttpServletResponse());
    assertNotNull(request.getSession().getAttribute("message"));
}
```

Listing 4.47

```
public ModelAndView onSubmit(HttpServletRequest request,
                             HttpServletResponse response, Object
command,
                             BindException errors)
    throws Exception {
    if (log.isDebugEnabled()) {
        log.debug("entering 'onSubmit' method...");
    }

    User user = (User) command;

    if (request.getParameter("delete") != null) {
        userManager.removeUser(user.getId().toString());
        request.getSession().setAttribute("message",
            getMessageSourceAccessor().
                getMessage("user.deleted",
                    new Object[] {
                        user.getFirstName() + ' ' +
                        user.getLastName() }));
    } else {
        userManager.saveUser(user);
        request.getSession().setAttribute("message",
            getMessageSourceAccessor().getMessage("user.saved",
                new Object[] {user.getFirstName() + ' ' +
                    user.getLastName() }));
    }

    return new ModelAndView(getSuccessView());
}
```

Listing 4.48

```
<input type="submit" class="button" name="save" value="Save"/>
<c:if test="${not empty param.id}">
  <input type="submit" class="button" name="delete" value="Delete"/>
</c:if>
<input type="submit" class="button" name="cancel" value="Cancel"/>
```

Listing 4.49

```
<form-validation>
  <formset>
    <form name="user">
      <field property="lastName" depends="required">
        <arg0 key="user.lastName"/>
      </field>
    </form>
  </formset>
</form-validation>
```

Listing 4.50

```
<html:javascript formName="user"/>
```

Listing 4.51

```
<form method="post" action="<c:url value="/editUser.html"/>"
  onsubmit="return validateUser(this)">
```

Listing 4.52

```
<input type="submit" class="button" name="cancel" value="Cancel"
  onclick="bCancel=true"/>
```

Listing 4.53

```
<bean id="validatorFactory"
  class="org.springframework.modules.commons.validator.DefaultValidatorFactory">
  <property name="validationConfigLocations">
    <list>
      <value>/WEB-INF/validator-rules.xml</value>
      <value>/WEB-INF/validation.xml</value>
    </list>
  </property>
</bean>
```

```
<bean id="beanValidator"
      class="org.springframework.validation.DefaultBeanValidator">
  <property name="validatorFactory" ref="validatorFactory"/>
</bean>
```

Listing 4.54

```
<property name="validator" ref="beanValidator"/>
```

Listing 4.55

```
<spring:bind path="user.*">
  <c:if test="{not empty status.errorMessages}">
    <div class="error">
      <c:forEach var="error" items="{status.errorMessages}">
        <c:out value="{error}" escapeXml="false"/><br/>
      </c:forEach>
    </div>
  </c:if>
</spring:bind>
```

Advanced Spring MVC

Code

Listing 5.1

```
<filter>
  <filter-name>sitemesh</filter-name>
  <filter-class>
    com.opensymphony.module.sitemesh.filter.PageFilter
  </filter-class>
</filter>
```

Listing 5.2

```
<filter-mapping>
  <filter-name>sitemesh</filter-name>
  <url-pattern>/*</url-pattern>
  <dispatcher>REQUEST</dispatcher>
  <dispatcher>FORWARD</dispatcher>
</filter-mapping>
```

Listing 5.3

```
<sitemesh>
  <page-parsers>
    <parser default="true"

class="com.opensymphony.module.sitemesh.parser.FastPageParser"/>
    <parser content-type="text/html"

class="com.opensymphony.module.sitemesh.parser.FastPageParser"/>
    <parser content-type="text/html;charset=ISO-8859-1"

class="com.opensymphony.module.sitemesh.parser.FastPageParser"/>
```

```

    </page-parsers>

    <decorator-mappers>
        <mapper

class="com.opensymphony.module.sitemesh.mapper.ConfigDecoratorMapper">
            <param name="config" value="/WEB-INF/decorators.xml"/>
        </mapper>
    </decorator-mappers>
</sitemesh>

```

Listing 5.4

```

<decorators defaultdir="/decorators">
    <decorator name="default" page="default.jsp">
        <pattern>*/</pattern>
    </decorator>
</decorators>

```

Listing 5.5

```

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
    "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<%@ include file="/taglibs.jsp"%>

<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en">
<head>
    <title><decorator:title default="MyUsers"/></title>
    <meta http-equiv="content-type" content="text/html; charset=utf-8"/>
    <link href="{ctx}/styles/global.css" type="text/css"
rel="stylesheet"/>
    <link href="{ctx}/images/favicon.ico" rel="SHORTCUT ICON"/>
    <decorator:head/>
    <!-- HTML & Design contributed by Boer Attila (http://www.calcium.ro)
-->
    <!-- Found at http://www.csszengarden.com/?cssfile=/083/
083.css&page=2 -->
</head>
<body>
<a name="top"></a>
<div id="container">
    <div id="intro">
        <div id="pageHeader">
            <h1><span>Welcome to Equinox</span></h1>
            <div id="logo" onclick="location.href='{<c:url value="/" />}'"
onkeypress="location.href='{<c:url value="/" />}'"></div>
            <h2><span>Spring Rocks!</span></h2>
        </div>

```



```

        <div id="quickSummary">
            <p>
                <strong>Equinox</strong> is a lightweight version of
                <a href="http://raibledesigns.com/appfuse">AppFuse</a>
designed
                to accelerate starting a webapp with the
                <a href="http://www.springframework.org">Spring
Framework</a>.
            </p>
            <p class="credit">
                <a href="http://www.csszengarden.com/?cssfile=/083/
083.css">
                Design and CSS</a> donated by <a href="http://
www.calcium.ro">
                Bo&eacute;r Attila</a>.
            </p>
        </div>

        <div id="content">
            <%@ include file="/messages.jsp"%>
            <decorator:body/>
        </div>
    </div>

    <div id="supportingText">
        <div id="underground">
            <decorator:getProperty property="page.underground"/>
        </div>
        <div id="footer"></div>

    </div>

    <div id="linkList">
        <div id="linkList2">
            </div>
        </div>

    </div>

</body>
</html>

```

Listing 5.6

```

    <content tag="underground">
        <h3>Additional Information</h3>
        <!-- more content here -->
    </content>

```

Listing 5.7

```
<%@ taglib uri="http://www.opensymphony.com/sitemesh/decorator"
    prefix="decorator" %>
```

Listing 5.8

```
<bean id="tilesConfigurer"
    class="org.springframework.web.servlet.view.tiles.TilesConfigurer">
    <property name="factoryClass"
        value="org.apache.struts.tiles.xmlDefinition.I18nFactorySet"/>
    <property name="definitions">
        <list>
            <value>/WEB-INF/tiles-config.xml</value>
        </list>
    </property>
</bean>
```

Listing 5.9

```
<bean id="viewResolver"
    class="org.springframework.web.servlet.view.InternalResourceViewResolver"
    >
    <property name="requestContextAttribute" value="rc"/>
    <property name="viewClass"
        value="org.springframework.web.servlet.view.tiles.TilesJstlView"/>
</bean>
```

Listing 5.10

```
        return new ModelAndView("userList", "users",
            mgr.getUsers());
```

Listing 5.11

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
    "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<%@ include file="/taglibs.jsp"%>

<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en">
<head>
    <title><tiles:getAsString name="title"/></title>
    <meta http-equiv="content-type" content="text/html; charset=utf-8"/>
    <link href="{ctx}/styles/global.css" type="text/css"
        rel="stylesheet"/>
```

```

    <link href="${ctx}/images/favicon.ico" rel="SHORTCUT ICON"/>
    <!-- HTML & Design contributed by Boer Attila (http://www.calcium.ro)
-->
    <!-- Found at http://www.csszengarden.com/?cssfile=/083/
083.css&page=2 -->
</head>
<body>
<a name="top"></a>
<div id="container">
    <div id="intro">
        <div id="pageHeader">
            <h1><span>Welcome to Equinox</span></h1>
            <div id="logo" onclick="location.href='<c:url value="/" />'"
                onkeypress="location.href='<c:url value="/" />'"></div>
            <h2><span>Spring Rocks!</span></h2>
        </div>

        <div id="quickSummary">
            <p>
                <strong>Equinox</strong> is a lightweight version of
                <a href="http://raibledesigns.com/appfuse">AppFuse</a>
designed
                to accelerate starting a webapp with the
                <a href="http://www.springframework.org">Spring
Framework</a>.
            </p>
            <p class="credit">
                <a href="http://www.csszengarden.com/?cssfile=/083/
083.css">
                Design and CSS</a> donated by <a href="http://
www.calcium.ro">
                Bo&eacute;r Attila</a>.
            </p>
        </div>

        <div id="content">
            <%@ include file="/messages.jsp"%>
            <tiles:insert attribute="content"/>
        </div>
    </div>

    <div id="supportingText">
        <div id="underground">
            <c:out value="${underground}" escapeXml="false"/>
        </div>
        <div id="footer"></div>

    </div>

    <div id="linkList">
        <div id="linkList2">

```

```

        </div>
    </div>

</div>

</body>
</html>

```

Listing 5.12

```

<%@ page language="java" errorPage="/error.jsp" %>
<%@ taglib uri="http://java.sun.com/jstl/core_rt" prefix="c" %>
<%@ taglib uri="http://java.sun.com/jstl/fmt" prefix="fmt" %>
<%@ taglib uri="http://www.springframework.org/tags/commons-validator"
    prefix="html" %>
<%@ taglib uri="http://www.springframework.org/tags" prefix="spring" %>
<%@ taglib uri="http://jakarta.apache.org/struts/tags-tiles"
    prefix="tiles" %>

```

Listing 5.13

```

<?xml version="1.0" encoding="UTF-8" ?>
<!DOCTYPE tiles-definitions PUBLIC
    "-//Apache Software Foundation//DTD Tiles Configuration 1.1//EN"
    "http://jakarta.apache.org/struts/dtds/tiles-config_1_1.dtd">

<tiles-definitions>
    <!-- base layout definition -->
    <definition name="baseLayout" path="/layouts/baseLayout.jsp">
        <put name="title" value="MyUsers"/>
    </definition>

    <!-- index definition -->
    <definition name="index" extends="baseLayout">
        <put name="title" value="MyUsers ~ Welcome"/>
        <put name="content" value="/index.jsp"/>
    </definition>

    <!-- user list definition -->
    <definition name="userList" extends="baseLayout">
        <put name="title" value="MyUsers ~ User List"/>
        <put name="content" value="/userList.jsp"/>
    </definition>

    <!-- user form definition -->
    <definition name="userForm" extends="baseLayout">

```

```

        <put name="title" value="MyUsers ~ User Details"/>
        <put name="content" value="/userForm.jsp"/>
    </definition>
</tiles-definitions>

```

Listing 5.14

```

<bean id="filenameController"
class="org.springframework.web.servlet.mvc.UrlFilenameViewContro
ller"/>

```

Listing 5.15

```

<prop key="/index.html">filenameController</prop>

```

Listing 5.16

```

<%@ include file="/taglibs.jsp"%>

<tiles:insert definition="index"/>

```

Listing 5.17

```

<definition name="index" extends="baseLayout">
    <put name="title" value="MyUsers ~ Welcome"/>
    <put name="content" value="/welcome.jsp"/>
</definition>

```

Listing 5.18

```

<filter>
    <filter-name>UrlRewriteFilter</filter-name>
    <filter-class>
        org.tuckey.web.filters.urlrewrite.UrlRewriteFilter
    </filter-class>
</filter>

```

Listing 5.19

```

<filter-mapping>
    <filter-name>UrlRewriteFilter</filter-name>
    <url-pattern>/*</url-pattern>
    <dispatcher>REQUEST</dispatcher>
    <dispatcher>FORWARD</dispatcher>
</filter-mapping>

```

Listing 5.20

```
<?xml version="1.0" encoding="utf-8"?>
<!DOCTYPE urlrewrite PUBLIC "-//tuckey.org//DTD UrlRewrite 1.0//EN"
    "http://tuckey.org/res/dtd/urlrewrite1.dtd">

<urlrewrite>
  <rule>
    <from>/${</from>
    <to type="forward">index.html</to>
  </rule>
  <rule>
    <from>/index.jsp</from>
    <to type="forward">index.html</to>
  </rule>
</urlrewrite>
```

Listing 5.21

```
<c:set var="underground" scope="request">
...
</c:set>
```

Listing 5.22

```
<c:out value="${underground}" escapeXml="false"/>
```

Listing 5.23

```
/**
 * Return whether or not this object can validate objects
 * of the given class.
 */
boolean supports(Class clazz);

/**
 * Validate an object, which must be of a class for which
 * the supports() method returned true.
 * @param obj Populated object to validate
 * @param errors Errors object we're building. May contain
 * errors for this field relating to types.
 */
void validate(Object obj, Errors errors);
```

Listing 5.24

```
<form method="post" action="<c:url value="/editUser.html"/>">
```

Listing 5.25

```
package org.appfuse.web;

// use your IDE to organize imports

public class UserValidator implements Validator {
    private Log log = LogFactory.getLog(UserValidator.class);
    public boolean supports(Class clazz) {
        return clazz.equals(User.class);
    }

    public void validate(Object obj, Errors errors) {
        if (log.isDebugEnabled()) {
            log.debug("entering 'validate' method...");
        }

        User user = (User) obj;

        ValidationUtils.rejectIfEmptyOrWhitespace(errors,
            "lastName", "errors.required", "Value required.");
    }
}
```

Listing 5.26

```
<bean id="userValidator"
class="org.appfuse.web.UserValidator"/>
```

Listing 5.27

```
<property name="validator" ref="userValidator"/>
```

Listing 5.28

```
<form method="post" action="<c:url value="/editUser.html"/>"
onsubmit="return validateUser(this)">
```

Listing 5.29

```
<html:javascript formName="user"
    staticJavaScript=false xhtml=true cdata=false/>
<script type="text/javascript"
    src="<c:url value="/scripts/validator.jsp"/>"></script>
```

Listing 5.30

```
<%@ page language="java" contentType="javascript/x-javascript" %>
<%@ taglib uri="http://www.springmodules.org/tags/commons-validator"
prefix="html" %>

<html:javascript dynamicJavascript="false" staticJavascript="true"/>
```

Listing 5.31

```
<target name="webdoclet"
  description="Generate web deployment descriptors">
  <taskdef name="webdoclet"
    classname="xdoclet.modules.web.WebDocletTask">
    <classpath>
      <path refid="xdoclet.classpath"/>
    </classpath>
  </taskdef>
  <webdoclet destdir="${webapp.target}/WEB-INF"
    force="${xdoclet.force}"
    mergedir="metadata/web"
    excludedtags="@version,@author"
    verbose="true">
    <fileset dir="src"/>
    <springvalidationxml/>
  </webdoclet>
</target>
```

Listing 5.32

```
/**
 * @spring.validator type="required"
 */
public void setLastName(String lastName) {
    this.lastName = lastName;
}
```

Listing 5.33

```
<property name="validators">
  <list>
    <ref bean="beanValidator"/>
    <ref bean="userValidator"/>
  </list>
</property>
```


Listing 5.34

```
public User saveUser(User user) {
    BindException errors = new BindException(user, "user");
    new UserValidator().validate(user, errors);
    if (errors.hasErrors()) {
        throw new RuntimeException("validation failed!", errors);
    }
    dao.saveUser(user);

    return user;
}
```

Listing 5.35

```
public void testWithValidationErrors() {
    user = new User();
    user.setFirstName("Bill");

    try {
        user = mgr.saveUser(user);
        fail("Validation exception not thrown!");
    } catch (Exception e) {
        log.debug(e.getCause().getMessage());
        assertNotNull(e.getCause());
    }
}
```

Listing 5.36

```
<error-page>
    <error-code>404</error-code>
    <location>/404.jsp</location>
</error-page>
```

Listing 5.37

```
<error-page>
    <error-code>500</error-code>
    <location>/error.jsp</location>
</error-page>
```

Listing 5.38

```
<error-page>
    <exception-type>
        org.appfuse.service.UserNotFound
    </exception-type>
    <location>/userNotFound.html</location>
</error-page>
```

Listing 5.39

```
<bean id="exceptionResolver"

class="org.springframework.web.servlet.handler.SimpleMappingExceptionRes
olver">
    <property name="exceptionMappings">
        <props>
            <prop key="org.springframework.dao.DataAccessException">

                dataAccessFailure
            </prop>
        </props>
    </property>
</bean>
```

Listing 5.40

```
assertNull(dao.getUser(user.getId()));
```

Listing 5.41

```
try {
    user = dao.getUser(user.getId());
    fail("User found in database");
} catch (DataAccessException dae) {
    log.debug("Expected exception: " + dae.getMessage());
    assertNotNull(dae);
}
```

Listing 5.42

```
public User getUser(Long id) {
    User user = (User) getHibernateTemplate().get(User.class, id);
    if (user == null) {
```

```

        throw new ObjectRetrievalFailureException(User.class, id);
    }
    return user;
}

```

Listing 5.43

```

<%@ include file="/taglibs.jsp" %>

<h3>Data Access Failure</h3>
<p>
    <c:out value="${requestScope.exception.message}"/>
</p>

<!--
<%
Exception ex = (Exception) request.getAttribute("exception");
ex.printStackTrace(new java.io.PrintWriter(out));
%>
-->

<a href="<c:url value='/'/>">&#171; Home</a>

```

Listing 5.44

```

<definition name="dataAccessFailure" extends="baseLayout">
    <put name="title" value="Data Access Failure"/>
    <put name="content" value="/dataAccessFailure.jsp"/>
</definition>

```

Listing 5.45

```

<bean id="multipartResolver"

class="org.springframework.web.multipart.commons.CommonsMultipartResolve
r"/>

```

Listing 5.46

```

package org.appfuse.web;

public class FileUpload {
    private byte[] file;

    public void setFile(byte[] file) {
        this.file = file;
    }
}

```

```
    public byte[] getFile() {  
        return file;  
    }  
}
```

Listing 5.47

```
package org.appfuse.web;  
  
// use your IDE to organize imports  
  
public class FileUploadController extends SimpleFormController {  
  
    private static Log log =  
LogFactory.getLog(FileUploadController.class);  
  
    public FileUploadController() {  
  
        super();  
  
        setCommandClass(FileUpload.class);  
    }  
  
    protected void initBinder(HttpServletRequest request,  
                               ServletRequestDataBinder binder)  
        throws ServletException {  
  
        binder.registerCustomEditor(byte[].class,  
                                     new ByteArrayMultipartFileEditor());  
    }  
  
    protected ModelAndView onSubmit(HttpServletRequest request,  
                                     HttpServletResponse response,
```

```

        Object command,

        BindException errors)

throws ServletException, IOException {

    FileUpload bean = (FileUpload) command;

    byte[] bytes = bean.getFile();

    // cast to multipart file so we can get additional information
    MultipartHttpServletRequest multipartRequest =
        (MultipartHttpServletRequest) request;
    CommonsMultipartFile file =
        (CommonsMultipartFile) multipartRequest.getFile("file");

    String uploadDir = getServletContext().getRealPath("/upload/");

    // Create the directory if it doesn't exist
    File dirPath = new File(uploadDir);

    if (!dirPath.exists()) {
        dirPath.mkdirs();
    }

    String sep = System.getProperty("file.separator");
    if (log.isDebugEnabled()) {
        log.debug("uploading to: " + uploadDir + sep +

```

```

        file.getOriginalFilename());

    }

    File uploadedFile = new File(uploadDir + sep +
                                   file.getOriginalFilename());

    FileCopyUtils.copy(bytes, uploadedFile);

    // set success message

    request.getSession().setAttribute("message", "Upload
completed.");

    String url = request.getContextPath() + "/upload/" +
                file.getOriginalFilename();

    Map model = new HashMap();

    model.put("filename", file.getOriginalFilename());

    model.put("url", url);

    return new ModelAndView(getSuccessView(), "model", model);
}
}

```

Listing 5.48

```

<bean id="fileUploadController"
      class="org.appfuse.web.FileUploadController">
  <property name="commandClass">
    <value>org.appfuse.web.FileUpload</value>
  </property>
  <property name="formView"><value>fileUpload</value></property>

```

```

        <property name="successView">
            <value>fileUpload</value>
        </property>
    </bean>

```

Listing 5.49

```

<prop key="/fileUpload.html">fileUploadController</prop>

```

Listing 5.50

```

<%@ include file="/taglibs.jsp"%>

<h3>File Upload</h3>

<c:if test="${not empty model.filename}">
    <p style="font-weight: bold">
        Uploaded file (click to view): <a
        href="${model.url}">${model.filename}</a>
    </p>
</c:if>

<p>Select a file to upload:</p>
<form method="post" action="fileUpload.html" enctype="multipart/form-
data">
    <input type="file" name="file"/><br/>
    <input type="submit" value="Upload" class="button"
        style="margin-top: 5px"/>
</form>

```

Listing 5.51

```

<definition name="fileUpload" extends="baseLayout">
    <put name="title" value="My Users ~ File Upload"/>
    <put name="content" value="/fileUpload.jsp"/>
</definition>

```

Listing 5.52

```

<security-constraint>
    <web-resource-collection>
        <url-pattern>/*</url-pattern>
    </web-resource-collection>

    <auth-constraint>
        <role-name>tomcat</role-name>
        <role-name>manager</role-name>
    </auth-constraint>
</security-constraint>

```

```

        </auth-constraint>
    </security-constraint>

    <login-config>
        <auth-method>BASIC</auth-method>
        <realm-name>My Users</realm-name>
    </login-config>

    <security-role>
        <role-name>tomcat</role-name>
    </security-role>

    <security-role>
        <role-name>manager</role-name>
    </security-role>

```

Listing 5.53

```

    <bean id="managersOnly"

class="org.springframework.web.servlet.handler.UserRoleAuthorizationInte
rceptor">
        <property name="authorizedRoles" value="manager"/>
    </bean>

```

Listing 5.54

```

    <bean id="managerMappings"

class="org.springframework.web.servlet.handler.SimpleUrlHandlerMapping">
        <property name="interceptors">
            <list>
                <ref bean="managersOnly"/>
            </list>
        </property>
        <property name="mappings">
            <props>
                <prop key="/fileUpload.html">
                    fileUploadController
                </prop>
            </props>
        </property>
    </bean>

```


Listing 5.55

```
<bean id="mailSender"
      class="org.springframework.mail.javamail.JavaMailSenderImpl">
    <property name="host" value="localhost"/>
</bean>
```

Listing 5.56

```
private MailSender mailSender;
private SimpleMailMessage message;

public void setMailSender(MailSender mailSender) {
    this.mailSender = mailSender;
}

public void setMessage(SimpleMailMessage message) {
    this.message = message;
}
```

Listing 5.57

```
<bean id="mailMessage"
      class="org.springframework.mail.SimpleMailMessage">
    <property name="from">
        <!-- The <value> and CDATA below must be on the same line -->
        <value><![CDATA[Uploader <springlive@sourcebeat.com>]]></
value>
    </property>
    <property name="subject">
        <value>File finished uploading</value>
    </property>
</bean>
```

Listing 5.58

```
<bean id="fileUploadController"
      class="org.appfuse.web.FileUploadController">
    <property name="formView" value="fileUpload"/>
    <property name="successView" value="fileUpload"/>
    <property name="mailSender" ref="mailSender"/>
    <property name="message" ref="mailMessage"/>
</bean>
```

Listing 5.59

```
// Notify user that file has finished uploading
SimpleMailMessage msg = new SimpleMailMessage(this.message);
msg.setTo("matt@raibledesigns.com");
msg.setText("File \"" + file.getOriginalFilename() +
           "\" has finished uploading.");
try {
    mailSender.send(msg);
} catch (MailException ex) {
    log.error(ex.getMessage());
}
```

View Options

Code

Listing 6.1

```
package org.appfuse.web;

// use your IDE to organize imports

public class UserWebTest extends WebTestCase {

    public UserWebTest(String name) {
        super(name);
        getTestContext().setBaseUrl("http://localhost:8080/myusers");
    }

    public void testWelcomePage() {
        beginAt("/");
        assertEquals("MyUsers ~ Welcome");
    }

    public void testAddUser() {
        beginAt("/editUser.html");
        assertEquals("MyUsers ~ User Details");
        setFormElement("id", "");
        setFormElement("firstName", "Spring");
        setFormElement("lastName", "User");
        submit("save");
        assertTextPresent("saved successfully");
    }

    public void testListUsers() {
        beginAt("/users.html");

        // check that table is present
        assertTablePresent("userList");
    }
}
```

```

//check that a set of strings are present somewhere in table
assertTextInTable("userList",
    new String[] {"Spring", "User"});
}

public void testEditUser() {
    beginAt("/editUser.html?id=" + getInsertedUserId());
    assertFormElementEquals("firstName", "Spring");
    submit("save");
    assertTitleEquals("MyUsers ~ User List");
}

public void testDeleteUser() {
    beginAt("/editUser.html?id=" + getInsertedUserId());
    assertTitleEquals("MyUsers ~ User Details");
    submit("delete");
    assertTitleEquals("MyUsers ~ User List");
}

/**
 * Convenience method to get the id of the inserted user
 * Assumes last user in table is "Spring User"
 */
protected String getInsertedUserId() {
    beginAt("/users.html");
    assertTablePresent("userList");
    assertTextInTable("userList", "Spring");
    String[][] sparseTableCellValues =
        getDialog().getSparseTableBySummaryOrId("userList");
    return sparseTableCellValues[sparseTableCellValues.length-1][0];
}
}

```

Listing 6.2

```
assertTablePresent("userList");
```

Listing 6.3

```
<table class="list" id="userList">
```

Listing 6.4

```

<batchtest todir="${test.dir}/data" unless="testcase">
  <fileset dir="${test.dir}/classes">

```

```

    <include name="**/*Test.class"/>
    <exclude name="**/*WebTest.class"/>
  </fileset>
</batchtest>

```

Listing 6.5

```

<target name="test-web" depends="compile"
  description="Runs tests that required a running server">
  <property name="testcase" value="WebTest"/>
  <antcall target="test"/>
</target>

```

Listing 6.6

```

<bean id="viewResolver"
  class="org.springframework.web.servlet.view.InternalResourceViewResolver"
  >
  <property name="requestContextAttribute" value="rc"/>
  <property name="viewClass"
    value="org.springframework.web.servlet.view.JstlView"/>
  <property name="prefix" value="/" />
  <property name="suffix" value=".jsp"/>
</bean>

```

Listing 6.7

```

<property name="prefix" value="/WEB-INF/jsp/" />

```

Listing 6.8

```

<property name="viewClass"
  value="org.springframework.web.servlet.view.JstlView"/>

```

Listing 6.9

```

<%@ include file="/taglibs.jsp"%>
<%@ taglib uri="http://displaytag.sf.net" prefix="display" %>

<head>
  <title>MyUsers ~ User List</title>
  <link href="styles/displaytag.css" type="text/css" rel="stylesheet"/>
</head>

<button onclick="location.href='editUser.html'">Add User</button>

```

```

<display:table name="${users}" class="list" requestURI="" id="userList"
  export="true">
  <display:column property="id" sort="true" href="editUser.html"
    paramId="id" paramProperty="id" titleKey="user.id"/>
  <display:column property="firstName" sort="true"
    titleKey="user.firstName"/>
  <display:column property="lastName" sort="true"
    titleKey="user.lastName"/>
  <display:setProperty name="basic.empty.showtable" value="true"/>
</display:table>

```

Listing 6.10

```
export.pdf=true
```

Listing 6.11

```

<property name="viewClass"
  value="org.springframework.web.servlet.view.tiles.TilesJstlView"/>

```

Listing 6.12

```

<bean id="tilesConfigurer"
  class="org.springframework.web.servlet.view.tiles.TilesConfigurer">
  <property name="factoryClass"
    value="org.apache.struts.tiles.xmlDefinition.I18nFactorySet"/>
  <property name="definitions">
    <list>
      <value>/WEB-INF/tiles-config.xml</value>
    </list>
  </property>
</bean>

```

Listing 6.13

```

<bean id="velocityConfig"
  class="org.springframework.web.servlet.view.velocity.Velocity
Configurer">
  <property name="resourceLoaderPath" value="/" />
</bean>

```

Listing 6.14

```
<bean id="viewResolver"
class="org.springframework.web.servlet.view.velocity.VelocityViewResolve
r">
    <property name="exposeSpringMacroHelpers" value="true"/>
    <property name="requestContextAttribute" value="rc"/>
    <property name="cache" value="true"/>
    <property name="prefix" value="/WEB-INF/velocity/">
    <property name="suffix" value=".vm"/>
</bean>
```

Listing 6.15

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
    "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<%@ include file="/taglibs.jsp"%>

<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en">
<head>
    <title>${title}</title>
    <meta http-equiv="content-type" content="text/html; charset=utf-8"/>
    <link href="${base}/styles/global.css" type="text/css"
rel="stylesheet"/>
    <link href="${base}/images/favicon.ico" rel="SHORTCUT ICON"/>
    ${head}
    ...
    <h1><span>Welcome to MyUsers</span></h1>
    <div id="logo" onclick="location.href='${base}'"
        onkeypress="location.href='${base}'"></div>
    <h2><span>Spring Rocks!</span></h2>
</div>
    ...
    <div id="content">
        #parse("/messages.vm")
        ${body}
    </div>
</div>

    <div id="supportingText">
        <div id="underground">${!{page.getProperty("page.underground")}}</
div>

        <div id="footer">
```

```

        ...
        <a href="http://bobby.watchfire.com/bobby/
bobbyServlet?URL=${request.requestURL}&output=Submit&gl=sec508&a
mp;test=" title="Check the accessibility of this site according to U.S.
Section 508">508</a> &middot;
        <a href="http://bobby.watchfire.com/bobby/
bobbyServlet?URL=${request.requestURL}&output=Submit&gl=wcag1-
aaa&test=" title="Check the accessibility of this site according to
WAI Content Accessibility Guidelines 1">aaa</a>
    </div>

```

Listing 6.16

```

## Success Messages
#if ($message)
    <div class="message">${message}</div>
    ${request.session.removeAttribute("message")}
#end

```

Listing 6.17

```

<decorators defaultdir="/decorators">
    <decorator name="default" page="default.vm">
        <pattern>/*</pattern>
    </decorator>
</decorators>

```

Listing 6.18

```

<servlet>
    <servlet-name>sitemesh-velocity</servlet-name>
    <servlet-
class>com.opensymphony.module.sitemesh.velocity.VelocityDecoratorServlet
</servlet-class>
</servlet>

<servlet-mapping>
    <servlet-name>sitemesh-velocity</servlet-name>
    <url-pattern>*.vm</url-pattern>
</servlet-mapping>

```

Listing 6.19: userList.vm

```

<title>MyUsers ~ User List</title>

<button onclick="location.href='editUser.html'">Add User</button>

```



```

<table class="list" id="userList">
<thead>
<tr>
    <th>${rc.getMessage("user.id")}</th>
    <th>${rc.getMessage("user.firstName")}</th>
    <th>${rc.getMessage("user.lastName")}</th>
</tr>
</thead>
<tbody>
#foreach ($user in $users)
#if ($velocityCount % 2 == 0) <tr class="even">
#else <tr class="odd">
#end
    <td><a href="editUser.html?id=${user.id}">${user.id}</a></td>
    <td>${user.firstName}</td>
    <td>${user.lastName}</td>
</tr>
#end
</tbody>
</table>

```

Listing 6.20: userForm.vm

```

<title>MyUsers ~ User Details</title>

#springBind("user.*")
#if ($status.error)
<div class="error">
    #foreach ($error in $status.errorMessages)
        ${error}<br/>
    #end
</div>
#end

<p>Please fill in user's information below:</p>

<form method="post" action="editUser.html">
#springBind("user.id")
<input type="hidden" name="id" value="${status.value}"/>
<table>
<tr>
    <th>${rc.getMessage("user.firstName")}</th>
    <td>
        #springBind("user.firstName")
        <input type="text" name="firstName" value="${status.value}"/>
        <span class="fieldError">${status.errorMessage}</span>
    </td>
</tr>
<tr>
    <th>${rc.getMessage("user.lastName")}</th>

```

```

        <td>
            #springBind("user.lastName")
            <input type="text" name="lastName" value="${status.value}"/>
            <span class="fieldError">${status.errorMessage}</span>
        </td>
    </tr>
    <tr>
        <td></td>
        <td>
            <input type="submit" class="button" name="save" value="Save"/>
            #if ($user.id)
                <input type="submit" class="button" name="delete" value="Delete"/>
            >
            #end
            <input type="submit" class="button" name="cancel" value="Cancel"/>
        </td>
    </tr>
</table>
</form>

```

Listing 6.21

```

#set($springXhtmlCompliant = true)

<form method="post" action="#springUrl('editUser.html')">
    #springFormHiddenInput("user.id" '')
    <table>
    <tr>
        <th><label for="firstName">
            #springMessage("user.firstName"):</label></th>
        <td>
            #springFormInput("user.firstName" 'id="firstName"')
            #springShowErrors("<br/>" "fieldError")
        </td>
    </tr>
    <tr>
        <th><label for="lastName">
            #springMessage("user.lastName"):</label></th>
        <td>
            #springFormInput("user.lastName" 'id="lastName"')
            #springShowErrors("<br/>" "fieldError")
        </td>
    </tr>
    </table>
</form>

```

Listing 6.22: fileUpload.vm

```
<h3>File Upload</h3>

#if ($model.filename)
<p style="font-weight: bold">
    Uploaded file (click to view):
    <a href="{model.url}">${model.filename}</a>
</p>
#end

<p>Select a file to upload:</p>
<form method="post" action="#springUrl('fileUpload.html')"
    enctype="multipart/form-data">
    <input type="file" name="file"/><br/>
    <input type="submit" value="Upload" class="button"
        style="margin-top: 5px"/>
</form>
```

Listing 6.23: dataAccessFailure.vm

```
<h3>Data Access Failure</h3>
<p>
    ${exception}
</p>

<a href="{rc.contextPath}">&#171; Home</a>
```

Listing 6.24

```
<bean id="freemarkerConfig"
class="org.springframework.web.servlet.view.freemarker.FreeMarkerConfigu
rer">
    <property name="templateLoaderPath" value="/" />
</bean>
```

Listing 6.25

```
<bean id="viewResolver"
class="org.springframework.web.servlet.view.freemarker.FreeMarkerViewRes
olver">
    <property name="exposeSpringMacroHelpers" value="true"/>
    <property name="requestContextAttribute" value="rc"/>
    <property name="prefix" value="/WEB-INF/freemarker/" />
    <property name="suffix" value=".ftl"/>
</bean>
```

Listing 6.26

```
<div id="content">
    <#include "/messages.ftl"/>
    ${body}
</div>
</div>

<div id="supportingText">
    <div id="underground">
        <#if page.getProperty("page.underground")?exists>
            ${page.getProperty("page.underground")}
        </#if>
    </div>
    ...
    <a href="http://bobby.watchfire.com/bobby/
bobbyServlet?URL=${Request.requestURL}&output=Submit&gl=sec508&a
mp;test=" title="Check the accessibility of this site according to U.S.
Section 508">508</a> &middot;
    <a href="http://bobby.watchfire.com/bobby/
bobbyServlet?URL=${Request.requestURL}&output=Submit&gl=wcag1-
aaa&test=" title="Check the accessibility of this site according to
WAI Content Accessibility Guidelines 1">aaa</a>
    ...

```

Listing 6.27

```
<!-- Success Messages -->
<#if message?exists>
    <div class="message">${message}</div>
</#if>

```

Listing 6.28

```
<decorators defaultdir="/decorators">
    <decorator name="default" page="default.ftl">
        <pattern>*/</pattern>
    </decorator>
</decorators>

```

Listing 6.29

```
package org.appfuse.web;

// use your IDE to organize imports

public class MessageFilter implements Filter {
    private static Log log = LogFactory.getLog(MessageFilter.class);

```

```

public void doFilter(ServletRequest req, ServletResponse res,
                    FilterChain chain)
    throws IOException, ServletException {
    HttpServletRequest request = (HttpServletRequest) req;

    // grab messages from the session and put them into request
    // this is so they're not lost in a redirect
    Object message = request.getSession().getAttribute("message");
    if (message != null) {
        request.setAttribute("message", message);
        request.getSession().removeAttribute("message");
    }

    // set the requestURL as a request attribute for templates
    // esp. freemarker, which doesn't allow request.getRequestURL()
    request.setAttribute("requestURL", request.getRequestURL());

    chain.doFilter(req, res);
}

public void init(FilterConfig filterConfig) {}

public void destroy() {}
}

```

Listing 6.30

```

<filter>
  <filter-name>messageFilter</filter-name>
  <filter-class>org.appfuse.web.MessageFilter</filter-class>
</filter>

```

Listing 6.31

```

<filter-mapping>
  <filter-name>messageFilter</filter-name>
  <url-pattern>/*</url-pattern>
</filter-mapping>

```

Listing 6.32

```

<servlet>
  <servlet-name>sitemesh-freemarker</servlet-name>
  <servlet-
class>com.opensymphony.module.sitemesh.freemarker.FreemarkerDecoratorSer
vlet</servlet-class>
  <init-param>

```

```

        <param-name>TemplatePath</param-name>
        <param-value></param-value>
    </init-param>
    <init-param>
        <param-name>default_encoding</param-name>
        <param-value>ISO-8859-1</param-value>
    </init-param>
</servlet>

<servlet-mapping>
    <servlet-name>sitemesh-freemarker</servlet-name>
    <url-pattern>*.ftl</url-pattern>
</servlet-mapping>

```

Listing 6.33: userList.ftl

```

<title>MyUsers ~ User List</title>

<button onclick="location.href='editUser.html'">Add User</button>

<table class="list" id="userList">
<thead>
<tr>
    <th>${rc.getMessage("user.id")}</th>
    <th>${rc.getMessage("user.firstName")}</th>
    <th>${rc.getMessage("user.lastName")}</th>
</tr>
</thead>
<tbody>
<#list users as user>
<#if user.index % 2 == 0> <tr class="even">
<#else> <tr class="odd">
</#if>
    <td><a href="editUser.html?id=${user.id}">${user.id}</a></td>
    <td>${user.firstName}</td>
    <td>${user.lastName}</td>
</tr>
</#list>
</tbody>
</table>

```

Listing 6.34: userForm.ftl

```

<#import "/spring.ftl" as spring/>

<title>MyUsers ~ User Details</title>

<@spring.bind "user.*"/>
<#if spring.status.error>

```

```

<div class="error">
  <#list spring.status.errorMessages as error>
    ${error}<br/>
  </#list>
</div>
</#if>

<p>Please fill in user's information below:</p>

<form method="post" action="editUser.html">
  <@spring.bind "user.id"/>
  <input type="hidden" name="id"
    value="${spring.status.value?default('')}"/>
  <table>
  <tr>
    <th><label for="firstName">
      ${rc.getMessage("user.firstName")}</label>:</th>
    <td>
      <@spring.bind "user.firstName"/>
      <input type="text" name="${spring.status.expression}"
        value="${spring.status.value?default('')}"/>
      <span class="fieldError">${spring.status.errorMessage}</span>
    </td>
  </tr>
  <tr>
    <th><label for="lastName">
      ${rc.getMessage("user.lastName")}</label>:</th>
    <td>
      <@spring.bind "user.lastName"/>
      <input type="text" name="${spring.status.expression}"
        value="${spring.status.value?default('')}"/>
      <span class="fieldError">${spring.status.errorMessage}</span>
    </td>
  </tr>
  <tr>
    <td></td>
    <td>
      <input type="submit" class="button" name="save" value="Save"/>
      <#if user.id?exists>
        <input type="submit" class="button" name="delete"
          value="Delete"/>
      </#if>
      <input type="submit" class="button" name="cancel"
        value="Cancel"/>
    </td>
  </tr>
</table>
</form>

```

Listing 6.35

```
<#assign xhtmlCompliant = true in spring>

<form method="post" action="@spring.url '/editUser.html'/">
<@spring.formHiddenInput "user.id"/>
<table>
<tr>
    <th><label for="firstName">
        <@spring.message "user.firstName"/></label>:</th>
    <td>
        <@spring.formInput "user.firstName", 'id="firstName"'/>
        <span class="fieldError">${spring.status.errorMessage}</span>
    </td>
</tr>
<tr>
    <th><label for="lastName">
        <@spring.message "user.lastName"/></label>:</th>
    <td>
        <@spring.formInput "user.lastName", 'id="lastName"'/>
        <span class="fieldError">${spring.status.errorMessage}</span>
    </td>
</tr>
</table>
```

Listing 6.36: fileUpload.ftl

```
<#import "/spring.ftl" as spring/>

<h3>File Upload</h3>

<#if model?exists>
<p style="font-weight: bold">
    Uploaded file (click to view): <a
href="${model.url}">${model.filename}</a>
</p>
</#if>

<p>Select a file to upload:</p>
<form method="post" action="@spring.url '/fileUpload.html'/"
enctype="multipart/form-data">
    <input type="file" name="file"/><br/>
    <input type="submit" value="Upload" class="button"
        style="margin-top: 5px"/>
</form>
```


Listing 6.37: dataAccessFailure.ftl

```
<h3>Data Access Failure</h3>
<p>
    ${exception}
</p>

<a href="${rc.contextPath}">&#171; Home</a>
```

Listing 6.38

```
public void testGetUsersAsXML() throws Exception {
    UserController c = (UserController) ctx.getBean("userController");
    MockHttpServletRequest request = new MockHttpServletRequest();
    request.addParameter("report", "XML");
    ModelAndView mav =
        c.handleRequest(request, (HttpServletResponse) null);
    assertEquals(mav.getViewName(), "userListXML");
}
```

Listing 6.39

```
public ModelAndView handleRequest(HttpServletRequest request,
                                HttpServletResponse response)
    throws Exception {
    if (log.isDebugEnabled()) {
        log.debug("entering 'handleRequest' method...");
    }

    String viewName = "userList";
    if (request != null &&
        request.getParameter("report") != null) {
        viewName += request.getParameter("report");
    }

    return new ModelAndView(viewName, "users", mgr.getUsers());
}
```

Listing 6.40

```
public String getFullName() {
    return firstName + ' ' + lastName;
}
```

Listing 6.41

```
package org.appfuse.web;

// user your IDE to organize imports

public class UserXMLViewTest extends TestCase {
    private static Log log = LogFactory.getLog(UserXMLViewTest.class);

    public void testXMLCreation() throws Exception {
        // setup a user to print out as XML
        User user = new User();
        user.setFirstName("James");
        user.setLastName("Strachan");
        List users = new ArrayList();
        users.add(user);
        Map model = new HashMap();
        model.put("users", users);

        // invoke the XsltView and call its 'createDomNode' method
        UserXMLView feed = new UserXMLView();
        Source source = feed.createXsltSource(model, "users",
            new MockHttpServletRequest(),
            new MockHttpServletResponse());
        Node node = ((DOMSource) source).getNode();
        assertEquals(node.getFirstChild().toString(),
            "<users><user>James Strachan</user></users>");
    }
}
```

Listing 6.42

```
package org.appfuse.web;

// use your IDE to organize imports

public class UserXMLView extends AbstractXsltView {
    protected Source createXsltSource(Map model, String rootName,
        HttpServletRequest request,
        HttpServletResponse response)

    throws Exception {
        Document doc = DocumentHelper.createDocument();
        Element root = doc.addElement(rootName);
        doc.setRootElement(root);

        List users = (List) model.get("users");
        for (Iterator it = users.iterator(); it.hasNext();) {
            User user = (User) it.next();
            root.addElement("user").addText(user.getFullName());
        }
    }
}
```

```

    }

    response.setContentType("text/xml");

    return new DOMSource(new DOMWriter().write(doc));
}
}

```

Listing 6.43

```

<?xml version="1.0"?>
<xsl:stylesheet version="1.0"
  xmlns:xsl="http://www.w3.org/1999/XSL/Transform">
  <xsl:output method="xml" omit-xml-declaration="no"/>

  <xsl:template match="/">
    <users>
      <xsl:for-each select="users/user">
        <user><xsl:value-of select="."/></user>
      </xsl:for-each>
    </users>
  </xsl:template>
</xsl:stylesheet>

```

Listing 6.44

```

<bean id="reportViewResolver"

class="org.springframework.web.servlet.view.ResourceBundleViewResolver">
  <property name="order" value="1"/>
</bean>

```

Listing 6.45

```

userListXML.class=org.appfuse.web.UserXMLView
userListXML.stylesheetLocation=/WEB-INF/xsl/users.xsl
userListXML.root=users

```

Listing 6.46

```

<p style="text-align: right; margin-bottom: -10px">
<strong>Export Options:</strong>
  <a href="?report=XML">XML</a> &middot;
  <a href="?report=Excel">Excel</a> &middot;
  <a href="?report=PDF">PDF</a>
</p>

```

Listing 6.47

```
package org.appfuse.web;

// use your IDE to organize imports

public class UserExcelView extends AbstractExcelView {

    protected void buildExcelDocument(Map model, HSSFWorkbook wb,
                                       HttpServletRequest req,
                                       HttpServletResponse resp)
        throws Exception {

        HSSFSheet sheet = wb.createSheet("My Users");
        sheet.setDefaultColumnWidth((short) 12);

        List users = (List) model.get("users");

        for (int i = 0; i < users.size(); i++) {
            HSSFCell cell = getCell(sheet, i, 0);
            setText(cell, ((User) users.get(i)).getFullName());
        }
    }
}
```

Listing 6.48

```
userListExcel.class=org.appfuse.web.UserExcelView
```

Listing 6.49

```
package org.appfuse.web;

// use your IDE to organize imports

public class UserPDFView extends AbstractPdfView {

    protected void buildPdfDocument(Map model, Document doc,
                                     PdfWriter writer,
                                     HttpServletRequest req,
                                     HttpServletResponse resp)
        throws Exception {

        List users = (List) model.get("users");

        for (int i = 0; i < users.size(); i++) {
            String fullName = ((User) users.get(i)).getFullName();
        }
    }
}
```

```
        doc.add(new Paragraph(fullName));
    }
}
```

Listing 6.50

```
userListPDF.class=org.appfuse.web.UserPDFView
```



Persistence Strategies

Code

Listing 7.1

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE hibernate-mapping PUBLIC
    "-//Hibernate/Hibernate Mapping DTD 2.0//EN"
    "http://hibernate.sourceforge.net/hibernate-mapping-2.0.dtd">

<hibernate-mapping>
    <class name="org.appfuse.model.User" table="app_user">

        <id name="id" column="id" unsaved-value="0">
            <generator class="increment"/>
        </id>
        <property name="firstName" column="first_name"
            not-null="true"/>
        <property name="lastName" column="last_name" not-null="true"/>

    </class>
</hibernate-mapping>
```

Listing 7.2

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE beans PUBLIC "-//SPRING//DTD BEAN//EN"
    "http://www.springframework.org/dtd/spring-beans.dtd">

<beans>
    <bean id="dataSource"
class="org.springframework.jdbc.datasource.DriverManagerDataSource">
        <property name="driverClassName">
            <value>org.hsqldb.jdbcDriver</value>
        </property>
        <property name="url">
```

```

        <value>jdbc:hsqldb:db/appfuse</value>
    </property>
    <property name="username"><value>sa</value></property>
    <property name="password"><value></value></property>
</bean>

<!-- Add additional bean definitions here -->
</beans>

```

Listing 7.3

```

<bean id="dataSource"
    class="org.springframework.jndi.JndiObjectFactoryBean">
    <property name="jndiName">
        <value>java:comp/env/jdbc/appfuse</value>
    </property>
</bean>

```

Listing 7.4

```

<bean id="sessionFactory"

class="org.springframework.orm.hibernate.LocalSessionFactoryBean">
    <property name="dataSource"><ref bean="dataSource"/></property>
    <property name="mappingResources">
        <list>
            <value>org/appfuse/model/User.hbm.xml</value>
        </list>
    </property>
    <property name="hibernateProperties">
    <props>
        <prop key="hibernate.dialect">
            net.sf.hibernate.dialect.HSQLDialect
        </prop>
        <prop key="hibernate.hbm2ddl.auto">update</prop>
    </props>
    </property>
</bean>

```

Listing 7.5

```

<bean id="transactionManager"

class="org.springframework.orm.hibernate.HibernateTransactionManager">
    <property name="sessionFactory">
        <ref local="sessionFactory"/>
    </property>
</bean>

```


Listing 7.6

```
package org.appfuse.dao.hibernate;

// use your IDE to organize imports

public class UserDAOHibernate extends HibernateDaoSupport implements
UserDAO {

    public List getUsers() {
        return getHibernateTemplate().find("from User");
    }

    public User getUser(Long id) {
        User user = (User) getHibernateTemplate().get(User.class, id);
        if (user == null) {
            throw new ObjectRetrievalFailureException(User.class, id);
        }
        return user;
    }

    public void saveUser(User user) {
        getHibernateTemplate().saveOrUpdate(user);
    }

    public void removeUser(Long id) {
        Object user = getHibernateTemplate().load
            (User.class, id);
        getHibernateTemplate().delete(user);
    }
}
```

Listing 7.7

```
<bean id="userDAO" class="org.appfuse.dao.hibernate.UserDAOHibernate">
<property name="sessionFactory">
    <ref local="sessionFactory"/>
</property>
</bean>
```

Listing 7.8

```
if (logger.isDebugEnabled()) {
    logger.debug("User's id set to: " + user.getId());
}
```

Listing 7.9

```
public void testGetUsers() {
    // add a record to the database so we have something to work with
    user = new User();
    user.setFirstName("Rod");
    user.setLastName("Johnson");
    dao.saveUser(user);

    List users = dao.getUsers();
    assertTrue(users.size() >= 1);
    assertTrue(users.contains(user));
}
```

Listing 7.10

```
<property name="driverClassName">
    <value>com.mysql.jdbc.Driver</value>
</property>
<property name="url">
    <value>jdbc:mysql://localhost/myusers</value>
</property>
<property name="username"><value>root</value></property>
<property name="password"><value></value></property>
```

Listing 7.11

```
<prop key="hibernate.dialect">
    net.sf.hibernate.dialect.MySQLDialect
</prop>
```

Listing 7.12

```
mysqladmin -u root -p create myusers
```

Listing 7.13

```
<class name="org.appfuse.model.User" table="app_user">
    <cache usage="read-write"/>
    <id name="id" column="id" unsaved-value="0">
```

Listing 7.14

```
<ehcache>
  <!-- Only needed if overFlowToDisk="true" -->
  <diskStore path="java.io.tmpdir"/>
  <!-- Required element -->
  <defaultCache
    maxElementsInMemory="10000"
    eternal="false"
    timeToIdleSeconds="120"
    timeToLiveSeconds="120"
    overflowToDisk="true"/>

  <!-- Cache settings per class -->
  <cache name="org.appfuse.model.User"
    maxElementsInMemory="1000"
    eternal="false"
    timeToIdleSeconds="300"
    timeToLiveSeconds="600"
    overflowToDisk="true"/>
</ehcache>
```

Listing 7.15

```
<logger name="net.sf.ehcache">
  <level value="DEBUG"/>
</logger>
```

Listing 7.16

```
<filter>
  <filter-name>hibernateFilter</filter-name>
  <filter-class>org.springframework.orm.hibernate.support.OpenSession
    InViewFilter</filter-class>
</filter>
<filter-mapping>
  <filter-name>hibernateFilter</filter-name>
  <url-pattern>*.html</url-pattern>
</filter-mapping>
```

Listing 7.17

```
[junit] net.sf.hibernate.LazyInitializationException: Failed to lazily
initialize a collection - no session or session was closed
```

Listing 7.18

```
protected void setUp() throws Exception {
    // the following is necessary for lazy loading
    sf = (SessionFactory) ctx.getBean("sessionFactory");
    // open and bind the session for this test thread.
    Session s = sf.openSession();
    TransactionSynchronizationManager
        .bindResource(sf, new SessionHolder(s));

    // setup code here
}

protected void tearDown() throws Exception {
    // unbind and close the session.
    SessionHolder holder = (SessionHolder)
        TransactionSynchronizationManager.getResource(sf);
    Session s = holder.getSession();
    s.flush();
    TransactionSynchronizationManager.unbindResource(sf);
    SessionFactoryUtils.closeSessionIfNecessary(s, sf);

    // teardown code here
}
```

Listing 7.19

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE sqlMap PUBLIC "-//ibatis.com//DTD SQL Map 2.0//EN"
    "http://www.ibatis.com/dtd/sql-map-2.dtd">

<sqlMap namespace="UserSQL">
    <insert id="addUser" parameterClass="org.appfuse.model.User">
        insert into app_user (id, first_name, last_name)
        values (#id#, #firstName#, #lastName#);
        <selectKey resultClass="java.lang.Long" keyProperty="id" >
            select last_insert_id();
        </selectKey>
    </insert>

    <update id="updateUser" parameterClass="org.appfuse.model.User">
        update app_user set first_name = #firstName#,
        last_name = #lastName#
        where id = #id#;
    </update>

    <select id="getUser" parameterClass="java.lang.Long"
        resultClass="org.appfuse.model.User">
        select id, first_name as firstName, last_name as lastName
```

```

        from app_user where id=#id#;
    </select>

    <select id="getUsers" resultClass="org.appfuse.model.User">
        select id, first_name as firstName, last_name as lastName
        from app_user;
    </select>

    <delete id="deleteUser" parameterClass="java.lang.Long">
        delete from app_user where id = #id#;
    </delete>
</sqlMap>

```

Listing 7.20

```

<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE sqlMapConfig PUBLIC
    "-//ibatis.com//DTD SQL Map Config 2.0//EN"
    "http://www.ibatis.com/dtd/sql-map-config-2.dtd">

<sqlMapConfig>
    <settings enhancementEnabled="true" maxTransactions="5"
        maxRequests="32" maxSessions="10"/>

    <sqlMap resource="org/appfuse/dao/ibatis/UserSQL.xml" />
</sqlMapConfig>

```

Listing 7.21

```

<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE beans PUBLIC "-//SPRING//DTD BEAN//EN"
    "http://www.springframework.org/dtd/spring-beans.dtd">

<beans>
    <bean id="dataSource"

class="org.springframework.jdbc.datasource.DriverManagerDataSource">
        <property name="driverClassName">
            <value>com.mysql.jdbc.Driver</value>
        </property>
        <property name="url">
            <value>jdbc:mysql://localhost/myusers</value>
        </property>
        <property name="username"><value>root</value></property>
        <property name="password"><value></value></property>
    </bean>
    <!-- Add bean definitions here -->
</beans>

```

Listing 7.22

```
<bean id="sqlMapClient"
      class="org.springframework.orm.ibatis.SqlMapClientFactoryBean">
  <property name="configLocation">
    <value>classpath:/org/appfuse/dao/ibatis/sql-map-config.xml
    </value>
  </property>
  <property name="dataSource"><ref bean="dataSource"/></property>
</bean>
```

Listing 7.23

```
<bean id="transactionManager"
      class="org.springframework.jdbc.datasource.
        DataSourceTransactionManager">
  <property name="dataSource"><ref bean="dataSource"/></property>
</bean>
```

Listing 7.24

```
package org.appfuse.dao.ibatis;

// use your IDE to organize imports

public class UserDaoIbatis extends SqlMapClientDaoSupport
    implements UserDao {

    public List getUsers() {
        return getSqlMapClientTemplate()
            .queryForList("getUsers", null);
    }

    public User getUser(Long id) {
        User user = (User) getSqlMapClientTemplate()
            .queryForObject("getUser", id);

        if (user == null) {
            throw new ObjectRetrievalFailureException(User.class, id);
        }

        return user;
    }

    public void saveUser(User user) {
        if (user.getId() == null) {
            // Use iBatis's <selectKey> feature, which is db-specific
            Long id = (Long) getSqlMapClientTemplate()
```

```

        .insert("addUser", user);
        user.setId(id);
        logger.info("new User id set to: " + id);
    } else {
        getSqlMapClientTemplate().update("updateUser", user);
    }
}

public void removeUser(Long id) {
    getSqlMapClientTemplate().update("deleteUser", id);
}
}

```

Listing 7.25

```

<bean id="userDAO" autowire="byName"
      class="org.appfuse.dao.ibatis.UserDAOiBatis"/>
<!-- injected property: sqlMapClient -->

```

Listing 7.26

```

create table app_user (id bigint not null auto_increment,
first_name varchar(50), last_name varchar(50),
primary key (id));

```

Listing 7.27

```

<!-- Copy hibernate mapping files to ${build.dir}/classes -->
<copy todir="${build.dir}/classes">
    <fileset dir="${src.dir}" includes="**/*.hbm.xml"/>
</copy>

```

Listing 7.28

```

<!-- Copy XML files to ${build.dir}/classes -->
<copy todir="${build.dir}/classes">
    <fileset dir="${src.dir}" includes="**/*.xml"/>
</copy>

```

Listing 7.29

```

<sqlMap namespace="UserSQL">
    <cacheModel id="userCache" type="LRU">
        <flushInterval hours="24"/>
        <property name="size" value="1000"/>
    </cacheModel>
    <insert id="addUser" parameterClass="org.appfuse.model.User">

```

Listing 7.30

```
<select id="getUser" parameterClass="java.lang.Long"
    resultClass="org.appfuse.model.User"
    cacheModel="userCache">
    select id, first_name as firstName, last_name as lastName
    from app_user where id=#id#;
</select>
```

Listing 7.31

```
<insert id="addUser" parameterClass="org.appfuse.model.User">
    insert into app_user (id, first_name, last_name)
    values (#id#, #firstName#, #lastName#);
    <selectKey resultClass="java.lang.Long" keyProperty="id" >
        select last_insert_id();
    </selectKey>
</insert>
```

Listing 7.32

```
create table app_user (id integer identity, first_name varchar(50),
last_name varchar(50));
```

Listing 7.33

```
public void saveUser(User user) {
    if (user.getId() == null) {
        MySQLMaxValueIncrementer incrementer =
            new MySQLMaxValueIncrementer(
                getDataSource(), "user_sequence", "value");
        Long id = new Long(incrementer.nextLongValue());
        user.setId(id);

        getSqlMapClientTemplate().insert("addUser", user);
        logger.info("new User id set to: " + id);
    } else {
        getSqlMapClientTemplate().update("updateUser", user);
    }
}
```

Listing 7.34

```
create table user_sequence (value bigint auto_increment,
    primary key (value));
insert into user_sequence values(0);
```


Listing 7.35

```
private DataFieldMaxValueIncrementer incrementer;

public void setIncrementer(DataFieldMaxValueIncrementer incrementer) {
    this.incrementer = incrementer;
}
```

Listing 7.36

```
<bean id="incrementer"
      class="org.springframework.jdbc.support.incrementer.MySQLMaxValue
        Incrementer">
  <property name="dataSource"><ref local="dataSource"/></property>
  <property name="incrementerName">
    <value>user_sequence</value>
  </property>
  <property name="columnName"><value>value</value></property>
</bean>
```

Listing 7.37

```
public List getList() {
    return getJdbcTemplate().queryForList("select * from tablename");
}
```

Listing 7.38

```
public class UsersQuery extends MappingSqlQuery {
    public UsersQuery(DataSource ds) {
        super(ds, "SELECT * FROM app_user");
        compile();
    }
    protected Object mapRow(ResultSet rs, int rowNum)
    throws SQLException {
        User user = new User();
        user.setId(new Long(rs.getLong("id")));
        user.setFirstName(rs.getString("first_name"));
        user.setLastName(rs.getString("last_name"));
        return user;
    }
}
```

Listing 7.39

```
public List getUsers() {  
    return new UsersQuery(getDataSource()).execute();  
}
```

Listing 7.40

```
public UserQuery(DataSource ds) {  
    super(ds, "SELECT * FROM app_user WHERE id = ?");  
    super.declareParameter(new SqlParameter("id", Types.INTEGER));  
    compile();  
}
```

Pass in the id parameter using an Object array:

```
public User getUser(Long id) {  
    List users = new UserQuery(getDataSource())  
        .execute(new Object[]{id});  
    if (users.isEmpty()) {  
        throw new ObjectRetrievalFailureException(User.class, id);  
    }  
    return (User) users.get(0);  
}
```

Listing 7.41

```
public class UserUpdate extends SqlUpdate {  
    public UserUpdate(DataSource ds) {  
        super(ds, "INSERT INTO app_user (id, first_name, last_name)  
values  
        (?, ?, ?)");  
        declareParameter(new SqlParameter("id", Types.BIGINT));  
        declareParameter(new SqlParameter("first_name", Types.VARCHAR));  
        declareParameter(new SqlParameter("last_name", Types.VARCHAR));  
        compile();  
    }  
}
```

Listing 7.42

```
Object[] params =  
    new Object[] {user.getId(), user.getFirstName(),  
        user.getLastName()};  
new UserUpdate(getDataSource()).update(params);
```

Listing 7.43

```
getJdbcTemplate().update(  
    "UPDATE app_user SET first_name = ?, last_name = ? WHERE id = ?",  
    new Object[] {user.getFirstName(), user.getLastName(), user.getId()});
```

Listing 7.44

```
String sql = "INSERT INTO app_user (id, first_name, last_name) ";  
    sql += "values (?, ?, ?)";  
SqlUpdate su = new SqlUpdate(getDataSource(), sql);  
su.declareParameter(new SqlParameter("id", Types.BIGINT));  
su.declareParameter(new SqlParameter("first_name", Types.VARCHAR));  
su.declareParameter(new SqlParameter("last_name", Types.VARCHAR));  
su.compile();  
  
Object[] params = new Object[]  
    {user.getId(), user.getFirstName(), user.getLastName()};  
  
su.update(params);
```

Listing 7.45

```
KeyHolder keys = new GeneratedKeyHolder();  
su.update(params, keys);  
user.setId(new Long(keys.getKey().longValue()));
```

Listing 7.46

```
<bean id="dataSource"  
    class="org.springframework.jdbc.datasource.DriverManagerDataSource">  
    <property name="driverClassName">  
        <value>com.mysql.jdbc.Driver</value>  
    </property>  
    <property name="url">  
        <value>jdbc:mysql://localhost/myusers</value>  
    </property>  
    <property name="username"><value>root</value></property>  
    <property name="password"><value></value></property>  
</bean>
```

Listing 7.47

```
<bean id="transactionManager"
class="org.springframework.jdbc.datasource.DataSourceTransactionManager"
">
    <property name="dataSource"><ref bean="dataSource"/></property>
</bean>
```

Listing 7.48

```
package org.appfuse.dao.jdbc;

// use your IDE to organize imports

public class UserDaoJdbc extends JdbcDaoSupport implements UserDao {

    public List getUsers() {
        return new UsersQuery(getDataSource()).execute();
    }

    public User getUser(Long id) {
        List users = new UserQuery(getDataSource())
            .execute(new Object[]{id});
        if (users.isEmpty()) {
            throw new ObjectRetrievalFailureException(User.class, id);
        }
        return (User) users.get(0);
    }

    public void saveUser(User user) {
        if (user.getId() == null) {
            String sql = "INSERT INTO app_user (id, first_name, ";
            sql += " last_name) values (?, ?, ?)";
            SqlUpdate su = new SqlUpdate(getDataSource(), sql);
            su.declareParameter(new SqlParameter("id", Types.BIGINT));
            su.declareParameter(new SqlParameter("first_name",
                Types.VARCHAR));
            su.declareParameter(new SqlParameter("last_name",
                Types.VARCHAR));
            su.compile();

            Object[] params = new Object[]
                {user.getId(), user.getFirstName(),
                user.getLastName()};

            KeyHolder keys = new GeneratedKeyHolder();
            su.update(params, keys);
            user.setId(new Long(keys.getKey().longValue()));
        }
    }
}
```

```

        if (logger.isDebugEnabled()) {
            logger.info("user's id is: " + user.getId());
        }
    } else {
        getJdbcTemplate().update("UPDATE app_user SET first_name = ?,
                                last_name = ? WHERE id = ?",
                                new Object[] {user.getFirstName(), user.getLastName(),
                                                user.getId()});
    }
}

public void removeUser(Long id) {
    getJdbcTemplate().update("DELETE FROM app_user WHERE id = ?",
                            new Object[] {id});
}

// Query to get a single User
class UserQuery extends MappingSqlQuery {
    public UserQuery(DataSource ds) {
        super(ds, "SELECT * FROM app_user WHERE id = ?");
        super.declareParameter(new SqlParameter("id",
Types.INTEGER));
        compile();
    }
    protected Object mapRow(ResultSet rs, int rowNum)
    throws SQLException {
        User user = new User();
        user.setId(new Long(rs.getLong("id")));
        user.setFirstName(rs.getString("first_name"));
        user.setLastName(rs.getString("last_name"));
        return user;
    }
}

// Query to get a list of User objects
class UsersQuery extends MappingSqlQuery {
    public UsersQuery(DataSource ds) {
        super(ds, "SELECT * FROM app_user");
        compile();
    }
    protected Object mapRow(ResultSet rs, int rowNum)
    throws SQLException {
        User user = new User();
        user.setId(new Long(rs.getLong("id")));
        user.setFirstName(rs.getString("first_name"));
        user.setLastName(rs.getString("last_name"));
        return user;
    }
}
}

```

Listing 7.49

```
create table app_user (id bigint not null auto_increment,  
    first_name varchar(50), last_name varchar(50),  
    primary key (id));
```

Listing 7.50

```
public void testSaveUser() throws Exception {  
    user = new User();  
    user.setFirstName("Rod");  
    user.setLastName("Johnson");  
  
    dao.saveUser(user);  
    assertTrue("primary key assigned", user.getId() != null);  
    log.info(user);  
    assertTrue(user.getFirstName() != null);  
}
```

Listing 7.51

```
<target name="enhance" description="JPOX enhancement"  
    depends="compile">  
    <taskdef name="jpoxenhancer"  
        classname="org.jpox.enhancer.tools.EnhancerTask"  
        classpathref="classpath"/>  
  
    <jpoxenhancer dir="${build.dir}/classes" failonerror="true"  
        fork="true" verbose="true" check="true">  
        <classpath>  
            <path refid="classpath"/>  
            <path location="${build.dir}/classes"/>  
            <path location="web/WEB-INF/classes"/>  
        </classpath>  
    </jpoxenhancer>  
</target>  
    Change the "test" and "deploy" targets to depend on "enhance".  
<target name="test" depends="enhance"  
    description="Runs JUnit tests">  
    ...  
<target name="war" depends="enhance"  
    description="Packages app as WAR">
```

Listing 7.52

```
<?xml version="1.0"?>
<!DOCTYPE jdo PUBLIC
    "-//Sun Microsystems, Inc.//DTD Java Data Objects Metadata 2.0//EN"
    "http://java.sun.com/dtd/jdo_2_0.dtd">
<jdo>
    <package name="org.appfuse.model">
        <class name="User" identity-type="application"
            objectid-class="org.appfuse.model.PrimaryKey"
            table="app_user">
            <field name="id" primary-key="true"/>
            <field name="firstName" persistence-modifier="persistent">
                <column name="first_name" jdbc-type="VARCHAR"
length="50"/>
            </field>
            <field name="lastName" persistence-modifier="persistent">
                <column name="last_name" jdbc-type="VARCHAR" length="50"/
>
                </field>
            <extension vendor-name="jpox" key="use-poid-generator"
                value="true"
            />
        </class>
    </package>
</jdo>
```

Listing 7.53

```
<extension vendor-name="jpox" key="use-poid-generator" value="true"/>
```

Listing 7.54

```
<!-- Copy XML files to ${build.dir}/classes -->
<copy todir="${build.dir}/classes">
    <fileset dir="${src.dir}" includes="**/*.xml"/>
    <!-- Copy JDO mapping files -->
    <fileset dir="${src.dir}" includes="**/*.jdo"/>
</copy>
```

Listing 7.55

```
package org.appfuse.model;

// organize imports with your IDE

public class PrimaryKey extends BaseObject implements Serializable {
    public Long id;
```

```

    public PrimaryKey() {}

    public PrimaryKey(String value) {
        id = Long.valueOf(value);
    }
}

```

Listing 7.56

```

<bean id="persistenceManagerFactory"
class="org.springframework.orm.jdo.LocalPersistenceManagerFactoryBean">
    <property name="jdoProperties">
        <props>
            <prop key="javax.jdo.PersistenceManagerFactoryClass">
                org.jpox.PersistenceManagerFactoryImpl
            </prop>
            <prop key="javax.jdo.option.ConnectionDriverName">
                com.mysql.jdbc.Driver
            </prop>
            <prop key="javax.jdo.option.ConnectionUserName">root</prop>
            <prop key="javax.jdo.option.ConnectionPassword"></prop>
            <prop key="javax.jdo.option.ConnectionURL">
                jdbc:mysql://localhost/myusers
            </prop>
            <prop key="org.jpox.autoCreateSchema">true</prop>
            <prop key="org.jpox.validateTables">false</prop>
            <prop key="javax.jdo.option.NontransactionalRead">
                true
            </prop>
        </props>
    </property>
</bean>

```

Listing 7.57

```

<bean id="transactionManager"
    class="org.springframework.orm.jdo.JdoTransactionManager">
    <property name="persistenceManagerFactory">
        <ref bean="persistenceManagerFactory"/>
    </property>
</bean>

```


Listing 7.58

```
package org.appfuse.dao.jdo;

// user your IDE to organize imports

public class UserDaoJdo extends JdoDaoSupport implements UserDao {

    public List getUsers() {
        Collection users = getJdoTemplate().find(User.class);
        users = getPersistenceManager().detachCopyAll(users);
        return new ArrayList(users);
    }

    public User getUser(Long id) {
        User user = (User) getJdoTemplate().getObjectById(User.class,
id);
        if (user == null) {
            throw new ObjectRetrievalFailureException(User.class, id);
        }
        return (User) getPersistenceManager().detachCopy(user);
    }

    public void saveUser(User user) {
        if (user.getId() == null) {
            getJdoTemplate().makePersistent(user);
        } else {
            getPersistenceManager().attachCopy(user, true);
        }
    }

    public void removeUser(Long id) {
        getJdoTemplate().deletePersistent(
            getJdoTemplate().getObjectById(User.class, id));
    }
}
```

Listing 7.59

```
<bean id="userDAO" class="org.appfuse.dao.jdo.UserDaoJdo">
    <property name="persistenceManagerFactory">
        <ref bean="persistenceManagerFactory"/>
    </property>
</bean>
```

Listing 7.60

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE beans PUBLIC "-//SPRING//DTD BEAN//EN"
    "http://www.springframework.org/dtd/spring-beans.dtd">

<beans>
    <!-- userDAO for testing UserDAOJdo -->
    <bean id="userDAO"

class="org.springframework.transaction.interceptor.TransactionProxy
    FactoryBean">
        <property name="transactionManager">
            <ref bean="transactionManager"/>
        </property>
        <property name="target">
            <bean class="org.appfuse.dao.jdo.UserDAOJdo"
                autowire="byName"/>
        </property>
        <property name="transactionAttributes">
            <props>
                <prop key="*">PROPAGATION_REQUIRED</prop>
            </props>
        </property>
    </bean>
</beans>
```

Listing 7.61

```
protected void setUp() throws Exception {
    String[] paths =
        {"org/appfuse/dao/applicationContext-test.xml"};
    ctx = new ClassPathXmlApplicationContext(paths, ctx);
    dao = (UserDAO) ctx.getBean("userDAO");
}
```

Listing 7.62

```
<!-- Copy JDO mapping files -->
<fileset dir="${src.dir}" includes="**/*.jdo"/>
</copy>
<!-- Copy overriding test files -->
<copy todir="${test.dir}/classes">
<fileset dir="${test.src}" includes="**/*.xml"/>
</copy>
</target>
```

Listing 7.63

```
<descriptor-repository version="1.0">
  <jdbc-connection-descriptor jcd-alias="dataSource"
    default-connection="true" useAutoCommit="1" platform="MySQL">
    <sequence-manager
      className="org.apache.obj.broker.util.sequence.
        SequenceManagerNativeImpl"/>
    </jdbc-connection-descriptor>

  <class-descriptor class="org.appfuse.model.User" table="app_user">
    <field-descriptor name="id" column="id" primaryKey="true"
      autoincrement="true" access="readonly"/>
    <field-descriptor name="firstName" column="first_name"/>
    <field-descriptor name="lastName" column="last_name"/>
  </class-descriptor>
</descriptor-repository>
```

Listing 7.64

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE beans PUBLIC "-//SPRING//DTD BEAN//EN"
  "http://www.springframework.org/dtd/spring-beans.dtd">

<beans>
  <bean id="dataSource" class="org.springframework.jdbc.datasource.
    DriverManagerDataSource">
    <property name="driverClassName">
      <value>com.mysql.jdbc.Driver</value>
    </property>
    <property name="url">
      <value>jdbc:mysql://localhost/myusers</value>
    </property>
    <property name="username"><value>root</value></property>
    <property name="password"><value></value></property>
  </bean>
</beans>
```

Listing 7.65

```
<bean id="transactionManager"
  class="org.springframework.orm.obj.PersistenceBrokerTransaction
    Manager"/>
```

Listing 7.66

```
<bean id="objConfigurer"
  class="org.springframework.orm.obj.support.LocalObjConfigurer"/>
```

Listing 7.67

```
repositoryFile=org/appfuse/model/repository.xml
```

Listing 7.68

```
ConnectionFactoryClass=org.springframework.orm.objb.support.LocalDataSourceConnectionFactory
```

Listing 7.69

```
ObjectCacheClass=org.apache.objb.broker.cache.ObjectCachePerBrokerImpl
```

Listing 7.70

```
package org.appfuse.dao.objb;

// organize imports with your IDE

public class UserDAOObjb extends PersistenceBrokerDaoSupport implements
    UserDao {

    public List getUsers() {
        return new ArrayList(getPersistenceBrokerTemplate().
            getCollectionByQuery(new QueryByCriteria(User.class)));
    }

    public User getUser(Long id) {
        Criteria criteria = new Criteria();
        criteria.addEqualTo("id", id);
        User user = (User) getPersistenceBrokerTemplate().
            getObjectByQuery(new QueryByCriteria(User.class,
                criteria));
        if (user == null) {
            throw new ObjectRetrievalFailureException(User.class, id);
        }
        return user;
    }

    public void saveUser(User user) {
        getPersistenceBrokerTemplate().store(user);
    }

    public void removeUser(Long id) {
        getPersistenceBrokerTemplate().delete(getUser(id));
    }
}
```

Listing 7.71

```
<bean id="userDAO" class="org.appfuse.dao.ojb.UserDAOObjb"/>
```

Listing 7.72

```
create table app_user (id bigint not null auto_increment,  
first_name varchar(50), last_name varchar(50),  
primary key (id));
```

Listing 7.73

```
protected void setUp() throws Exception {  
    //String[] paths = {"org/appfuse/dao/applicationContext-test.xml"};  
    //ctx = new ClassPathXmlApplicationContext(paths, ctx);  
    dao = (UserDAO) ctx.getBean("userDAO");  
}
```

Listing 7.74

```
<property name="target">  
    <bean class="org.appfuse.dao.ojb.UserDAOObjb"/>  
</property>
```



Testing Spring Applications

Code

Listing 8.1

```
package org.appfuse.dao;

import junit.framework.TestCase;

import org.appfuse.model.User;

public class SimpleTest extends TestCase {

    public void testGetFullName() {
        User user = new User();
        user.setFirstName("Jack");
        user.setLastName("Raible");
        assertEquals(user.getLastName(), "Jack Raible");
    }
}
```

Listing 8.2

```
assertEquals(user.getFullName(), "Jack Raible");
```

Listing 8.3

```
package org.appfuse.dao;

// use your IDE to organize imports

public class UserDAOTest extends BaseDAOTestCase {

    private User user = null;
    private UserDAO dao = null;
```

```

protected void setUp() throws Exception {
    dao = (UserDAO) ctx.getBean("userDAO");
}

protected void tearDown() throws Exception {
    dao = null;
}

public void testGetUsers() {
    // add a record to the database so we have something to work with
    user = new User();
    user.setFirstName("Rod");
    user.setLastName("Johnson");
    dao.saveUser(user);

    List users = dao getUsers();
    assertTrue(users.size() >= 1);
    assertTrue(users.contains(user));
}

public void testSaveUser() throws Exception {
    user = new User();
    user.setFirstName("Rod");
    user.setLastName("Johnson");

    dao.saveUser(user);
    assertTrue("primary key assigned", user.getId() != null);
    log.info(user);
    assertTrue(user.getFirstName() != null);
}

public void testAddAndRemoveUser() throws Exception {
    user = new User();
    user.setFirstName("Bill");
    user.setLastName("Joy");

    dao.saveUser(user);

    assertTrue(user.getId() != null);
    assertTrue(user.getFirstName().equals("Bill"));

    if (log.isDebugEnabled()) {
        log.debug("removing user...");
    }

    dao.removeUser(user.getId());

    try {
        user = dao.getUser(user.getId());
        fail("User found in database");
    }
}

```



```

    } catch (DataAccessException dae) {
        log.debug("Expected exception: " + dae.getMessage());
        assertTrue(dae != null);
    }
}

```

Listing 8.4

```

protected ApplicationContext ctx = null;

public BaseDAOTestCase() {
    String[] paths = { "/WEB-INF/applicationContext*.xml" };
    ctx = new ClassPathXmlApplicationContext(paths);
}

```

Listing 8.5

```

public BaseDAOTestCase() {
    String[] paths = { "web/WEB-INF/applicationContext*.xml" };
    ctx = new FileSystemXmlApplicationContext(paths);
}

```

Listing 8.6

```

<classpathentry excluding="WEB-INF/classes/" kind="src" path="web"/>

```

Listing 8.7

```

<classpath>
    <path refid="classpath"/>
    <path location="${build.dir}/classes"/>
    <path location="${test.dir}/classes"/>
    <path location="web/WEB-INF/classes"/>
    <path location="web"/>
</classpath>

```

Listing 8.8

```

<?xml version="1.0" encoding="utf-8"?>
<dataset>
    <table name='app_user'>
        <column>id</column>
        <column>first_name</column>
        <column>last_name</column>
    <row>

```

```
<value>1</value>
<value>Rod</value>
<value>Johnson</value>
</row>
</table>
</dataset>
```

Listing 8.9

```
private IDatabaseConnection conn = null;
private IDataset dataSet = null;
```

Listing 8.10

```
protected void setUp() throws Exception {
    DataSource ds = (DataSource) ctx.getBean("dataSource");
    conn = new DatabaseConnection(ds.getConnection());
    dataSet = new XmlDataSet(new FileInputStream(
        "test/data/sample-data.xml"));
    // clear table and insert only sample data
    DatabaseOperation.CLEAN_INSERT.execute(conn, dataSet);
}
```

Listing 8.11

```
protected void tearDown() throws Exception {
    // clear out database
    DatabaseOperation.DELETE.execute(conn, dataSet);
    conn.close();
    conn = null;
}
```

Listing 8.12

```
protected void setUp() throws Exception {
    super.setUp();
    dao = (UserDAO) ctx.getBean("userDAO");
}

protected void tearDown() throws Exception {
    super.tearDown();
    dao = null;
}
```

Listing 8.13

```
public void testGetUsers() {
    List users = dao.getUsers();
    assertTrue(users.size() == 1);
    User user = (User) users.get(0);
    assertEquals(user.getFullName(), "Rod Johnson");
}
```

Listing 8.14

```
<bean id="dataSource"
      class="org.springframework.jdbc.datasource.DriverManagerDataSource">
  <property name="driverClassName">
    <value>${jdbc.driverClassName}</value>
  </property>
  <property name="url"><value>${jdbc.url}</value></property>
  <property name="username">
    <value>${jdbc.username}</value>
  </property>
  <property name="password">
    <value>${jdbc.password}</value>
  </property>
</bean>
```

Listing 8.15

```
jdbc.driverClassName=com.mysql.jdbc.Driver
jdbc.url=jdbc:mysql://localhost/myusers
jdbc.username=root
jdbc.password=
```

Listing 8.16

```
<bean id="propertyConfigurer"
      class="org.springframework.beans.factory.config.PropertyPlaceholder
      Configurer">
  <property name="location">
    <value>classpath:jdbc.properties</value>
  </property>
</bean>
```

Listing 8.17

```
<property name="locations">
  <list>
    <value>classpath:jdbc.properties</value>
  </list>
</property>
```

Listing 8.18

```
<property file="build.properties"/>
<property file="web/WEB-INF/classes/jdbc.properties"/>
```

Listing 8.19

```
<sql driver="${jdbc.driverClassName}" url="${jdbc.url}"
userid="${jdbc.username}" password="${jdbc.password}">
```

Listing 8.20

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE beans PUBLIC "-//SPRING//DTD BEAN//EN"
    "http://www.springframework.org/dtd/spring-beans.dtd">

<beans>
  <!-- userDao for testing UserDaoJdo -->
  <bean id="userDAO"
class="org.springframework.transaction.interceptor.TransactionProxy
    FactoryBean">
    <property name="transactionManager"><ref
bean="transactionManager"/></property>
    <property name="target">
      <bean class="org.appfuse.dao.ojb.UserDAOObjb"
autowire="byName"/>
    </property>
    <property name="transactionAttributes">
      <props>
        <prop key="*">PROPAGATION_REQUIRED</prop>
      </props>
    </property>
  </bean>
</beans>
```

Listing 8.21

```
protected void setUp() throws Exception {
    super.setUp();
    String[] paths =
        {"org/appfuse/dao/applicationContext-test.xml"};
    ctx = new ClassPathXmlApplicationContext(paths, ctx);
    dao = (UserDAO) ctx.getBean("userDAO");
}
```

Listing 8.22

```
<bean id="dataSource"
    class="org.springframework.jndi.JndiObjectFactoryBean">
    <property name="jndiName">
        <value>java:comp/env/jdbc/myusers</value>
    </property>
</bean>
```

Listing 8.23

```
try {
    SimpleNamingContextBuilder builder =
        SimpleNamingContextBuilder.emptyActivatedContextBuilder();

    DriverManagerDataSource ds = new DriverManagerDataSource();
    ds.setDriverClassName("com.mysql.jdbc.Driver");
    ds.setUrl("jdbc:mysql://localhost/myusers");
    ds.setUsername("root");
    ds.setPassword("");
    builder.bind("java:comp/env/jdbc/myusers", ds);
} catch (NamingException ne) {
    // do nothing, test will fail on its own
}
```

Listing 8.24

```
<Context path="/myusers" docBase="myusers" debug="0" reloadable="true">
    <Resource name="jdbc/myusers" auth="Container"
        type="javax.sql.DataSource"/>
    <ResourceParams name="jdbc/myusers">
        <parameter>
            <name>factory</name>
            <value>org.apache.commons.dbcp.BasicDataSourceFactory</value>
        </parameter>
        <parameter><name>maxActive</name><value>30</value></parameter>
        <parameter><name>maxIdle</name><value>5</value></parameter>
        <parameter><name>maxWait</name><value>5000</value></parameter>
```

```

<parameter><name>username</name><value>root</value></parameter>
<parameter><name>password</name><value></value></parameter>
<parameter>
    <name>driverClassName</name>
    <value>com.mysql.jdbc.Driver</value>
</parameter>
<parameter>
    <name>defaultAutoCommit</name>
    <value>true</value>
</parameter>
<parameter>
    <name>url</name>
    <value>jdbc:mysql://localhost/myusers</value>
</parameter>
</ResourceParams>
</Context>

```

Listing 8.25

```

<resource-ref>
    <description>Connection Pool</description>
    <res-ref-name>jdbc/myusers</res-ref-name>
    <res-type>javax.sql.DataSource</res-type>
    <res-auth>Container</res-auth>
</resource-ref>

```

Listing 8.26

```
protected static ApplicationContext ctx = null;
```

Listing 8.27

```
private static UserDao dao = null;
```

Listing 8.28

```

static {
    String[] paths = {"WEB-INF/applicationContext*.xml",
                      "/org/appfuse/dao/applicationContext-test.xml"};
    ctx = new ClassPathXmlApplicationContext(paths);
}

```

Listing 8.29

```
public void setUp() throws Exception {
    super.setUp();
    String[] paths =
        {"org/appfuse/dao/applicationContext-test.xml"};
    ctx = new ClassPathXmlApplicationContext(paths, ctx);
    dao = (UserDAO) ctx.getBean("userDAO");
}
```

Listing 8.30

```
private UserManager mgr = new UserManagerImpl();
private MockControl control;
private UserDAO mockDAO;
protected void setUp() throws Exception {
    // Create a MockControl
    control = MockControl.createControl(UserDAO.class);
    // Get the mock
    mockDAO = (UserDAO) control.getMock();

    // Set required dependencies
    mgr.setValidator(new UserValidator());
    mgr.setUserDAO(mockDAO);
}

protected void testGetUser() {
    // Set expected behavior
    mockDAO.removeUser(new Long(1));
    control.setVoidCallable();

    // Active the mock
    control.replay();

    // Execute method to test
    mgr.removeUser("1");

    // Verify methods called
    control.verify();
}
```

Listing 8.31

```
package org.appfuse.service;

// use your IDE to organize imports

public class UserManagerEMTest extends TestCase {
```

```

private static Log log =
    LogFactory.getLog(UserManagerEMTest.class);
private UserManager mgr = new UserManagerImpl();
private MockControl control;
private UserDao mockDAO;

protected void setUp() throws Exception {
    control = MockControl.createControl(UserDao.class);
    mockDAO = (UserDao) control.getMock();
    mgr.setValidator(new UserValidator());
    mgr.setUserDAO(mockDAO);
}

public void testAddAndRemoveUser() throws Exception {
    User user = new User();
    user.setFirstName("Easter");
    user.setLastName("Bunny");

    // set expected behavior on dao
    mockDAO.saveUser(user);
    control.setVoidCallable();

    // switch from record to playback
    control.replay();

    user = mgr.saveUser(user);
    assertEquals(user.getFullName(), "Easter Bunny");
    control.verify();

    if (log.isDebugEnabled()) {
        log.debug("removing user...");
    }

    // set userId since Hibernate doesn't do it
    String userId = "1";
    user.setId(new Long(userId));

    // reset to record state
    control.reset();
    mockDAO.removeUser(new Long(1));
    control.setVoidCallable();
    control.replay();

    mgr.removeUser(userId);

    control.verify();

    try {
        // reset to record state
        control.reset();
        control.expectAndThrow(mockDAO.getUser(user.getId()),

```



```

        new ObjectRetrievalFailureException(
            User.class, user.getId());
    // switch to playback
    control.replay();

    user = mgr.getUser(userId);

    control.verify();
    fail("User '" + userId + "' found in database");
} catch (DataAccessException dae) {
    log.debug("Expected exception: " + dae.getMessage());
    assertNotNull(dae);
}
}
}

```

Listing 8.32

```

private UserManager mgr = new UserManagerImpl();
private Mock mockDAO;

protected void setUp() throws Exception {
    // Create a Mock
    mockDAO = new Mock(UserDAO.class);

    // Set dependencies
    mgr.setValidator(new UserValidator());
    mgr.setUserDAO((UserDAO) mockDAO.proxy());
}

protected void testGetUser() {
    // Set expected behavior
    mockDAO.expects(once()).method("getUser")
        .with( eq(new Long(1)));

    // Execute method to test
    mgr.removeUser("1");

    // Verify expectations
    mockDAO.verify();
}

```

Listing 8.33

```

package org.appfuse.service;

// use your IDE to organize imports

public class AssignIdStub implements Stub {

```

```

    public StringBuffer describeTo(StringBuffer buffer) {
        return buffer.append("assigns random id to an object");
    }

    public Object invoke(Invocation invocation) throws Throwable {
        Long id = new Long(new Random().nextInt(10));
        ((User) invocation.parameterValues.get(0)).setId(id);

        return null;
    }
}

```

Listing 8.34

```

package org.appfuse.service;

// use your IDE to organize imports

public class UserManagerJMTest extends MockObjectTestCase {
    private static Log log =
        LogFactory.getLog(UserManagerJMTest.class);
    private UserManager mgr = new UserManagerImpl();
    private Mock mockDAO;

    protected void setUp() throws Exception {
        mockDAO = new Mock(UserDAO.class);
        mgr.setValidator(new UserValidator());
        mgr.setUserDAO((UserDAO) mockDAO.proxy());
    }

    public void testAddAndRemoveUser() throws Exception {
        User user = new User();
        user.setFirstName("Easter");
        user.setLastName("Bunny");

        // set expected behavior on dao
        mockDAO.expects(once()).method("saveUser")
            .with( same(user) ).will(assignId());

        user = mgr.saveUser(user);

        // verify expectations
        mockDAO.verify();

        assertEquals(user.getFullName(), "Easter Bunny");

        assertTrue(user.getId() != null);

        if (log.isDebugEnabled()) {

```

```

        log.debug("removing user...");
    }

    String userId = user.getId().toString();

    mockDAO.expects(once()).method("removeUser")
        .with( eq(new Long(userId)) );

    mgr.removeUser(userId);

    // verify expectations
    mockDAO.verify();

    try {
        // set expectations
        Throwable ex =
            new ObjectRetrievalFailureException(
                User.class, user.getId());

        mockDAO.expects(once()).method("getUser")
            .with( eq(new Long(userId)))
            .will(throwException(ex));

        user = mgr.getUser(userId);

        // verify expectations
        mockDAO.verify();
        fail("User '" + userId + "' found in database");
    } catch (DataAccessException dae) {
        log.debug("Expected exception: " + dae.getMessage());
        assertNotNull(dae);
    }
}

private Stub assignId() {
    return new AssignIdStub();
}
}

```

Listing 8.35

```

package org.appfuse.web;

// use your IDE to organize imports

public class UserControllerTest extends TestCase {
    private static Log log =
        LogFactory.getLog(UserControllerTest.class);

    private XmlWebApplicationContext ctx;

```

```

    public void setUp() {
        String[] paths = {"/WEB-INF/applicationContext*.xml",
                           "/WEB-INF/action-servlet.xml"};
        ctx = new XmlWebApplicationContext();
        ctx.setConfigLocations(paths);
        ctx.setServletContext(new MockServletContext(""));
        ctx.refresh();
    }

    public void testGetUsers() throws Exception {
        UserController c = (UserController)
ctx.getBean("userController");
        ModelAndView mav = c.handleRequest((HttpServletRequest) null,
                                           (HttpServletResponse) null);
        Map m = mav.getModel();
        assertNotNull(m.get("users"));
        assertEquals(mav.getViewName(), "userList");
    }
}

```

Listing 8.36

```

package org.appfuse.web;

// use your IDE to organize imports

public class UserControllerEMTest extends TestCase {
    private MockControl control = null;
    private UserManager mockManager = null;
    private UserController c = new UserController();

    protected void setUp() throws Exception {
        control = MockControl.createControl(UserManager.class);
        mockManager = (UserManager) control.getMock();
        c.setUserManager(mockManager);
    }

    public void testGetUsers() throws Exception {
        // set expected behavior on manager
        mockManager.getUsers();
        control.setReturnValue(new ArrayList());

        // switch from record to playback
        control.replay();

        ModelAndView mav =
            c.handleRequest((HttpServletRequest) null,
                           (HttpServletResponse) null);
        Map m = mav.getModel();
    }
}

```

```

        assertNotNull(m.get("users"));
        assertEquals(mav.getViewName(), "userList");

        // verify getUsers() was called on manager
        control.verify();
    }
}

```

Listing 8.37

```

package org.appfuse.web;

// use your IDE to organize imports

public class UserControllerJMTTest extends MockObjectTestCase {
    private UserController c = new UserController();
    private Mock mockManager = null;

    protected void setUp() throws Exception {
        mockManager = new Mock(UserManager.class);
        c.setUserManager((UserManager) mockManager.proxy());
    }

    public void testGetUsers() throws Exception {
        // set expected behavior on manager
        mockManager.expects(once()).method("getUsers")
            .will(returnValue(new ArrayList()));

        ModelAndView mav =
            c.handleRequest((HttpServletRequest) null,
                           (HttpServletResponse) null);
        Map m = mav.getModel();
        assertNotNull(m.get("users"));
        assertEquals(mav.getViewName(), "userList");

        // verify expectations
        mockManager.verify();
    }
}

```

Listing 8.38

```

package org.appfuse.web;

// use your IDE to organize imports

public class UserFormControllerTest extends TestCase {
    private static Log log =
        LogFactory.getLog(UserFormControllerTest.class);
}

```

```

private XmlWebApplicationContext ctx = null;
private UserFormController c = null;
private MockHttpServletRequest request = null;
private ModelAndView mv = null;
private User user = null;

protected void setUp() throws Exception {
    super.setUp();
    String[] paths = {"/WEB-INF/applicationContext*.xml",
        "/WEB-INF/action-servlet.xml"};
    ctx = new XmlWebApplicationContext();
    ctx.setConfigLocations(paths);
    ctx.setServletContext(new MockServletContext(""));
    ctx.refresh();
    c = (UserFormController) ctx.getBean("userFormController");
    // add a test user to the database
    UserManager mgr = (UserManager) ctx.getBean("userManager");
    user = new User();
    user.setFirstName("Matt");
    user.setLastName("Raible");
    user = mgr.saveUser(user);
}

public void testEdit() throws Exception {
    log.debug("testing edit...");
    request = new MockHttpServletRequest("GET", "/editUser.html");
    request.addParameter("id", user.getId().toString());
    mv = c.handleRequest(request, new MockHttpServletResponse());
    assertEquals("userForm", mv.getViewName());
}

public void testSave() throws Exception {
    request = new MockHttpServletRequest("POST", "/editUser.html");
    request.addParameter("id", user.getId().toString());
    request.addParameter("firstName", user.getFirstName());
    request.addParameter("lastName", "Updated Last Name");
    mv = c.handleRequest(request, new MockHttpServletResponse());
    Errors errors =
        (Errors) mv.getModel().get(BindException.ERROR_KEY_PREFIX +
"user");
    assertNull(errors);
    assertNotNull(request.getSession().getAttribute("message"));
}

public void testRemove() throws Exception {
    request = new MockHttpServletRequest("POST", "/editUser.html");
    request.addParameter("delete", "");
    request.addParameter("id", user.getId().toString());

```

```

        mv = c.handleRequest(request, new MockHttpServletResponse());
        assertNotNull(request.getSession().getAttribute("message"));
    }
}

```

Listing 8.39

```

package org.appfuse.web;

// use your IDE to organize imports

public class UserFormControllerJMTest extends MockObjectTestCase {
    private static Log log =
        LoggerFactory.getLog(UserFormControllerJMTest.class);
    private UserFormController c = new UserFormController();
    private MockHttpServletRequest request = null;
    private ModelAndView mv = null;
    private User user = new User();
    private Mock mockManager = null;

    protected void setUp() throws Exception {
        super.setUp();
        mockManager = new Mock(UserManager.class);

        // manually set properties (dependencies) on userFormController
        c.setUserManager((UserManager) mockManager.proxy());
        c.setFormView("userForm");

        // set context with messages avoid NPE when controller calls
        // getMessageSourceAccessor().getMessage()
        StaticApplicationContext ctx = new StaticApplicationContext();
        Map properties = new HashMap();
        properties.put("basename", "messages");
        ctx.registerSingleton("messageSource",
                               ResourceBundleMessageSource.class,
                               new MutablePropertyValues(properties));

        ctx.refresh();
        c.setApplicationContext(ctx);

        // setup user values
        user.setId(new Long(1));
        user.setFirstName("Matt");
        user.setLastName("Raible");
    }

    public void testEdit() throws Exception {
        log.debug("testing edit...");

        // set expected behavior on manager
        mockManager.expects(once()).method("getUser")

```

```

        .will(returnValue(new User()));

        request = new MockHttpServletRequest("GET", "/editUser.html");
        request.addParameter("id", user.getId().toString());
        mv = c.handleRequest(request, new MockHttpServletResponse());
        assertEquals("userForm", mv.getViewName());

        // The getCommandName() method is available in Spring 1.1.1+
        User editUser = (User) mv.getModel().get(c.getCommandName());
        assertEquals(editUser.getFullName(), "Matt Raible");
        // verify expectations
        mockManager.verify();
    }

    public void testSave() throws Exception {
        // set expected behavior on manager
        // called by formBackingObject()
        mockManager.expects(once()).method("getUser")
            .will(returnValue(user));

        User savedUser = user;
        savedUser.setLastName("Updated Last Name");
        // called by onSubmit()
        mockManager.expects(once()).method("saveUser")
            .with(eq(savedUser));

        request = new MockHttpServletRequest("POST", "/editUser.html");
        request.addParameter("id", user.getId().toString());
        request.addParameter("firstName", user.getFirstName());
        request.addParameter("lastName", "Updated Last Name");
        mv = c.handleRequest(request, new MockHttpServletResponse());
        Errors errors =
            (Errors) mv.getModel().get(BindException.ERROR_KEY_PREFIX +
"user");
        assertNull(errors);
        assertNotNull(request.getSession().getAttribute("message"));

        // verify expectations
        mockManager.verify();
    }

    public void testRemove() throws Exception {
        // set expected behavior on manager
        // called by formBackingObject()
        mockManager.expects(once()).method("getUser")
            .will(returnValue(user));
        // called by onSubmit()
        mockManager.expects(once()).method("removeUser").with(eq("1"));

        request = new MockHttpServletRequest("POST", "/editUser.html");
        request.addParameter("delete", "");
    }

```



```

        request.addParameter("id", user.getId().toString());
        mv = c.handleRequest(request, new MockHttpServletResponse());
        assertNotNull(request.getSession().getAttribute("message"));

        // verify expectations
        mockManager.verify();
    }
}

```

Listing 8.40

```

<target name="test-cactus" depends="war"
    description="Runs Cactus tests in-container">

    <!-- Define Cactus Tasks -->
    <taskdef resource="cactus.tasks" classpathref="classpath"/>

    <cactifywar srcfile="${dist.dir}/${webapp.name}.war"
        destfile="${dist.dir}/${webapp.name}-cactus.war">
        <classes dir="${test.dir}/classes"/>
        <classes dir="test" includes="cactus.properties"/>
        <!-- If you use EasyMock or Spring Mocks in a Cactus test
            it needs to be included in the WAR -->
        <lib dir="web/WEB-INF/lib" includes="*mock.jar"/>
        <servletredirector/>
    </cactifywar>

    <mkdir dir="${test.dir}/data/tomcat"/>

    <cactus warfile="${dist.dir}/${webapp.name}-cactus.war"
        printsummary="yes" failureproperty="tests.failed">
        <classpath>
            <path refid="classpath"/>
            <path location="${build.dir}/classes"/>
            <path location="${test.dir}/classes"/>
        </classpath>
        <containerset>
            <tomcat5x dir="${tomcat.home}" if="tomcat.home"
                port="8080" todir="${test.dir}/data/tomcat"/>
        </containerset>
        <formatter type="xml"/>
        <formatter type="brief" usefile="false"/>
        <batchtest todir="${test.dir}/data" if="testcase">
            <fileset dir="${test.dir}/classes">
                <include name="**/*${testcase}*" />
                <exclude name="**/*TestCase.class" />
            </fileset>
        </batchtest>
        <batchtest todir="${test.dir}/data" unless="testcase">
            <fileset dir="${test.dir}/classes">

```

```

        <include name="**/*Cactus*Test.class*" />
    </fileset>
</batchtest>
</cactus>
<fail if="tests.failed">Cactus test(s) failed.</fail>
</target>

```

Listing 8.41

```

package org.appfuse.web;

// use your IDE to organize imports

public class UserCactusTest extends ServletTestCase {
    private static Log log = LogFactory.getLog(UserCactusTest.class);
    private UserController list = new UserController();
    private UserFormController form = new UserFormController();

    protected void setUp() throws Exception {
        super.setUp();
        ApplicationContext ctx =
            WebApplicationContextUtils
                .getRequiredWebApplicationContext(
                    session.getServletContext());
        UserManager userManager =
            (UserManager) ctx.getBean("userManager");
        list.setUserManager(userManager);
        form.setUserManager(userManager);
        // needed to prevent NPE with getMessageSourceAccessor()
        form.setApplicationContext(ctx);
    }

    public void beginAddUser(WebRequest wRequest) {
        wRequest.addParameter("firstName", "Dion", "post");
        wRequest.addParameter("lastName", "Almaer", "post");
    }

    public void testAddUser() throws Exception {
        form.handleRequest(request, response);
        assertTrue(request.getSession().getAttribute("message") !=
null);
    }

    public void testUserList() throws Exception {
        ModelAndView mav = list.handleRequest(request, response);
        Map m = mav.getModel();
        assertNotNull(m.get("users"));
        assertEquals(mav.getViewName(), "userList");
    }
}

```

Listing 8.42

```
# Web app Context under which our application to test runs
cactus.contextURL=http://localhost:8080/myusers-cactus

# Default Servlet Redirector Name. Used by ServletTestCase test cases.
cactus.servletRedirectorName=ServletRedirector
```

Listing 8.43

```
...
public class UserCactusTest extends ServletTestCase {
    private static Log log = LogFactory.getLog(UserCactusTest.class);
    private UserController list = null;
    private UserFormController form = null;

    protected void setUp() throws Exception {
        super.setUp();
        ApplicationContext ctx =
            WebApplicationContextUtils
                .getRequiredWebApplicationContext(
                    session.getServletContext());
        list = (UserController) ctx.getBean("userController");
        form = (UserFormController) ctx.getBean("userFormController");
    }

    public void beginAddUser(WebRequest wRequest) {
        ...
    }
}
```

Listing 8.44

```
package org.appfuse.web;

// user your IDE to organize imports

public class UserCactusEMTest extends ServletTestCase {
    private UserController list = new UserController();
    private UserFormController form = new UserFormController();
    private MockControl control = null;
    private UserManager mockManager = null;

    protected void setUp() throws Exception {
        control = MockControl.createControl(UserManager.class);
        mockManager = (UserManager) control.getMock();
        list.setUserManager(mockManager);
        form.setUserManager(mockManager);

        // needed to prevent NPE with getMessageSourceAccessor()
    }
}
```

```

        ApplicationContext ctx =
            WebApplicationContextUtils
                .getRequiredWebApplicationContext(
                    session.getServletContext());
        form.setApplicationContext(ctx);
    }

    public void beginAddUser(WebRequest wRequest) {
        wRequest.addParameter("firstName", "Dion", "post");
        wRequest.addParameter("lastName", "Almaer", "post");
    }

    public void testAddUser() throws Exception {
        // set expected behavior on manager
        User user = new User();
        user.setFirstName("Dion");
        user.setLastName("Almaer");
        mockManager.saveUser(user);
        control.setReturnValue(user);

        // switch from record to playback
        control.replay();

        form.handleRequest(request, response);

        // verify saveUser() was called
        control.verify();
        assertTrue(request.getSession()
            .getAttribute("message") != null);
    }

    public void testUserList() throws Exception {
        // set expected behavior on manager
        control.expectAndReturn(mockManager.getUsers(),
            new ArrayList());
        // switch from record to playback
        control.replay();
        ModelAndView mav = list.handleRequest(request, response);

        // verify getUsers() was called
        control.verify();

        Map m = mav.getModel();
        assertNotNull(m.get("users"));
        assertEquals(mav.getViewName(), "userList");
    }
}

```

Listing 8.45

```
package org.appfuse.web;

// organize imports using your IDE

public class FileUploadControllerTest extends TestCase {
    private static Log log =
        LogFactory.getLog(FileUploadControllerTest.class);
    private XmlWebApplicationContext ctx = null;
    private FileUploadController fileUpload = null;

    public void setUp() {
        String[] paths = {"/WEB-INF/applicationContext*.xml",
            "/WEB-INF/action-servlet.xml"};
        ctx = new XmlWebApplicationContext();
        ctx.setConfigLocations(paths);
        ServletContext servletContext =
            new MockServletContext("file:web");

        ctx.setServletContext(servletContext);
        ctx.refresh();
        fileUpload = (FileUploadController)
            ctx.getBean("fileUploadController");

        // Create a mailSender that'll use Dumbster's ports
        JavaMailSenderImpl mailSender = new JavaMailSenderImpl();
        mailSender.setHost("localhost");
        mailSender.setPort(2525);
        fileUpload.setMailSender(mailSender);
    }

    // continued below
}
```

Listing 8.46

```
String uploadDir = getServletContext().getRealPath("/upload/");
```

Listing 8.47

```
public void testUpload() throws Exception {
    log.debug("testing upload...");
    MockHttpServletRequest request =
        new MockHttpServletRequest("POST", "/fileUpload.html");

    MockCommonsMultipartResolver resolver =
        new MockCommonsMultipartResolver();
    ctx.getServletContext();
}
```

```

request.setContentType("multipart/form-data");
request.addHeader("Content-type", "multipart/form-data");
assertTrue(resolver.isMultipart(request));
    MultipartHttpServletRequest multipartRequest =
        resolver.resolveMultipart(request);

// setup a simple mail server using Dumbster
SimpleSmtplibServer server = SimpleSmtplibServer.start(2525);

ModelAndView mav =
    fileUpload.handleRequest(multipartRequest,
                            new MockHttpServletRequest());

server.stop();
// the getReceieved() method is spelled wrong in the API. ;-)
assertEquals(1, server.getReceievedEmailSize());

log.debug("model: " + mav.getModel());

assertNotNull(request.getSession().getAttribute("message"));

// ensure the file got uploaded
Resource uploadedFile =
    ctx.getResource("file:web/upload/test.xml");
assertTrue(uploadedFile.exists());

// delete the upload directory
Resource uploadDir = ctx.getResource("file:web/upload");
uploadedFile.getFile().delete();
uploadDir.getFile().delete();
assertFalse(uploadDir.exists());
}

public static class MockCommonsMultipartResolver extends
CommonsMultipartResolver {

    private boolean empty;

    protected void setEmpty(boolean empty) {
        this.empty = empty;
    }

    protected DiskFileUpload newFileUpload() {
        return new DiskFileUpload() {
            public List parseRequest(HttpServletRequest request) {
                if (request instanceof MultipartHttpServletRequest) {
                    throw new IllegalStateException(
                        "Already a multipart request");
                }
                List fileItems = new ArrayList();

```

```

        MockFileItem fileItem = new MockFileItem(
            "file", "text/html", empty ? "" :
            "test.xml", empty ? "" : "<root/>");
        fileItems.add(fileItem);
        return fileItems;
    }
};
}
}
}

```

Listing 8.48

```

package org.appfuse.web;

// use your IDE to organize imports

public class UserWebTest extends WebTestCase {

    public UserWebTest(String name) {
        super(name);
        getTestContext().setBaseUrl("http://localhost:8080/myusers");
    }

    public void testWelcomePage() {
        beginAt("/");
        assertEquals("MyUsers ~ Welcome");
    }

    public void testAddUser() {
        beginAt("/editUser.html");
        assertEquals("MyUsers ~ User Details");
        setFormElement("firstName", "Spring");
        setFormElement("lastName", "User");
        submit("save");
        assertTextPresent("saved successfully");
    }

    public void testListUsers() {
        beginAt("/users.html");

        // check that table is present
        assertTablePresent("userList");

        //check that a set of strings are present somewhere in table
        assertTextInTable("userList",
            new String[] {"Spring", "User"});
    }

    public void testEditUser() {

```

```

        beginAt("/editUser.html?id=" + getInsertedUserId());
        assertFormElementEquals("firstName", "Spring");
        submit("save");
        assertTitleEquals("MyUsers ~ User List");
    }

    public void testDeleteUser() {
        beginAt("/editUser.html?id=" + getInsertedUserId());
        assertTitleEquals("MyUsers ~ User Details");
        submit("delete");
        assertTitleEquals("MyUsers ~ User List");
    }

    /**
     * Convenience method to get the id of the inserted user
     * Assumes last inserted user is "Spring User"
     */
    public String getInsertedUserId() {
        String[] paths = {"WEB-INF/applicationContext*.xml"};
        ApplicationContext ctx =
            new ClassPathXmlApplicationContext(paths);
        List users = ((UserDAO) ctx.getBean("userDAO")).getUsers();
        // assumed that user inserted in testAddUser() is last user
        return "" + (User) users.get(users.size() - 1).getId();
    }
}

```

Listing 8.49

```

# Page titles
index.title=MyUsers ~ Welcome
userList.title=MyUsers ~ User List
userForm.title=MyUsers ~ User Details

```

Listing 8.50

```

<title>${rc.getMessage("userList.title")}</title>

```

Listing 8.51

```

<title>${rc.getMessage("userForm.title")}</title>

```

Listing 8.52

```

<title>${rc.getMessage("index.title")}</title>

```


Listing 8.53

```
<prop key="/index.html">filenameController</prop>
```

Listing 8.54

```
<filter>
  <filter-name>rewriteFilter</filter-name>
  <filter-class>
    org.tuckey.web.filters.urlrewrite.UrlRewriteFilter
  </filter-class>
</filter>
```

Listing 8.55

```
<filter-mapping>
  <filter-name>rewriteFilter</filter-name>
  <url-pattern>/*</url-pattern>
</filter-mapping>
```

Listing 8.56

```
<urlrewrite>
  <rule>
    <from>/${</from>
    <to type="forward">index.html</to>
  </rule>
  <rule>
    <from>/index.vm</from>
    <to type="forward">index.html</to>
  </rule>
</urlrewrite>
```

Listing 8.57

```
package org.appfuse.web;

import java.util.List;
import java.util.Locale;

import net.sourceforge.jwebunit.WebTestCase;

import org.apache.commons.logging.Log;
import org.apache.commons.logging.LogFactory;
import org.appfuse.dao.UserDAO;
import org.appfuse.model.User;
import org.springframework.context.ApplicationContext;
```

```

import
org.springframework.context.support.ClassPathXmlApplicationContext;

public class UserWebTest extends WebTestCase {

    public UserWebTest(String name) {
        super(name);
        getTestContext().setBaseUrl("http://localhost:8080/myusers");
        getTestContext().setResourceBundleName("messages");
        getTestContext().setLocale(Locale.ENGLISH);
    }

    public void testWelcomePage() {
        beginAt("/");
        //assertTitleEquals("MyUsers ~ Welcome");
        assertTitleEqualsKey("index.title");
    }

    public void testAddUser() {
        beginAt("/editUser.html");
        assertTitleEquals("MyUsers ~ User Details");
        setFormElement("firstName", "Spring");
        setFormElement("lastName", "User");
        submit("save");
        //assertTextPresent("saved successfully");
        assertTitleEqualsKey("userList.title");
    }

    public void testListUsers() {
        beginAt("/users.html");

        // check that table is present
        assertTablePresent("userList");
        //check that a set of strings are present somewhere in table
        assertTextInTable("userList",
            new String[] {"Spring", "User"});
    }

    public void testEditUser() {
        beginAt("/editUser.html?id=" + getInsertedUserId());
        assertFormElementEquals("firstName", "Spring");
        submit("save");
        //assertTitleEquals("MyUsers ~ User List");
        assertTitleEqualsKey("userList.title");
    }

    public void testDeleteUser() {
        beginAt("/editUser.html?id=" + getInsertedUserId());
        //assertTitleEquals("MyUsers ~ User Details");
        assertTitleEqualsKey("userForm.title");
        submit("delete");
    }
}

```

```

        //assertTitleEquals("MyUsers ~ User List");
        assertTitleEqualsKey("userList.title");
    }

    /**
     * Convenience method to get the id of the inserted user
     * Assumes last inserted user is "Spring User"
     */

    public String getInsertedUserId() {
        String[] paths = {"/WEB-INF/applicationContext*.xml"};
        ApplicationContext ctx =
            new ClassPathXmlApplicationContext(paths);
        List users = ((UserDAO) ctx.getBean("userDAO")).getUsers();
        // assumed that user inserted in testAddUser() is last user
        return "" + (User) users.get(users.size() - 1).getId();
    }
}

```

Listing 8.58

```
getTestContext().setLocale(Locale.GERMAN);
```

Listing 8.59

```
getTestContext().getWebClient().setHeaderField("Accept-Language", "de");
```

Listing 8.60

```

<target name="test-tomcat" depends="war"
    description="Runs jWebUnit tests in a running server">
    <taskdef resource="cargo.tasks" classpathref="classpath"/>
    <cargo-tomcat5x homeDir="${tomcat.home}"
        output="${test.dir}/cargo.log"
        workingDir="${test.dir}/tomcat5x" action="start">
        <war warFile="${dist.dir}/${webapp.name}.war"/>
    </cargo-tomcat5x>
    <antcall target="test-web"/>
</target>

```

Listing 8.61

```

<p style="text-align: right; margin-bottom: -10px">
<strong>Export Options:</strong>

```

```

<a href="?report=XML">XML</a> &middot;
<a href="?report=Excel">Excel</a> &middot;
<a href="?report=PDF">PDF</a>
</p>

```

Listing 8.62

```

<target name="test-canoo" depends="deploy"
  description="Runs Canoo WebTests in Tomcat to test JSPs">

  <taskdef file="webtestTasks.properties">
    <classpath>
      <path refid="classpath"/>
      <!-- for log4j.xml -->
      <path location="web/WEB-INF/classes"/>
    </classpath>
  </taskdef>

  <mkdir dir="${test.dir}/data"/>
  <!-- Delete old results file if it exists -->
  <delete file="${test.dir}/data/web-tests-result.xml"/>

  <property name="testcase" value="run-all-tests"/>
  <ant antfile="test/web-tests.xml" target="${testcase}"/>
</target>

```

Listing 8.63

```

<config host="localhost" port="8080"
  protocol="http" basepath="${webapp.name}"
  resultpath="${test.dir}/data"
  resultfile="web-tests-result.xml"
  summary="true" saveresponse="true"/>

```

Listing 8.64

```

<!DOCTYPE project [
  <!ENTITY config SYSTEM "file:./config.xml">
]>
<project basedir="." default="run-all-tests">
  <!-- Include messages.properties so we can test against
    keys, rather than values -->
  <property file="web/WEB-INF/classes/messages.properties"/>

  <!-- runs all targets -->
  <target name="run-all-tests" depends="UserTests"
    description="Call and executes all test cases (targets)">
    <echo>All UserTests passed!</echo>
  </target>

```

```

</target>

<!-- Verify adding a user, viewing and deleting a user works -->
<target name="UserTests"
    description="Adds a new user profile">
    <canoo name="userTests">
        &config;
        <steps>
            <!-- View add screen -->
            <invoke url="/editUser.html"/>
            <verifytitle text="\${userForm.title}"/>
            <!-- Enter data and save -->
            <setinputfield name="firstName" value="Test"/>
            <setinputfield name="lastName" value="Name"/>
            <clickbutton label="Save"/>
            <!-- View user list -->
            <verifytitle text="\${userList.title}"/>
            <!-- Verify PDF contains new user -->
            <clicklink label="PDF"/>
            <verifyPdfText text="Test Name"/>
            <!-- Delete first user in table -->
            <invoke url="/users.html"/>
            <clicklink href="editUser.html"/>
            <verifytitle text="\${userForm.title}"/>
            <clickbutton label="Delete"/>
            <verifytitle text="\${userList.title}"/>
        </steps>
    </canoo>
</target>
</project>

```

Listing 8.65

```

<property name="order"><value>1</value></property>

```

Listing 8.66

```

<logger name="com.canoo">
    <level value="WARN"/>
</logger>

```

Listing 8.67

```

<target name="test-view" depends="war"
    description="Runs canoo tests in a running server">
    <taskdef resource="cargo.tasks" classpathref="classpath"/>
    <cargo-tomcat5x homeDir="\${tomcat.home}"
        output="\${test.dir}/cargo.log"

```

```
        workingDir="${test.dir}/tomcat5x" action="start">
        <war warFile="${dist.dir}/${webapp.name}.war"/>
    </cargo-tomcat5x>
    <antcall target="test-canoo"/>
</target>
```

AOP

Code

Listing 9.1

```
<bean id="loggingInterceptor"  
      class="org.springframework.aop.interceptor.TraceInterceptor"/>
```

Listing 9.2

```
<property name="preInterceptors">  
  <list>  
    <ref bean="loggingInterceptor"/>  
  </list>  
</property>
```

Listing 9.3

```
<logger name="org.springframework.aop.interceptor">  
  <level value="DEBUG"/>  
</logger>
```

Listing 9.4

```
<bean id="dataManipulationPointcut"  
      class="org.springframework.aop.support.JdkRegexpMethodPointcut">  
  <property name="patterns">  
    <list>  
      <value>.*save.*</value>  
      <value>.*remove.*</value>  
    </list>  
  </property>  
</bean>
```

Listing 9.5

```
<bean id="notificationAdvisor"
      class="org.springframework.aop.support.RegexpMethodPointcutAdvisor">
  <property name="advice">
    <ref bean="notificationInterceptor"/>
  </property>
  <property name="pattern">
    <value>.*saveUser</value>
  </property>
</bean>
```

Listing 9.6

```
<bean id="businessObject"
      class="org.springframework.aop.framework.ProxyFactoryBean">
  <property name="target">
    <bean class="org.appfuse.service.BusinessObject"/>
  </property>
  <property name="interceptorNames">
    <list><ref bean="loggingInterceptor"/></list>
  </property>
</bean>
```

Listing 9.7

```
<bean id="managerProxyCreator"
      class="org.springframework.aop.framework.autoproxy.BeanNameAuto
      ProxyCreator">
  <property name="beanNames"><value>*Manager</value></property>
  <property name="interceptorNames">
    <list>
      <value>loggingInterceptor</value>
    </list>
  </property>
</bean>
```

Listing 9.8

```
<bean id="autoProxyCreator"
      class="org.springframework.aop.framework.autoproxy.DefaultAdvisorAuto
      ProxyCreator"/>
```


Listing 9.9

```
<bean id="txProxyTemplate" abstract="true"
      class="org.springframework.transaction.interceptor.TransactionProxy
        FactoryBean">
  <property name="transactionManager">
    <ref bean="transactionManager"/>
  </property>
  <property name="transactionAttributes">
    <props>
      <prop key="save*">PROPAGATION_REQUIRED</prop>
      <prop key="remove*">PROPAGATION_REQUIRED</prop>
      <prop key="*">PROPAGATION_REQUIRED,readOnly</prop>
    </props>
  </property>
  <property name="preInterceptors">
    <list>
      <ref bean="cacheInterceptor"/>
    </list>
  </property>
</bean>

<bean id="userManager" parent="txProxyTemplate">
  <property name="target">
    <bean class="org.appfuse.service.impl.UserManagerImpl">
      <property name="userDAO"><ref bean="userDAO"/></property>
      <property name="validator">
        <ref bean="userValidator"/>
      </property>
    </bean>
  </property>
</bean>
```

Listing 9.10

```
public void testGetUserPerformance() {
    user = new User();
    user.setFirstName("Easter");
    user.setLastName("Bunny");

    user = mgr.saveUser(user);

    String name = "getUser";
    Stopwatch sw = new Stopwatch(name);
    sw.start(name);
    log.debug("Begin timing of method '" + name + "'");

    for (int i=0; i < 200; i++) {
        mgr.getUser(user.getId().toString());
    }
}
```

```

    }

    sw.stop();
    log.info(sw.shortSummary());
    log.debug("End timing of method '" + name + "'");
}

```

Listing 9.11

```

package org.appfuse.service.impl;

// use your IDE to organize imports

public class BaseManager {
    // Adding this log variable will allow children to re-use it
    protected final Log log = LogFactory.getLog(getClass());
    protected Map cache;

    protected void putIntoCache(String key, Object value) {
        if (cache == null) {
            cache = new HashMap();
        }
        cache.put(key, value);
    }

    protected void removeFromCache(String key) {
        if (cache != null) {
            cache.remove(key);
        }
    }
}

```

Listing 9.12

```

public User getUser(String userId) {
    return dao.getUser(Long.valueOf(userId));
}

public User saveUser(User user) {
    dao.saveUser(user);
    return user;
}

public void removeUser(String userId) {
    dao.removeUser(Long.valueOf(userId));
}

```

Listing 9.13

```
public User getUser(String userId) {
    // check cache for user
    User user = (User) cache.get(userId);
    if (user == null) {
        // user not in cache, fetch from database
        user = dao.getUser(Long.valueOf(userId));
        super.putIntoCache(userId, user);
    }
    return user;
}

public User saveUser(User user) {
    dao.saveUser(user);
    // update cache with saved user
    super.putIntoCache(String.valueOf(user.getId()), user);
    return user;
}

public void removeUser(String userId) {
    dao.removeUser(Long.valueOf(userId));
    // remove user from cache
    super.removeFromCache(userId);
}
```

Listing 9.14

```
package org.appfuse.aop;

// use your IDE to organize imports

public class CacheInterceptor implements MethodInterceptor {
    private final Log log = LogFactory.getLog(
        (CacheInterceptor.class);

    public Object invoke(MethodInvocation invocation) throws Throwable {
        String name = invocation.getMethod().getName();
        Object returnValue;

        // check cache before executing method
        if (name.indexOf("get") > -1 && !name.endsWith("s")) {
            String id = (String) invocation.getArguments()[0];
            returnValue = cache.get(id);
            if (returnValue == null) {
                // user not in cache, proceed
                returnValue = invocation.proceed();
                putIntoCache(id, returnValue);
            }
            return returnValue;
        }
        return invocation.proceed();
    }
}
```

```

        else {
            //log.debug("retrieved object id '" + id + "' from
cache");
        }
    } else {
        returnValue = invocation.proceed();

        // update cache after executing method
        if (name.indexOf("save") > -1) {
            Method getId = returnValue.getClass().getMethod("getId",
new
            Class[]{});
            Long id = (Long) getId.invoke(returnValue, new
Object[]{});
            putIntoCache(String.valueOf(id), returnValue);
        } else if (name.indexOf("remove") > -1) {
            String id = (String) invocation.getArguments()[0];
            removeFromCache(String.valueOf(id));
        }
    }
    return returnValue;
}

protected Map cache;

protected void putIntoCache(String key, Object value) {
    if (cache == null) {
        cache = new HashMap();
    }
    cache.put(key, value);
}

protected void removeFromCache(String key) {
    if (cache != null) {
        cache.remove(key);
    }
}
}

```

Listing 9.15

```
<bean id="cacheInterceptor" class="org.appfuse.aop.CacheInterceptor"/>
```

Listing 9.16

```
<property name="preInterceptors">
```

```

    <list>
        <ref bean="cacheInterceptor"/>
    </list>
</property>

```

Listing 9.17

```

protected void setUp() throws Exception {
    String[] paths = {"/WEB-INF/applicationContext*.xml"};
    ctx = new ClassPathXmlApplicationContext(paths);
    mgr = (UserManager) ctx.getBean("userManager");

    // Modify the mailSender bean to use Dumbster's ports
    JavaMailSenderImpl mailSender =
        (JavaMailSenderImpl) ctx.getBean("mailSender");
    mailSender.setPort(2525);
}

public void testAddAndRemoveUser() throws Exception {
    user = new User();
    user.setFirstName("Easter");
    user.setLastName("Bunny");

    // setup a simple mail server using Dumbster
    SimpleSmtpServer server = SimpleSmtpServer.start(2525);

    user = mgr.saveUser(user);

    server.stop();
    assertEquals(1, server.getReceivedEmailSize()); // spelling is
correct
    SntpMessage sentMessage =
        (SntpMessage) server.getReceivedEmail().next();
    assertTrue(sentMessage.getBody().indexOf("Easter Bunny") != -1);
    log.debug(sentMessage);

    assertTrue(user.getId() != null);
}

```

Listing 9.18

```

package org.appfuse.aop;

// organize imports using your IDE

public class NotificationInterceptor implements MethodInterceptor {
    private final Log log =
        LoggerFactory.getLog(NotificationInterceptor.class);
    private MailSender mailSender;
    private SimpleMailMessage message;
}

```

```

public void setMailSender(MailSender mailSender) {
    this.mailSender = mailSender;
}

public void setMessage(SimpleMailMessage message) {
    this.message = message;
}

public Object invoke(MethodInvocation invocation) throws Throwable {
    User user = (User) invocation.getArguments()[0];
    if (user.getId() == null) {
        if (log.isDebugEnabled()) {
            log.debug("detected new user...");
        }
        Object returnValue = invocation.proceed();
        StringBuffer sb = new StringBuffer(100);
        sb.append("A new account has been created for " + user.getFullName());
        sb.append("\n\nView this users information at:\n\t ");
        sb.append("http://localhost:8080/myusers/editUser.html?id=" + user.getId());
        message.setText(sb.toString());

        mailSender.send(message);
    }

    return user;
}
}

```

Listing 9.19

```

<bean id="notificationInterceptor"
    class="org.appfuse.aop.NotificationInterceptor">
    <property name="mailSender"><ref bean="mailSender"/></property>
    <property name="message"><ref bean="accountMessage"/></property>
</bean>

<bean id="mailSender"
    class="org.springframework.mail.javamail.JavaMailSenderImpl">
    <property name="host"><value>localhost</value></property>
</bean>

<bean id="accountMessage" singleton="false"
    class="org.springframework.mail.SimpleMailMessage">
    <property name="to">
        <value><![CDATA[MyUser Admin <youremailhere>]]></value>
    </property>
    <property name="from">

```

```
        <value><![CDATA[MyUsers <mattr@sourcebeat.com>]]></value>
    </property>
    <property name="subject">
        <value>MyUsers Account Information</value>
    </property>
</bean>
```

Listing 9.20

```
<bean id="notificationAdvisor"
    class="org.springframework.aop.support.RegexpMethodPointcutAdvisor">
    <property name="advice">
        <ref bean="notificationInterceptor"/>
    </property>
    <property name="pattern">
        <value>.*saveUser</value>
    </property>
</bean>
```

Listing 9.21

```
<property name="preInterceptors">
    <list>
        <!--ref bean="loggingInterceptor"/-->
        <!--ref bean="cacheInterceptor"/-->
        <ref bean="notificationAdvisor"/>
    </list>
</property>
```



Transactions

Code

Listing 10.1

```
[mysqld]  
default-table-type=innodb
```

Listing 10.2

```
jdbc.driverClassName=com.mysql.jdbc.Driver  
jdbc.url=jdbc:mysql://localhost/myusers?autoReconnect=true  
jdbc.username=root  
jdbc.password=hibernate.dialect=net.sf.hibernate.dialect.MySQLDialect
```

Listing 10.3

```
jdbc.driverClassName=org.postgresql.Driver  
jdbc.url=jdbc:postgres://localhost/myusers  
jdbc.username=postgres  
jdbc.password=postgres hibernate.dialect=net.sf.hibernate.dialect  
.PostgreSQLDialect
```

Listing 10.4

```
public void testAddUser() throws Exception {  
    user = new User();  
    user.setFirstName("Easter");  
    user.setLastName("Bunny");  
  
    mgr.saveUser(user);  
  
    assertTrue(user.getId() != null);  
}
```

Listing 10.5

```
<bean id="userManager"
      class="org.appfuse.service.impl.UserManagerImpl">
    <property name="userDAO"><ref bean="userDAO"/></property>
</bean>
```

Listing 10.6

```
<property name="lastName" column="last_name" not-null="true"
unique="true"
/>
```

Listing 10.7

```
public void saveUser(User user) {
    String lastName = user.getLastName();
    for (int i=0; i < 3; i++) {
        user.setId(null);
        user.setLastName(lastName + i);
        // make the last one a duplicate record
        if (i == 2) {
            user.setLastName(lastName + 0);
        }
        log.debug("entering record " + user);
        dao.saveUser(user);
    }
}
```

Listing 10.8

```
public void saveUser(User user) {

    UserTransaction tx = null;
    try {
        InitialContext ctx = new InitialContext();
        tx = (javax.transaction.UserTransaction)
            ctx.lookup( "java:comp/UserTransaction" );

        // begin transaction
        tx.begin();
        dao.saveUser(user);

        // commit transactions
        tx.commit();
    } catch (NamingException ne) {
        log.error("NamingException occurred: " + ne.getMessage());
    } catch (SystemException se) {
```

```

        log.error("SystemException occurred: " + se.getMessage());
    } catch (NotSupportedException nse) {
        log.error("NotSupported!");
    } catch (SecurityException e) {
        log.error("How many of these are there?");
        e.printStackTrace();
    } catch (IllegalStateException e) {
        log.error("Is this the longest catch block ever?");
        e.printStackTrace();
    } catch (RollbackException e) {
        e.printStackTrace();
    } catch (HeuristicMixedException e) {
        e.printStackTrace();
    } catch (HeuristicRollbackException e) {
        e.printStackTrace();
    }
}

```

Listing 10.9

```

<bean id="userManager" class="org.appfuse.service.impl.UserManagerImpl">
    <property name="transactionManager">
        <ref bean="transactionManager"/>
    </property>
    <property name="userDAO"><ref bean="userDAO"/></property>
</bean>

```

Listing 10.10

```

private TransactionTemplate txTemplate;

public void setTransactionManager(PlatformTransactionManager txManager)
{
    this.txTemplate = new TransactionTemplate(txManager);
    // set propagation behavior, isolation level etc. template
    // we don't need to since REQUIRED is the default
}

```

Listing 10.11

```

public void saveUser(final User user) {
    txTemplate.execute(new TransactionCallbackWithoutResult() {
        public void doInTransactionWithoutResult(TransactionStatus
status) {
            // do business logic
            String lastName = user.getLastName();

            for (int i = 0; i < 3; i++) {

```

```

        // can't just set user.id to null or Hibernate
        // complains that identifier was changed
        User u = new User();
        u.setFirstName(user.getFirstName());
        u.setLastName(user.getLastName() + i);

        // make the last one a duplicate record
        if (i == 2) {
            u.setLastName(lastName + 0);
        }

        if (log.isDebugEnabled()) {
            log.debug("entering record " + u);
        }

        dao.saveUser(u);
    }
}
});
}

```

Listing 10.12

```

<logger name="org.springframework.transaction">
    <level value="DEBUG"/>
</logger>

```

Listing 10.13

```

private PlatformTransactionManager transactionManager;

public void setTransactionManager(PlatformTransactionManager txManager)
{
    this.transactionManager = txManager;
}

```

Listing 10.14

```

public void saveUser(final User user) {
    DefaultTransactionDefinition txDef =
        new DefaultTransactionDefinition();
    // set propagation behavior, isolation level etc. template
    // we don't need to since REQUIRED (the default) is good enough

    TransactionStatus status =
        transactionManager.getTransaction(txDef);

    try {

```

```

        // do business logic
        String lastName = user.getLastName();

        for (int i = 0; i < 3; i++) {
            // can't just set user.id to null or Hibernate
            // complains that identifier was changed
            User u = new User();
            u.setFirstName(user.getFirstName());
            u.setLastName(user.getLastName() + i);

            // make the last one a duplicate record
            if (i == 2) {
                u.setLastName(lastName + 0);
            }

            if (log.isDebugEnabled()) {
                log.debug("entering record " + u);
            }

            dao.saveUser(u);
        }
    } catch (DataAccessException ex) {
        transactionManager.rollback(status);
        throw ex;
    }
    transactionManager.commit(status);
}

```

Listing 10.15

```

public void saveUser(final User user) {
    // do business logic
    String lastName = user.getLastName();

    for (int i = 0; i < 3; i++) {
        // can't just set user.id to null or Hibernate
        // complains that identifier was changed
        User u = new User();
        u.setFirstName(user.getFirstName());
        u.setLastName(user.getLastName() + i);

        // make the last one a duplicate record
        if (i == 2) {
            u.setLastName(lastName + 0);
        }

        if (log.isDebugEnabled()) {
            log.debug("entering record " + u);
        }
    }
}

```

```

        dao.saveUser(u);
    }
}

```

Listing 10.16

```

<bean id="userManager"
    class="org.springframework.transaction.interceptor.TransactionProxy
        FactoryBean">
    <property name="transactionManager"><ref bean="transactionManager"/>
    </property>
    <property name="target"><ref bean="userManagerTarget"/></property>
    <property name="transactionAttributes">
        <props>
            <prop key="save*">PROPAGATION_REQUIRED</prop>
            <prop key="remove*">PROPAGATION_REQUIRED</prop>
            <prop key="*">PROPAGATION_REQUIRED,readOnly</prop>
        </props>
    </property>

    <bean id="userManagerTarget" class="org.appfuse.service.impl.
        UserManagerImpl">
        <property name="userDAO"><ref bean="userDAO"/></property>
    </bean>

```

Listing 10.17

```

<bean id="userManager"
    class="org.springframework.transaction.interceptor.TransactionProxy
        FactoryBean">
    <property name="transactionManager">
        <ref bean="transactionManager"/>
    </property>
    <property name="target">
        <bean class="org.appfuse.service.impl.UserManagerImpl">
            <property name="userDAO"><ref bean="userDAO"/></property>
        </bean>
    </property>
    <property name="transactionAttributes">
        <props>
            <prop key="save*">PROPAGATION_REQUIRED</prop>
            <prop key="remove*">PROPAGATION_REQUIRED</prop>
            <prop key="*">PROPAGATION_REQUIRED,readOnly</prop>
        </props>
    </property>
</bean>

```

Listing 10.18

```
<prop key="save*">PROPAGATION_REQUIRED, ISOLATION_REPEATABLE_READ</prop>
```

Listing 10.19

```
<prop key="save*">PROPAGATION_REQUIRED, +UserExistsException</prop>
```

Listing 10.20

```
<bean id="txProxyTemplate" abstract="true"
    class="org.springframework.transaction.interceptor.TransactionProxy
        FactoryBean">
    <property name="transactionManager">
        <ref bean="transactionManager"/>
    </property>
    <property name="transactionAttributes">
        <props>
            <prop key="save*">PROPAGATION_REQUIRED</prop>
            <prop key="remove*">PROPAGATION_REQUIRED</prop>
            <prop key="*">PROPAGATION_REQUIRED, readOnly</prop>
        </props>
    </property>
</bean>
```

Listing 10.21

```
<bean id="userManager" parent="txProxyTemplate">
    <property name="target">
        <bean class="org.appfuse.service.impl.UserManagerImpl">
            <property name="userDAO"><ref bean="userDAO"/></property>
        </bean>
    </property>
</bean>
```

Listing 10.22

```
<bean name="txAttributes"
    class="org.springframework.transaction.interceptor.NameMatchTransaction
        AttributeSource">
    <property name="properties">
        <props>
            <prop key="save*">PROPAGATION_REQUIRED</prop>
            <prop key="remove*">PROPAGATION_REQUIRED</prop>
        </props>
    </property>
</bean>
```

```

        <prop key="*">PROPAGATION_REQUIRED,readOnly</prop>
    </props>
</property>
</bean>

```

Listing 10.23

```

<bean id="userManager"
    class="org.springframework.transaction.interceptor.Transaction
        ProxyFactoryBean">
    <property name="transactionManager">
        <ref bean="transactionManager"/>
    </property>
    <property name="target">
        <bean class="org.appfuse.service.impl.UserManagerImpl">
            <property name="userDAO">
                <ref bean="userDAO"/>
            </property>
        </bean>
    </property>
    <property name="transactionAttributeSource"><ref
bean="txAttributes"/>
        </property>
</bean>

```

Listing 10.24

```

<target name="compileAttributes"
    description="compiles classes enhanced with commons attributes">
    <taskdef resource="org/apache/commons/attributes/
anttasks.properties">
        <classpath refid="classpath"/>
    </taskdef>
    <!-- Compile to a temp directory -->
    <mkdir dir="${build.dir}/attributes"/>
    <attribute-compiler destdir="${build.dir}/attributes">
        <fileset dir="${src.dir}" includes="**/*ManagerImpl.java"/>
    </attribute-compiler>
</target>

```

Listing 10.25

```

<target name="compile" depends="compileAttributes"
    description="Compile main source tree java files">
    <mkdir dir="${build.dir}/classes"/>
    <javac destdir="${build.dir}/classes" debug="true"
        deprecation="false" optimize="false" failonerror="true">
        <src path="${src.dir}"/>
    </javac>
</target>

```



```

        <src path="${build.dir}/attributes"/>
        <classpath refid="classpath"/>
    </javac>
    ...
</target>

```

Listing 10.26

```

<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE beans PUBLIC "-//SPRING//DTD BEAN//EN"
    "http://www.springframework.org/dtd/spring-beans.dtd">

<beans>
    <!--
        This bean is a post-processor that will automatically apply
        relevant advisors any bean in child factories.
    -->
    <bean id="autoProxyCreator"
class="org.springframework.aop.framework.
    autoproxy.DefaultAdvisorAutoProxyCreator"/>

    <!--
        This bean specifies the implementation to use for reading
attributes
    -->
    <bean id="transactionAttributeSource" class="org.springframework.
        transaction.interceptor.AttributesTransactionAttributeSource"
        autowire="constructor"/>

    <bean id="transactionInterceptor" class="org.springframework.
        transaction.interceptor.TransactionInterceptor"
        autowire="byType"/>

    <!--
        AOP advisor that will provide declarative transaction
        management on attributes. It's possible to add arbitrary
        custom Advisor implementations as well, and they will also
        be evaluated and applied automatically.
    -->
    <bean id="transactionAdvisor"
class="org.springframework.transaction.
        interceptor.TransactionAttributeSourceAdvisor"
        autowire="constructor"/>

    <!-- Commons Attributes Attributes implementation. -->
    <bean id="attributes"
        class="org.springframework.metadata.commons.CommonsAttributes"/>
</beans>
Add a DefaultTransactionAttribute element to the javadoc of the
saveUser()

```

```
method:
/**
 * @@org.springframework.transaction.interceptor.DefaultTransaction
 * Attribute()
 */
```

Listing 10.27

```
<bean id="userManager" class="org.appfuse.service.impl.UserManagerImpl">
  <property name="userDAO"><ref bean="userDAO"/></property>
</bean>
```

Listing 10.28

```
<bean id="transactionAttributeSource"
  class="org.springframework.transaction.annotations.Annotations
    TransactionAttributeSource"
  autowire="constructor"/>
```

Listing 10.29

```
<bean id="attributes"
  class="org.springframework.metadata.annotations.AnnotationsAttributes"/>
```

Listing 10.30

```
@org.springframework.transaction.annotations.TxAttribute
public void saveUser(final User user) {
```

Web Framework Integration

Code

Listing 11.1

```
<context-param>
  <param-name>contextConfigLocation</param-name>
  <param-value>/WEB-INF/applicationContext*.xml</param-value>
</context-param>
```

Listing 11.2

```
<listener>
  <listener-class>
    org.springframework.web.context.ContextLoaderListener
  </listener-class>
</listener>
```

Listing 11.3

```
WebApplicationContext ctx =
    WebApplicationContextUtils.getWebApplicationContext(servletContext);
```

Listing 11.4

```
ServletContext servletContext =
    request.getSession().getServletContext();
```

Listing 11.5

```
<?xml version="1.0"?>
<!DOCTYPE faces-config PUBLIC
    "-//Sun Microsystems, Inc.//DTD JavaServer Faces Config 1.1//EN"
    "http://java.sun.com/dtd/web-facesconfig_1_1.dtd">

<faces-config>
    <!-- Spring VariableResolver for JSF -->
    <application>
        <variable-
resolver>org.springframework.web.jsf.DelegatingVariable
        Resolver</variable-resolver>
        <locale-config>
            <default-locale>en</default-locale>
            <supported-locale>en</supported-locale>
            <supported-locale>es</supported-locale>
        </locale-config>
        <message-bundle>messages</message-bundle>
    </application>

    <navigation-rule>
        <from-view-id>/userList.jsp</from-view-id>
        <navigation-case>
            <from-outcome>add</from-outcome>
            <to-view-id>/userForm.jsp</to-view-id>
        </navigation-case>
    </navigation-rule>

    <managed-bean>
        <managed-bean-name>userList</managed-bean-name>
        <managed-bean-class>org.appfuse.web.UserList</managed-bean-
class>
        <managed-bean-scope>request</managed-bean-scope>
        <managed-property>
            <property-name>userManager</property-name>
            <value>#{userManager}</value>
        </managed-property>
    </managed-bean>
</faces-config>
```

Listing 11.6

```
<managed-property>
    <property-name>userManager</property-name>
    <value>#{userManager}</value>
</managed-property>
```

Listing 11.7

```
<faces-config>
  <application>
    <variable-resolver>
      org.springframework.web.jsf.DelegatingVariableResolver
    </variable-resolver>
    <locale-config>
      <default-locale>en</default-locale>
      <supported-locale>en</supported-locale>
      <supported-locale>es</supported-locale>
    </locale-config>
    <message-bundle>messages</message-bundle>
  </application>
```

Listing 11.8

```
<managed-bean>
  <managed-bean-name>userList</managed-bean-name>
  <managed-bean-class>org.appfuse.web.UserList</managed-bean-class>
  <managed-bean-scope>request</managed-bean-scope>
  <managed-property>
    <property-name>userManager</property-name>
    <value>#{userManager}</value>
  </managed-property>
</managed-bean>
```

Listing 11.9

```
ApplicationContext ctx = FacesContextUtils.getWebApplicationContext(
    FacesContext.getCurrentInstance());
```

Listing 11.10

```
public ActionForward execute(ActionMapping mapping,
    ActionForm form,
    HttpServletRequest request,
    HttpServletResponse response)
    throws Exception;
```

Listing 11.11

```
<plug-in className="org.springframework.web.struts.ContextLoaderPlugIn">
  <set-property property="contextConfigLocation"
```

```
        value="/WEB-INF/applicationContext-hibernate.xml,  
            /WEB-INF/applicationContext.xml,  
            /WEB-INF/action-servlet.xml"/>  
</plug-in>
```

Listing 11.12

```
<action path="/users" .../>
```

Listing 11.13

```
<bean name="/users" .../>
```

Listing 11.14

```
<controller>  
    <set-property property="processorClass"  
  
value="org.springframework.web.struts.DelegatingRequestProcessor"/>  
</controller>
```

Listing 11.15

```
<action path="/user" type="org.appfuse.web.UserAction"/>  
<action path="/user"/>
```

Listing 11.16

```
<action path="/user"  
    type="org.springframework.web.struts.DelegatingActionProxy"  
    name="userForm" scope="request" parameter="method"  
    validate="false">  
    <forward name="list" path="/userList.jsp"/>  
    <forward name="edit" path="/userForm.jsp"/>  
</action>
```

Listing 11.17

```
<bean name="/user" singleton="false" autowire="byName"  
    class="org.appfuse.web.UserAction"/>
```

Listing 11.18

```
public class UserAction extends DispatchActionSupport {

    public ActionForward execute(ActionMapping mapping,
                                ActionForm form,
                                HttpServletRequest request,
                                HttpServletResponse response)
        throws Exception {
        if (log.isDebugEnabled()) {
            log.debug("entering 'delete' method...");
        }

        WebApplicationContext ctx = getWebApplicationContext();
        UserManager mgr = (UserManager) ctx.getBean("userManager");

        // talk to manager for business logic

        return mapping.findForward("success");
    }
}
```

Listing 11.19

```
<span jwcid="@Insert" value="ognl:user.name">Joe User</span>
```

Listing 11.20

```
ServletContext servletContext =
    getRequestCycle().getRequestContext().getServlet()
        .getServletContext();
WebApplicationContext appContext =
    WebApplicationContextUtils.getApplicationContext(servletContext);
UserManager userManager =
    (UserManager) appContext.getBean("userManager");
```

Listing 11.21

```
package org.appfuse.web;

// ...

public class BaseEngine extends org.apache.tapestry.engine.BaseEngine {

    public static final String APPLICATION_CONTEXT_KEY = "appContext";

    protected void setupForRequest(RequestContext context) {
        super.setupForRequest(context);
    }
}
```

```

        Map global = (Map) getGlobal();
        ApplicationContext ac = (ApplicationContext) global
            .get(APPLICATION_CONTEXT_KEY);

        ac = WebApplicationContextUtils
            .getWebApplicationContext(context
                .getServlet().getServletContext());
        global.put(APPLICATION_CONTEXT_KEY, ac);
    }
}

```

Listing 11.22

```

<application name="tapestry"
    engine-class="org.appfuse.web.BaseEngine">

```

Listing 11.23

```

package org.appfuse.web;

import org.appfuse.service.UserManager;

public abstract class UserList extends BasePage {
    public abstract UserManager getUserManager();
}

```

Listing 11.24

```

<property-specification name="userManager"
    type="org.appfuse.service.UserManager">
    global.appContext.getBean("userManager")
</property-specification>

```

Listing 11.25

```

public class UserAction extends ActionSupport {
    private UserManager mgr = null;
    private List users;

    public void setUserManager(UserManager userManager) {
        this.mgr = userManager;
    }

    public List getUsers() {
        return users;
    }
}

```



```

    }

    public String execute() {
        users = mgr.getUsers();
        return SUCCESS;
    }
}

<action name="users" class="org.appfuse.web.UserAction">
    <result name="success" type="dispatcher">userList.jsp</result>
</action>

```

Listing 11.26

```

<interceptor-stack name="defaultStack">
    <interceptor-ref name="servlet-config"/>
    <interceptor-ref name="prepare"/>
    <interceptor-ref name="static-params"/>
    <interceptor-ref name="params"/>
    <interceptor-ref name="conversionError"/>
</interceptor-stack>

```

Listing 11.27

```

<listener>
    <listener-class>com.opensymphony.xwork.spring.SpringObjectFactory
        Listener
    </listener-class>
</listener>

```

Listing 11.28

```

<bean id="springObjectFactory"
    class="com.opensymphony.xwork.spring.SpringObjectFactory"
    init-method="initObjectFactory"/>

```

Listing 11.29

```

<bean id="userAction" class="org.appfuse.web.UserAction"
    singleton="false">
    <property name="userManager"><ref bean="userManager"/></property>
</bean>

```

Listing 11.30

```
<action name="users" class="userAction" method="list">
    <result name="success" type="dispatcher">userList.jsp</result>
</action>
```

Listing 11.31

```
<package name="default" extends="webwork-default">
    <interceptors>
        <interceptor name="autowireDependencies"
            class="com.opensymphony.xwork.spring.interceptor.Action
                AutowiringInterceptor"/>
        <interceptor-stack name="defaultActionStack">
            <interceptor-ref name="defaultStack"/>
            <interceptor-ref name="autowireDependencies"/>
        </interceptor-stack>
    </interceptors>

    <default-interceptor-ref name="defaultActionStack"/>
```

Listing 11.32

```
<listener>
    <listener-
class>com.opensymphony.xwork.spring.SpringExternalReference
        ResolverSetupListener</listener-class>
</listener>
```

Listing 11.33

```
<package name="default" extends="webwork-default"

    <externalReferenceResolver="com.opensymphony.xwork.spring.SpringExternal
        ReferenceResolver">
```

Listing 11.34

```
<interceptors>
    <interceptor name="referenceResolver" class="com.opensymphony.xwork.
        interceptor.ExternalReferencesInterceptor"/>
    <interceptor-stack name="defaultReferenceStack">
        <interceptor-ref name="defaultStack"/>
        <interceptor-ref name="referenceResolver"/>
    </interceptor-stack>
</interceptors>
```

```
</interceptor-stack>
</interceptors>

<default-interceptor-ref name="defaultReferenceStack"/>
```

Listing 11.35

```
<action name="users" class="org.appfuse.web.UserAction" method="list">
  <external-ref name="userManager">userManager</external-ref>
  <result name="success" type="dispatcher">userList.jsp</result>
</action>
```



Security

Code

Listing 12.1

```
<security-constraint>
  <web-resource-collection>
    <web-resource-name>Secure Area</web-resource-name>
    <url-pattern>/*</url-pattern>
  </web-resource-collection>

  <auth-constraint>
    <role-name>*</role-name>
  </auth-constraint>
</security-constraint>

<login-config>
  <auth-method>BASIC</auth-method>
  <realm-name>Protected Area</realm-name>
</login-config>
```

Listing 12.2

```
<security-role>
  <role-name>ROLE_NAME</role-name>
</security-role>
```

Listing 12.3

```
<auth-constraint>
  <role-name>administrator</role-name>
  <role-name>user</role-name>
</auth-constraint>
```

Listing 12.4

```
Authorization: Basic dG9tY2F0OnRvbWNhdA==
```

Listing 12.5

```
<login-config>
  <auth-method>DIGEST</auth-method>
  <realm-name>Protected Area</realm-name>
</login-config>
```

Listing 12.6

```
Authorization: Digest username="tomcat", realm="Protected Area",
nonce="935d7777d258321c0447286691d44a61", uri="/equinox-security/",
response="88196a77de6948312810f963ff3d3fe8",
opaque="fcfb979b1cbc533dfacc6d71a843db79", qop=auth, nc=00000001,
cnonce="01b6730aae57c007"
```

Listing 12.7

```
<form id="loginForm" method="post" action="j_security_check">
  <p>
    Username: <input type="text" name="j_username" id="j_username"/>
  <br/>
    Password: <input type="password" name="j_password"
id="j_password"/>
    <button type="submit">Login</button>
  </p>
</form>
```

Listing 12.8

```
<p>
  Login failed - please <a href="index.jsp">try again</a>.
</p>
```

Listing 12.9

```
<login-config>
  <auth-method>FORM</auth-method>
  <form-login-config>
    <form-login-page>/login.jsp</form-login-page>
    <form-error-page>/loginError.jsp</form-error-page>
  </form-login-config>
</login-config>
```

Listing 12.10

```
<% session.invalidate(); %>
<p>
    You have successfully logged out.
    <a href="index.jsp">Login</a> again.
</p>
```

Listing 12.11

```
<c:if test="${pageContext.request.remoteUser != null}">
<a href="logout.jsp" style="color: white; position: absolute; top: 5px;
right: 5px">Logout</a>
</c:if>
```

Listing 12.12

```
<welcome-file-list>
    <welcome-file>index.jsp</welcome-file>
</welcome-file-list>
```

Listing 12.13

```
<c:redirect url="/users.html"/>
```

Listing 12.14

```
<form-login-config>
    <form-login-page>/login.jsp</form-login-page>
    <form-error-page>/login.jsp?error=true</form-error-page>
</form-login-config>
```

Listing 12.15

```
<c:if test="${param.error != null}">
    <div class="error">
        Invalid username and/or password, please try again.
    </div>
</c:if>
```

Listing 12.16

```
<%@ include file="/taglibs.jsp"%>

<c:import url="login.jsp">
  <c:param name="error" value="true"/>
</c:import>
```

Listing 12.17

```
<!-- Allow anyone to access passwordHint and signup -->
<security-constraint>
  <web-resource-collection>
    <web-resource-name>Unrestricted</web-resource-name>
    <url-pattern>/passwordHint.html</url-pattern>
    <url-pattern>/signup.html</url-pattern>
    <http-method>POST</http-method>
    <http-method>GET</http-method>
  </web-resource-collection>
</security-constraint>
```

Listing 12.18

```
<Context path="/equinox-security" docBase="equinox-security" debug="99"
  reloadable="true" antiJARLocking="true" antiResourceLocking="true">

  <Realm className="org.apache.catalina.realm.JDBCRealm" debug="99"
    driverName="com.mysql.jdbc.Driver"
    connectionURL="jdbc:mysql://localhost/equinox?autoReconnect=true"
    connectionName="root" connectionPassword=""
    userTable="users" userNameCol="username"
    userCredCol="password"
    userRoleTable="user_roles" roleNameCol="rolename"/>
</Context>
```

Listing 12.19

```
<Resource name="UserDatabase" auth="Container"
  type="org.apache.catalina.UserDatabase"
  description="User database that can be updated and saved"
  factory="org.apache.catalina.users.MemoryUserDatabaseFactory"
  pathname="conf/tomcat-users.xml" />
...
<!-- Default realm setting -->
<Realm className="org.apache.catalina.realm.UserDatabaseRealm"
  resourceName="UserDatabase" />
```


Listing 12.20

```
<user username="spring" password="rocks" roles="admin,user"/>
```

Listing 12.21

```
<user username="tomcat" password="tomcat" roles="tomcat"/>
```

Listing 12.22

```
<Realm className="org.apache.catalina.realm.  
UserDatabaseRealm"  
resourceName="UserDatabase" digest="SHA"/>
```

Listing 12.23

```
java -cp "$CATALINA_HOME/server/lib/catalina.jar;$CATALINA_HOME/bin/  
jmx.jar;$CATALINA_HOME/bin/commons-logging-api.jar"  
org.apache.catalina.realm.RealmBase -a SHA tomcat
```

Listing 12.24

```
tomcat:536c0b339345616c1b33caf454454d8b8a190d6c
```

Listing 12.25

```
<Context path="/equinox-security" docBase="equinox-security" debug="99"  
reloadable="true" antiJARLocking="true" antiResourceLocking="true">  
  
  <Realm className="org.apache.catalina.realm.JDBCRealm" debug="99"  
    driverName="com.mysql.jdbc.Driver" debug="SHA"  
    connectionURL="jdbc:mysql://localhost/equinox"  
    connectionName="root" connectionPassword=""  
    userTable="users" userNameCol="username"  
    userCredCol="password"  
    userRoleTable="user_roles" roleNameCol="rolename"/>  
</Context>
```

Listing 12.26

```
CREATE DATABASE IF NOT EXISTS equinox;  
use equinox;  
  
DROP TABLE IF EXISTS users;  
CREATE TABLE users (
```

```

username varchar(50) NOT NULL,
password varchar(50) NOT NULL,
PRIMARY KEY (username)
) TYPE=InnoDB;

INSERT INTO users (username, password) VALUES ('springlive',
2a9152cff1d25b5bbaa3e5fbc7acdc6905c9f251');

DROP TABLE IF EXISTS user_roles;
CREATE TABLE user_roles (
username varchar(50) NOT NULL,
rolename varchar(20) NOT NULL,
PRIMARY KEY (username,rolename)
) TYPE=InnoDB;

INSERT INTO user_roles (username, rolename) VALUES
('springlive', 'subscriber');
```

Listing 12.27

```

includeC:/Tools/openldap/etc/schema/core.schema
includeC:/Tools/openldap/etc/schema/cosine.schema
includeC:/Tools/openldap/etc/schema/inetorgperson.schema
```

Listing 12.28

```

databasebdb
suffix      "dc=springlive,dc=com"
rootdn      "cn=Manager,dc=springlive,dc=com"
rootpw      secret
directoryC:/Tools/openldap/var/openldap-data
```

Listing 12.29

```

# Define top-level entry
dn: dc=springlive,dc=com
objectClass: dcObject
objectClass: organization
o: Spring Live
dc: springlive

# Define Manager Role to authenticate with
dn: cn=Manager,dc=springlive,dc=com
objectclass: organizationalRole
cn: Manager
description: Directory Manager
```

```

# Define an entry to contain users
# searches for users are based on this entry
dn: ou=users,dc=springlive,dc=com
objectClass: organizationalUnit
ou: users

# Define a user entry for "springlive" user
dn: uid=springlive,ou=users,dc=springlive,dc=com
objectClass: inetOrgPerson
uid: springlive
sn: Live
cn: Spring Live
userPassword: springlive

# Define an entry to contain LDAP groups (a.k.a. roles)
# searches for roles are based on this entry
dn: ou=groups,dc=springlive,dc=com
objectClass: organizationalUnit
ou: groups

# Define an entry for the "developer" role
dn: cn=developer,ou=groups,dc=springlive,dc=com
objectClass: groupOfUniqueNames
cn: developer
uniqueMember: uid=springlive,ou=users,dc=springlive,dc=com

```

Listing 12.30

```

c:/tools/openldap/ldapadd -x -D "cn=Manager,dc=springlive,dc=com" -W -f
metadata/ldif/entries.ldif

```

Listing 12.31

```

<Realm className="org.apache.catalina.realm.JNDIRealm" debug="99"
  connectionName="cn=Manager,dc=springlive,dc=com"
  connectionPassword="secret"
  connectionURL="ldap://localhost:389" userPassword="userPassword"
  userPattern="uid={0},ou=users,dc=springlive,dc=com"
  roleBase="ou=groups,dc=springlive,dc=com"
  roleName="cn" roleSearch="(uniqueMember={0})" />

```

Listing 12.32

```

<user-data-constraint>
  <transport-guarantee>CONFIDENTIAL</transport-guarantee>
</user-data-constraint>

```

Listing 12.33

```
<tag:secure/>
<tag:secure mode="unsecured"/>
```

Listing 12.34

```
// If using https, switch to http
String redirectString =
    SslUtil.getRedirectString(request, config.getServletContext(),
                             false); // false for http, true for https

if (redirectString != null) {
    // Redirect the page to the desired URL
    response.sendRedirect(response.encodeRedirectURL(redirectString));
    return;
}
```

Listing 12.35

```
Authentication auth =
    SecureContextUtils.getSecureContext().getAuthentication();
```

Listing 12.36

```
<logger name="net.sf.acegisecurity">
    <level value="INFO"/>
</logger>
```

Listing 12.37

```
<filter>
    <filter-name>securityFilter</filter-name>
    <filter-class>net.sf.acegisecurity.util.FilterToBeanProxy</
filter-class>
    <init-param>
        <param-name>targetClass</param-name>
        <param-value>net.sf.acegisecurity.util.FilterChainProxy</
param-value>
    </init-param>
</filter>
```

Listing 12.38

```
<filter-mapping>
  <filter-name>securityFilter</filter-name>
  <url-pattern>/*</url-pattern>
</filter-mapping>
```

Listing 12.39

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE beans PUBLIC "-//SPRING//DTD BEAN//EN"
    "http://www.springframework.org/dtd/spring-beans.dtd">

<beans>
  <!-- ===== FILTER CHAIN ===== -->
  <bean id="filterChainProxy"
    class="net.sf.acegisecurity.util.FilterChainProxy">
    <property name="filterInvocationDefinitionSource">
      <value>
        CONVERT_URL_TO_LOWERCASE_BEFORE_COMPARISON
        PATTERN_TYPE_APACHE_ANT
        /
        **=httpSessionContextIntegrationFilter,basicProcessingFilter,securityEnf
        orcementFilter
      </value>
    </property>
  </bean>

  <!-- ===== AUTHENTICATION ===== -->
  <bean id="authenticationManager"
    class="net.sf.acegisecurity.providers.ProviderManager">
    <property name="providers">
      <list>
        <ref local="daoAuthenticationProvider"/>
      </list>
    </property>
  </bean>

  <bean id="inMemoryDaoImpl"
    class="net.sf.acegisecurity.providers.dao.memory.InMemoryDaoImpl">
    <property name="userMap">
      <value>tomcat=tomcat,ROLE_USER</value>
    </property>
  </bean>

  <bean id="daoAuthenticationProvider"
    class="net.sf.acegisecurity.providers.dao.DaoAuthenticationProvider">
    <property name="authenticationDao">
      <ref local="inMemoryDaoImpl"/>
    </property>
  </bean>
```

```

        </property>
    </bean>

    <bean id="basicProcessingFilter"
        class="net.sf.acegisecurity.ui.basicauth.BasicProcessingFilter">
        <property name="authenticationManager">
            <ref local="authenticationManager"/>
        </property>
        <property name="authenticationEntryPoint">
            <ref local="basicProcessingFilterEntryPoint"/>
        </property>
    </bean>

    <bean id="basicProcessingFilterEntryPoint"
class="net.sf.acegisecurity.ui.basicauth.BasicProcessingFilterEntryPoint"
">
        <property name="realmName">
            <value>Protected Area</value>
        </property>
    </bean>

    <!-- ===== HTTP REQUEST SECURITY ===== -->
    <bean id="httpSessionContextIntegrationFilter"
class="net.sf.acegisecurity.context.HttpSessionContextIntegrationFilter"
>
        <property
name="context"><value>net.sf.acegisecurity.context.security.SecureContextImpl</value></property>
    </bean>

    <bean id="securityEnforcementFilter"
class="net.sf.acegisecurity.intercept.web.SecurityEnforcementFilter">
        <property name="filterSecurityInterceptor">
            <ref local="filterInvocationInterceptor"/>
        </property>
        <property name="authenticationEntryPoint">
            <ref local="basicProcessingFilterEntryPoint"/>
        </property>
    </bean>

    <bean id="httpRequestAccessDecisionManager"
        class="net.sf.acegisecurity.vote.AffirmativeBased">
        <property name="allowIfAllAbstainDecisions">
            <value>false</value>
        </property>
        <property name="decisionVoters">
            <list>
                <ref bean="roleVoter"/>
            </list>
        </property>
    </bean>

```

```

<bean id="roleVoter" class="net.sf.acegisecurity.vote.RoleVoter"/>

<!-- Note the order that entries are placed against the
      objectDefinitionSource is critical. The
      FilterSecurityInterceptor will work from the top of the list
      down to the FIRST pattern that matches the request URL.
      Accordingly, you should place MOST SPECIFIC (ie a/b/c/d.*)
      expressions first, with LEAST SPECIFIC (ie a/.* ) expressions
      last -->
<bean id="filterInvocationInterceptor"
class="net.sf.acegisecurity.intercept.web.FilterSecurityInterceptor">
  <property name="authenticationManager">
    <ref bean="authenticationManager"/>
  </property>
  <property name="accessDecisionManager">
    <ref local="httpRequestAccessDecisionManager"/>
  </property>
  <property name="objectDefinitionSource">
    <value>
      CONVERT_URL_TO_LOWERCASE_BEFORE_COMPARISON
      PATTERN_TYPE_APACHE_ANT
      /*.html*=ROLE_USER
      /*.jsp=ROLE_USER
    </value>
  </property>
</bean>
</beans>

```

Listing 12.40

```

/*.*.html*=ROLE_USER
/*.*.jsp=ROLE_USER

```

Listing 12.41

```

<property name="objectDefinitionSource">
  <value>
    CONVERT_URL_TO_LOWERCASE_BEFORE_COMPARISON
    PATTERN_TYPE_APACHE_ANT
    /admin/*=ROLE_ADMIN,ROLE_DBA
  </value>
</property>

```

Listing 12.42

```
<bean id="authenticationProcessingFilter"
class="net.sf.acegisecurity.ui.webapp.AuthenticationProcessingFilter">
<property name="authenticationManager">
    <ref local="authenticationManager"/></property>
<property name="authenticationFailureUrl">
    <value>/loginError.htm</value>
</property>
    <property name="defaultTargetUrl"><value>/</value></property>
<property name="filterProcessesUrl">
    <value>/j_security_check</value>
</property>
</bean>
```

Listing 12.43

```
<bean id="authenticationProcessingFilterEntryPoint"
class="net.sf.acegisecurity.ui.webapp.AuthenticationProcessingFilterEntr
ypoint">
    <property name="loginFormUrl"><value>/login.htm</value></property>
    <property name="forceHttps"><value>>false</value></property>
</bean>
```

Listing 12.44

```
/
**=httpSessionContextIntegrationFilter, authenticationProcessingFilter, se
curityEnforcementFilter
```

Listing 12.45

```
<bean id="securityEnforcementFilter"
    class="net.sf.acegisecurity.intercept.web.SecurityEnforcement
Filter">
<property name="filterSecurityInterceptor">
    <ref local="filterInvocationInterceptor"/>
</property>
<property name="authenticationEntryPoint">
    <ref local="authenticationProcessingFilterEntryPoint"/>
</property>
</bean>
```

Listing 12.46

```
<bean id="requestMethodsFilter"
class="net.sf.acegisecurity.wrapper.ContextHolderAwareRequestFilter"/>
```


Listing 12.47

```
...
authenticationProcessingFilter,requestMethodsFilter,securityEnforcementFi
lter
```

Listing 12.48

```
<% session.invalidate(); response.sendRedirect("index.jsp"); %>
```

Listing 12.49

```
<bean id="daoAuthenticationProvider"
    class="net.sf.acegisecurity.providers.dao.DaoAuthenticationProvider">
    <property name="authenticationDao">
        <ref local="inMemoryDaoImpl"/>
    </property>
</bean>
```

Listing 12.50

```
<bean id="authenticationManager"
    class="net.sf.acegisecurity.providers.ProviderManager">
    <property name="providers">
        <list>
            <ref local="daoAuthenticationProvider"/>
        </list>
    </property>
</bean>
```

Listing 12.51

```
<bean id="inMemoryDaoImpl"
    class="net.sf.acegisecurity.providers.dao.memory.InMemoryDaoImpl">
    <property name="userMap">
        <value>
            tomcat=tomcat,ROLE_USER
            springlive=springlive,ROLE_USER
        </value>
    </property>
</bean>
```

Listing 12.52

```
tomcat:536c0b339345616c1b33caf454454d8b8a190d6c
springlive:2a9152cff1d25b5bbaa3e5fbc7acdc6905c9f251
```

Listing 12.53

```
<bean id="inMemoryDaoImpl"
class="net.sf.acegisecurity.providers.dao.memory.InMemoryDaoImpl">
  <property name="userMap">
    <value>
      tomcat=536c0b339345616c1b33caf454454d8b8a190d6c,ROLE_USER
      springlive=2a9152cff1d25b5bbaa3e5fbc7acdc6905c9f251,ROLE_USER
    </value>
  </property>
</bean>
```

Listing 12.54

```
<bean id="daoAuthenticationProvider"
class="net.sf.acegisecurity.providers.dao.DaoAuthenticationProvider">
  <property name="authenticationDao">
    <ref local="inMemoryDaoImpl"/>
  </property>
  <property name="passwordEncoder">
    <ref local="passwordEncoder"/>
  </property>
</bean>
<bean id="passwordEncoder"
class="net.sf.acegisecurity.providers.encoding.ShaPasswordEncoder"/>
```

Listing 12.55

```
<bean id="jdbcDaoImpl"
class="net.sf.acegisecurity.providers.dao.jdbc.JdbcDaoImpl">
  <property name="dataSource"><ref bean="dataSource"/></property>
</bean>
```

Listing 12.56

```
"SELECT username,password,enabled FROM users WHERE username = ?";
"SELECT username,authority FROM authorities WHERE username = ?";
```

Listing 12.57

```
CREATE DATABASE IF NOT EXISTS equinox;
use equinox;

DROP TABLE IF EXISTS users;
CREATE TABLE users (
  username varchar(50) NOT NULL,
```

```

password varchar(50) NOT NULL,
enabled bit default 1,
PRIMARY KEY (username)
) TYPE=InnoDB;

INSERT INTO users (username, password) VALUES ('springlive',
8e86cbb973c3813c1d33b7525c0f00916595907a');

DROP TABLE IF EXISTS authorities;
CREATE TABLE authorities (
username varchar(50) NOT NULL,
authority varchar(20) NOT NULL,
PRIMARY KEY (username,authority)
) TYPE=InnoDB;

INSERT INTO authorities (username, authority) VALUES
('springlive', 'ROLE_USER');
```

Listing 12.58

```

<bean id="securityDataSource"
class="org.springframework.jdbc.datasource.DriverManagerDataSource">
<property name="driverClassName">
<value>com.mysql.jdbc.Driver</value>
</property>
<property name="url">
<value>jdbc:mysql://localhost/equinox</value>
</property>
<property name="username"><value>root</value></property>
<property name="password"><value></value></property>
</bean>
```

Listing 12.59

```

<bean id="jdbcDaoImpl"
class="net.sf.acegisecurity.providers.dao.jdbc.JdbcDaoImpl">
<property name="dataSource">
<ref bean="securityDataSource"/>
</property>
</bean>
```

Listing 12.60

```
<bean id="daoAuthenticationProvider"
class="net.sf.acegisecurity.providers.dao.DaoAuthenticationProvider">
<property name="authenticationDao">
    <ref local="jdbcDaoImpl"/>
</property>
    <property name="passwordEncoder">
        <ref local="passwordEncoder"/>
    </property>
</bean>
```

Listing 12.61

```
<bean id="jdbcDaoImpl"
    class="net.sf.acegisecurity.providers.dao.jdbc.JdbcDaoImpl">
<property name="dataSource">
    <ref bean="securitydataSource"/>
</property>
    <property name="usersByUsernameQuery">
        <value>SELECT username,password,enabled as 'true' FROM users
WHERE username = ?</value>
    </property>
    <property name="authoritiesByUsernameQuery">
        <value>SELECT username,rolename FROM user_roles WHERE username =
?</value>
    </property>
<property name="passwordEncoder">
    <ref local="passwordEncoder"/>
</property>
</bean>
```

Listing 12.62

```
<bean id="roleVoter" class="net.sf.acegisecurity.vote.RoleVoter">
    <property name="rolePrefix"><value></value></property>
</bean>
```

Listing 12.63

```
# Passwords for com.tagish.auth.FileLogin
jaasuser:5a105e8b9d40e1329780d62ea2265d8a:ROLE_USER
```

Listing 12.64

```
FileLogin
{
    com.tagish.auth.FileLogin required debug=true
    pwdFile="${catalina.home}/webapps/equinox-security/WEB-INF/
users.txt";
};
```

Listing 12.65

```
package org.appfuse.web;

// organize imports using your IDE

public class JaasAuthorityGranter implements AuthorityGranter {
private static Log log =
    LoggerFactory.getLog(JaasAuthorityGranter.class);

public String grant(Principal principal) {
    String role = null;

    // check for "GROUP" in principal.toString();
    if (principal.toString().indexOf("GROUP") > -1) {
        role = principal.getName();
    }

    return role;
}
}
```

Listing 12.66

```
<bean id="jaasAuthenticationProvider"
class="net.sf.acegisecurity.providers.jaas.JaasAuthenticationProvider">
    <property name="loginConfig">
        <value>/WEB-INF/login.conf</value>
    </property>
    <property name="loginContextName">
        <value>FileLogin</value>
    </property>
    <property name="callbackHandlers">
        <list>
            <bean
class="net.sf.acegisecurity.providers.jaas.JaasNameCallbackHandler"/>
            <bean
class="net.sf.acegisecurity.providers.jaas.JaasPasswordCallbackHandler"/>
        </list>
    </property>
</bean>
```

```

        </list>
    </property>
    <property name="authorityGranters">
        <list>
            <bean class="org.appfuse.web.JaasAuthorityGranter"/>
        </list>
    </property>
</bean>

```

Listing 12.67

```

<bean id="authenticationManager"
    class="net.sf.acegisecurity.providers.ProviderManager">
    <property name="providers">
        <list>
            <!--ref local="daoAuthenticationProvider"/-->
            <ref local="jaasAuthenticationProvider"/>
        </list>
    </property>
</bean>

```

Listing 12.68

```

admin:5a105e8b9d40e1329780d62ea2265d8a:ROLE_ADMIN:ROLE_USER

```

Listing 12.69

```

<bean id="securityInterceptor"

class="net.sf.acegisecurity.intercept.method.aopalliance.MethodSecurityI
nterceptor">
    <property name="authenticationManager">
        <ref bean="authenticationManager"/>
    </property>
    <property name="accessDecisionManager">
        <ref local="accessDecisionManager"/>
    </property>
    <property name="objectDefinitionSource">
        <value>
            org.appfuse.service.UserManager.getUser=ROLE_ADMIN
            org.appfuse.service.UserManager getUsers=ROLE_USER
            org.appfuse.service.UserManager.removeUser=ROLE_ADMIN
            org.appfuse.service.UserManager.saveUser=ROLE_ADMIN
        </value>
    </property>
</bean>

```

Listing 12.70

```
<bean id="userManager" parent="txProxyTemplate">
  <property name="target">
    <bean class="org.appfuse.service.impl.UserManagerImpl">
      <property name="userDAO"><ref bean="userDAO"/></property>
    </bean>
  </property>
  <property name="preInterceptors">
    <list>
      <ref bean="securityInterceptor"/>
    </list>
  </property>
</bean>
```

Listing 12.71

```
<h1>Access Denied</h1>
<p>You don't have permission to perform this operation.</p>
```

Listing 12.72

```
<error-page>
  <error-code>403</error-code>
  <location>/accessDenied.jsp</location>
</error-page>
```

Listing 12.73

```
<bean id="authenticationManager"
  class="net.sf.acegisecurity.providers.ProviderManager">
  <property name="providers">
    <list>
      <ref local="daoAuthenticationProvider"/>
      <ref local="jaasAuthenticationProvider"/>
    </list>
  </property>
</bean>
```

Listing 12.74

```
INSERT INTO users (username, password) VALUES
('superuser','8e67bb26b358e2ed20fe552ed6fb832f397a507d');
INSERT INTO users (username, password) VALUES
('readwrite','accf881e821ed62ca842d34164426a7d9215a948');
INSERT INTO users (username, password) VALUES
('nocando','3131a0e3591641dcd157f027b92d09f99af39312');

INSERT INTO authorities (username, authority) VALUES
('superuser','ROLE_USER');
INSERT INTO authorities (username, authority) VALUES
('readwrite','ROLE_USER');
INSERT INTO authorities (username, authority) VALUES
('nocando','ROLE_USER');
```

Listing 12.75

```
<!-- ===== ACL-BASED SECURITY ===== -->

<!-- ACL permission masks used by this application -->
<bean id="SimpleAclEntry.ADMINISTRATION"
class="org.springframework.beans.factory.config.FieldRetrievingFactoryBe
an">
    <property name="staticField">

<value>net.sf.acegisecurity.acl.basic.SimpleAclEntry.ADMINISTRATION</
value>
    </property>
</bean>

<bean id="SimpleAclEntry.READ_WRITE"
class="org.springframework.beans.factory.config.FieldRetrievingFactoryBe
an">
    <property name="staticField">

<value>net.sf.acegisecurity.acl.basic.SimpleAclEntry.READ_WRITE</value>
    </property>
</bean>

<bean id="SimpleAclEntry.READ"
class="org.springframework.beans.factory.config.FieldRetrievingFactoryBe
an">
    <property name="staticField">
        <value>net.sf.acegisecurity.acl.basic.SimpleAclEntry.READ</
value>
    </property>
</bean>
```



```

<!-- An access decision voter that reads ACL_USER_ADMIN settings -->
<bean id="aclUserAdminVoter"
    class="net.sf.acegisecurity.vote.BasicAclEntryVoter">
    <property name="aclManager"><ref local="aclManager"/></property>
    <property name="processConfigAttribute">
        <value>ACL_USER_ADMIN</value>
    </property>
    <property name="processDomainObjectClass">
        <value>org.appfuse.model.User</value>
    </property>
    <property name="requirePermission">
        <list>
            <ref local="SimpleAclEntry.ADMINISTRATION"/>
        </list>
    </property>
</bean>

<!-- An access decision voter that reads ACL_USER_UPDATE settings -->
<bean id="aclUserUpdateVoter"
    class="net.sf.acegisecurity.vote.BasicAclEntryVoter">
    <property name="aclManager"><ref local="aclManager"/></property>
    <property name="processConfigAttribute">
        <value>ACL_USER_UPDATE</value>
    </property>
    <property name="processDomainObjectClass">
        <value>org.appfuse.model.User</value>
    </property>
    <property name="requirePermission">
        <list>
            <ref local="SimpleAclEntry.ADMINISTRATION"/>
            <ref local="SimpleAclEntry.READ_WRITE"/>
        </list>
    </property>
</bean>

<!-- An access decision voter that reads ACL_USER_READ settings -->
<bean id="aclUserReadVoter"
    class="net.sf.acegisecurity.vote.BasicAclEntryVoter">
    <property name="aclManager"><ref local="aclManager"/></property>
    <property name="processConfigAttribute">
        <value>ACL_USER_READ</value>
    </property>
    <property name="processDomainObjectClass">
        <value>org.appfuse.model.User</value>
    </property>
    <property name="requirePermission">
        <list>
            <ref local="SimpleAclEntry.ADMINISTRATION"/>
            <ref local="SimpleAclEntry.READ_WRITE"/>
            <ref local="SimpleAclEntry.READ"/>
        </list>
    </property>
</bean>

```

```

    </property>
</bean>

<!-- An access decision manager used by the business objects -->
<bean id="aclAccessDecisionManager"
class="net.sf.acegisecurity.vote.AffirmativeBased">
    <property name="allowIfAllAbstainDecisions"><value>false</value></
property>
    <property name="decisionVoters">
        <list>
            <ref local="roleVoter"/>
            <ref local="aclUserAdminVoter"/>
            <ref local="aclUserUpdateVoter"/>
            <ref local="aclUserReadVoter"/>
        </list>
    </property>
</bean>

<!-- ===== ACCESS CONTROL LIST MANAGER DEFINITIONS ===== -->

<bean id="aclManager"
class="net.sf.acegisecurity.acl.AclProviderManager">
    <property name="providers">
        <list>
            <ref local="basicAclProvider"/>
        </list>
    </property>
</bean>

<bean id="basicAclProvider"
class="net.sf.acegisecurity.acl.basic.BasicAclProvider">
    <property name="basicAclDao">
        <ref local="basicAclExtendedDao"/>
    </property>
</bean>

<bean id="basicAclExtendedDao"
class="net.sf.acegisecurity.acl.basic.jdbc.JdbcExtendedDaoImpl">
    <property name="dataSource">
        <ref bean="securityDataSource"/>
    </property>
</bean>

<!-- ===== "AFTER INTERCEPTION" AUTHORIZATION ===== -->

<bean id="afterInvocationManager"
class="net.sf.acegisecurity.afterinvocation.AfterInvocationProviderManag
er">
    <property name="providers">
        <list>
            <ref local="afterAclRead"/>

```

```

        <ref local="afterAclCollectionRead"/>
    </list>
</property>
</bean>

<!-- Processes AFTER_ACL_COLLECTION_READ configuration settings -->
<bean id="afterAclCollectionRead"

class="net.sf.acegisecurity.afterinvocation.BasicAclEntryAfterInvocation
CollectionFilteringProvider">
    <property name="aclManager"><ref local="aclManager"/></property>
    <property name="requirePermission">
        <list>
            <ref local="SimpleAclEntry.ADMINISTRATION"/>
            <ref local="SimpleAclEntry.READ_WRITE"/>
            <ref local="SimpleAclEntry.READ"/>
        </list>
    </property>
</bean>

<!-- Processes AFTER_ACL_READ configuration settings -->
<bean id="afterAclRead"
class="net.sf.acegisecurity.afterinvocation.BasicAclEntryAfterInvocation
Provider">
    <property name="aclManager"><ref local="aclManager"/></property>
    <property name="requirePermission">
        <list>
            <ref local="SimpleAclEntry.ADMINISTRATION"/>
            <ref local="SimpleAclEntry.READ_WRITE"/>
            <ref local="SimpleAclEntry.READ"/>
        </list>
    </property>
</bean>

```

Listing 12.76

```

use equinox;

DROP TABLE IF EXISTS acl_object_identity;
CREATE TABLE acl_object_identity (
    id int(11) NOT NULL auto_increment,
    object_identity varchar(255) NOT NULL,
    parent_object int(11) default NULL,
    acl_class varchar(255) NOT NULL,
    PRIMARY KEY (id),
    UNIQUE KEY object_identity (object_identity)
) TYPE=InnoDB;

DROP TABLE IF EXISTS acl_permission;
CREATE TABLE acl_permission (

```

```

    id int(11) NOT NULL auto_increment,
    acl_object_identity int(11) NOT NULL,
    recipient varchar(100) NOT NULL,
    mask int(11) NOT NULL,
    PRIMARY KEY (id),
    UNIQUE KEY recipient (recipient)
) TYPE=InnoDB;

INSERT INTO acl_object_identity VALUES (1, 'org.appfuse.model.User:5',
null, 'net.sf.acegisecurity.acl.basic.SimpleAclEntry');
INSERT INTO acl_object_identity VALUES (2, 'org.appfuse.model.User:6',
null, 'net.sf.acegisecurity.acl.basic.SimpleAclEntry');

INSERT INTO acl_permission VALUES (null, 1, 'superuser', 1);
INSERT INTO acl_permission VALUES (null, 1, 'readwrite', 6);
INSERT INTO acl_permission VALUES (null, 1, 'nocando', 0);

--- Mask integer 0 = no permissions
--- Mask integer 1 = administer
--- Mask integer 2 = read
--- Mask integer 6 = read and write permissions
--- Mask integer 14 = read and write and create permissions

```

Listing 12.77

```

<bean id="securityInterceptor"

class="net.sf.acegisecurity.intercept.method.aopalliance.MethodSecurityI
nterceptor">
    <property name="authenticationManager">
        <ref bean="authenticationManager"/>
    </property>
    <property name="accessDecisionManager">
        <ref local="accessDecisionManager"/>
    </property>
    <property name="afterInvocationManager">
        <ref local="afterInvocationManager"/>
    </property>
    <property name="objectDefinitionSource">
        <value>
org.appfuse.service.UserManager.getUser=ROLE_USER,AFTER_ACL_READ
org.appfuse.service.UserManager.getUsers=ROLE_USER,AFTER_ACL_COLLECTION_
READ
org.appfuse.service.UserManager.removeUser=ROLE_ADMIN
org.appfuse.service.UserManager.saveUser=ROLE_ADMIN,ROLE_READWRITE
        </value>
    </property>
</bean>

```

Listing 12.78

```
mysql> select * from acl_permission where recipient = 'superuser';
+-----+-----+-----+-----+
| id | acl_object_identity | recipient | mask |
+-----+-----+-----+-----+
| 2 | | 1 | superuser | 1 |
+-----+-----+-----+-----+
```

Listing 12.79

```
<menu:useMenuDisplayer name="CoolMenu4"
permissions="rolesAdapter">
```

Listing 12.80

```
<%@ taglib uri="http://acegisecurity.sf.net/authz" prefix="authz" %>
```

Listing 12.81

```
<authz:authorize ifAnyGranted="ROLE_READWRITE,ROLE_ADMIN">
    <button onclick="location.href='editUser.html'">Add User</button>
</authz:authorize>
```

Listing 12.82

```
INSERT INTO authorities (username, authority) VALUES
('superuser','ROLE_ADMIN');
INSERT INTO authorities (username, authority) VALUES
('readwrite','ROLE_READWRITE');
```

Listing 12.83

```
<authz:authentication operation="principal"/>
```

Listing 12.84

```
${pageContext.request.remoteUser}
```

Listing 12.85

```
<c:if test="{not empty param.id}">
  <authz:acl domainObject="{user}" hasPermission="1">
    <input type="submit" class="button" name="delete" value="Delete"/>
  </authz:acl>
</c:if>
```

Listing 12.86

```
<bean id="loggerListener"
      class="net.sf.acegisecurity.providers.dao.event.LoggerListener"/>
```

Listing 12.87

```
<logger name="net.sf.acegisecurity.intercept.event.LoggerListener">
  <level value="WARN"/>
</logger>
```

Listing 12.A

```
<web-resource-collection>
<web-resource-name>
Secure Area</web-resource-name>
<url-pattern>*.html</url-pattern>
<url-pattern>*.jsp</url-pattern>
</web-resource-collection>
```

Advanced Form Processing

Code

Listing 13.1

```
private Date birthDate;
private Integer siblings;
private Double temperature;

public Date getBirthDate() {
    return birthDate;
}

public void setBirthDate(Date birthDate) {
    this.birthDate = birthDate;
}

public Integer getSiblings() {
    return siblings;
}

public void setSiblings(Integer siblings) {
    this.siblings = siblings;
}

public Double getTemperature() {
    return temperature;
}

public void setTemperature(Double temperature) {
    this.temperature = temperature;
}
```

Listing 13.2

```
<property name="birthDate" column="birth_date"/>
<property name="siblings" column="no_siblings"/>
<property name="temperature" column="temperature"/>
```

Listing 13.3

```
<tr>
  <th><fmt:message key="user.birthDate"/>:</th>
  <td>
    <spring:bind path="user.birthDate">
      <input type="text" name="${status.expression}"
        value="${status.value}" size="11"/>
      <span class="fieldError">${status.errorMessage}</span>
    </spring:bind>
  </td>
</tr>
<tr>
  <th><fmt:message key="user.siblings"/>:</th>
  <td>
    <spring:bind path="user.siblings">
      <input type="text" name="${status.expression}"
        value="${status.value}" size="5"/>
      <span class="fieldError">${status.errorMessage}</span>
    </spring:bind>
  </td>
</tr>
<tr>
  <th><fmt:message key="user.temperature"/>:</th>
  <td>
    <spring:bind path="user.temperature">
      <input type="text" name="${status.expression}"
        value="${status.value}" size="4"/> &deg;F
      <span class="fieldError">${status.errorMessage}</span>
    </spring:bind>
  </td>
</tr>
```

Listing 13.4

```
user.birthDate=Birth Date
user.siblings=No. of Siblings
user.temperature=Body Temperature
```


Listing 13.5

```
# -- conversion errors --
typeMismatch.java.lang.Integer={0} must be an integer.
typeMismatch.java.lang.Double={0} must be a double.
typeMismatch.java.util.Date={0} must be a date.
```

Listing 13.6

```
protected void initBinder(HttpServletRequest request,
                          ServletRequestDataBinder binder) {
    // convert java.lang.Long
    NumberFormat nf = NumberFormat.getNumberInstance();
    binder.registerCustomEditor(Long.class, null,
                                new CustomNumberEditor(Long.class, nf, true));

    // convert java.util.Date
    SimpleDateFormat dateFormat =
        new SimpleDateFormat("MM/dd/yyyy");
    dateFormat.setLenient(false);
    binder.registerCustomEditor(Date.class, null,
                                new CustomDateEditor(dateFormat, true));

    // convert java.lang.Integer
    binder.registerCustomEditor(Integer.class, null,
                                new CustomNumberEditor(Integer.class, nf, true));

    // convert java.lang.Double
    binder.registerCustomEditor(Double.class, null,
                                new CustomNumberEditor(Double.class, nf, true));
}
```

Listing 13.7

```
<head>
    <title>MyUsers ~ User Details</title>
    <link href="{ctx}/styles/calendar.css" type="text/css"
          rel="stylesheet"/>
    <script type="text/javascript"
            src="{ctx}/scripts/calendar.js"></script>
    <script type="text/javascript"
            src="{ctx}/scripts/calendar-setup.js"></script>
    <script type="text/javascript"
            src="{ctx}/scripts/lang/calendar-en.js"></script>
</head>
```

Listing 13.8

```
<input type="text" name="${status.expression}"
      id="${status.expression}" value="${status.value}" size="11"/>
<button id="birthDateCal" type="button" class="button">...</button>
```

Listing 13.9

```
<script type="text/javascript">
    Calendar.setup(
    {
        inputField : "birthDate",      // id of the input field
        ifFormat   : "%m/%d/%Y",      // the date format
        button     : "birthDateCal"    // id of the button
    }
    );
</script>
```

Listing 13.10

```
private String country;
public String getCountry() {
    return country;
}

public void setCountry(String country) {
    this.country = country;
}
```

Listing 13.11

```
<property name="country" column="country"/>
```

Listing 13.12

```
protected Map referenceData(HttpServletRequest request) {
    final Locale[] available = Locale.getAvailableLocales();

    List countries = new ArrayList();

    for (int i = 0; i < available.length; i++) {
        final String value = available[i].getCountry();
        final String name =
            available[i].getDisplayCountry(request.getLocale());

        if (!"".equals(value) && !"".equals(name)) {
            NameValue country = new NameValue(name, value);
        }
    }
}
```

```

        if (!countries.contains(country)) {
            countries.add(new NameValue(name, value));
        }
    }

    Collections.sort(countries, NameValue.CASE_INSENSITIVE_ORDER);
    Map countryMap = new HashMap();
    countryMap.put("countries", countries);

    return countryMap;
}

```

Listing 13.13

```

<tr>
    <th><fmt:message key="user.country"/></th>
    <td>
        <spring:bind path="user.country">
            <select name="${status.expression}">
                <option value=""></option>
                <c:forEach items="${countries}" var="country">
                    <option value="${country.value}"
                        <c:if test="${status.value ==
country.value}">selected="selected"</c:if>
                        ${country.name}
                    </option>
                </c:forEach>
            </select>
            <span class="fieldError">${status.errorMessage}</span>
        </spring:bind>
    </td>
</tr>

```

Listing 13.14

```

user.country=Country

```

Listing 13.15

```

<input type="hidden" name="_${status.expression}"/>
<input type="checkbox" name="${status.expression}" value="true"/>

```

Listing 13.16

```
private Boolean enabled;

public Boolean getEnabled() {
    return enabled;
}

public void setEnabled(Boolean enabled) {
    this.enabled = enabled;
}
```

Listing 13.17

```
<property name="enabled" column="enabled"/>
```

Listing 13.18

```
<tr>
  <th><fmt:message key="user.enabled"/>?</th>
  <td>
    <spring:bind path="user.enabled">
      <input type="hidden" name="_${status.expression}"/>
      <input type="checkbox" name="${status.expression}" value="true"
        <c:if test="${status.value}">checked="checked"</c:if>/>
      <span class="fieldError">${status.errorMessage}</span>
    </spring:bind>
  </td>
</tr>
```

Listing 13.19

```
user.enabled=Enabled
```

Listing 13.20

```
package org.appfuse.model;

public class Address extends BaseObject {
    private Long id;
    private String city;
    private String state;
    private String zip;

    // generate getters and setters with your IDE
}
```

Listing 13.21

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE hibernate-mapping PUBLIC
    "-//Hibernate/Hibernate Mapping DTD 2.0//EN"
    "http://hibernate.sourceforge.net/hibernate-mapping-2.0.dtd">

<hibernate-mapping>
    <class name="org.appfuse.model.Address" table="address">
        <id name="id" column="id">
            <generator class="increment"/>
        </id>
        <property name="city"/>
        <property name="state"/>
        <property name="zip"/>
    </class>
</hibernate-mapping>
```

Listing 13.22

```
<property name="mappingResources">
    <list>
        <value>org/appfuse/model/Address.hbm.xml</value>
        <value>org/appfuse/model/User.hbm.xml</value>
    </list>
</property>
```

Listing 13.23

```
private Address address;

public Address getAddress() {
    return address;
}

public void setAddress(Address address) {
    this.address = address;
}
```

Listing 13.24

```
<many-to-one name="address" class="org.appfuse.model.Address"
    column="address_id" cascade="all"/>
```

Listing 13.25

```
protected Object formBackingObject(HttpServletRequest request)
throws ServletException {
    String id = RequestUtils.getStringParameter(request, "id", "");
    User command = new User();
    if (!"".equals(id)) {
        command = userManager.getUser(id);
    }

    if (command.getAddress() == null) {
        command.setAddress(new Address());
    }
    return command;
}
```

Listing 13.26

```
<tr>
  <td colspan="2">
    <fieldset>
      <legend>Address</legend>
      <table>
        <tr>
          <th style="text-align: left">City</th>
          <th style="text-align: left">State</th>
          <th style="text-align: left">Zip</th>
        </tr>
        <tr>
          <spring:nestedPath path="user.address">
            <spring:bind path="id">
              <input type="hidden"
                name="${status.expression}"
                value="${status.value}"/>
            </spring:bind>
          <td>
            <spring:bind path="city">
              <input type="text"
                name="${status.expression}"
                value="${status.value}" size="15"/>
              <span class="fieldError">
                ${status.errorMessage}</span>
            </spring:bind>
          </td>
          <td>
            <spring:bind path="state">
              <input type="text" name="${status.expression}"
                value="${status.value}" size="2"/>
              <span class="fieldError">

```

```

        ${status.errorMessage}</span>
    </spring:bind>
</td>
<td>
    <spring:bind path="zip">
        <input type="text" name="${status.expression}"
            value="${status.value}" size="10"/>
        <span class="fieldError">
            ${status.errorMessage}</span>
    </spring:bind>
    </td>
</spring:nestedPath>
</tr>
</table>
</fieldset>
</td>
</tr>

```

Listing 13.27

```

    <field property="address.city" depends="required">
        <arg0 key="City" resource="false"/>
    </field>

```

Listing 13.28

```

package org.appfuse.web;

import java.beans.PropertyEditorSupport;
import org.appfuse.model.Address;
import org.springframework.util.StringUtils;

public class AddressEditor extends PropertyEditorSupport {
    /**
     * Parse the Address from the given text.
     * Assumes a format where the last comma separates the city and
     * state and the last space separates the state and zip
     */
    public void setAsText(String text) throws IllegalArgumentException
    {
        if (!StringUtils.hasText(text)) {
            // treat empty String as null value
            setValue(null);
        } else {
            String trim = text.trim();
            int lastComma = trim.lastIndexOf(",");
            int lastSpace = trim.lastIndexOf(" ");

            if ((lastComma >= 0) && (lastSpace >= 0) &&

```

```

        (lastSpace > lastComma)) {
            Address address = new Address();
            address.setCity(trim.substring(0, lastComma));
            address.setState(trim.substring(
                lastComma + 1, lastSpace).trim());
            address.setZip(trim.substring(lastSpace + 1));
            setValue(address);
        } else {
            throw new IllegalArgumentException(
                "Could not parse address: " + text);
        }
    }
}

/**
 * Format the Address as a one-line String
 */
public String getAsText() {
    Object value = getValue();

    if ((value != null) && (value instanceof Address)) {
        Address address = (Address) value;

        if (StringUtils.hasText(address.getCity())) {
            return address.getCity() + ", " + address.getState()
                + " " + address.getZip();
        }
    }

    return "";
}
}

```

Listing 13.29

```

protected void initBinder(HttpServletRequest request,
                          ServletRequestDataBinder binder) {
    ...

    // convert org.appfuse.model.Address
    binder.registerCustomEditor(Address.class,
        new AddressEditor());
}

```


Listing 13.30

```
<tr>
  <th>Address:</th>
  <td>
    <spring:bind path="user.address">
      <input type="text" name="${status.expression}"
        value="${status.value}" size="40"/>
      <span class="fieldError">${status.errorMessage}</span>
    </spring:bind>
  </td>
</tr>
```

Listing 13.31

```
package org.appfuse.model;

public class Phone extends BaseObject {
    private Long id;
    private String type;
    private String number;

    public Phone() {}

    public Phone(String type, String number) {
        this.type = type;
        this.number = number;
    }

    // generate getters and setters with your IDE
}
```

Listing 13.32

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE hibernate-mapping PUBLIC
    "-//Hibernate/Hibernate Mapping DTD 2.0//EN"
    "http://hibernate.sourceforge.net/hibernate-mapping-2.0.dtd">

<hibernate-mapping>
  <class name="org.appfuse.model.Phone" table="phone">
    <id name="id" column="id">
      <generator class="increment"/>
    </id>
    <property name="type"/>
    <property name="number"/>
  </class>
</hibernate-mapping>
```

Listing 13.33

```
<set name="phones" cascade="all">
    <key column="user_id"/>
    <one-to-many class="org.appfuse.model.Phone"/>
</set>
```

Listing 13.34

```
<property name="mappingResources">
    <list>
        <value>org/appfuse/model/Address.hbm.xml</value>
        <value>org/appfuse/model/Phone.hbm.xml</value>
        <value>org/appfuse/model/User.hbm.xml</value>
    </list>
</property>
```

Listing 13.35

```
<%--tr>
<th><fmt:message key="user.siblings"/>:</th>
....
<tr>
    <th><fmt:message key="user.enabled"/>?</th>
    <td>
        <spring:bind path="user.enabled">
            <input type="hidden" name="_${status.expression}"/>
            <input type="checkbox" name="${status.expression}" value="true"
                <c:if test="${status.value}">checked="checked"</c:if>/>
            <span class="fieldError">${status.errorMessage}</span>
        </spring:bind>
    </td>
</tr--%>
```

Listing 13.36

```
<%--tr>
    <th>Address:</th>
    <td>
        ...
    </td>
</tr--%>
```

Listing 13.37

```
<tr>
  <td colspan="2">
    <fieldset>
      <legend>Phone Numbers</legend>
      <table>
        <tr>
          <th style="text-align: right">Type</th>
          <th style="text-align: left">Number</th>
        </tr>
        <c:forEach var="no" items="${user.phones}"
          varStatus="s">
          <spring:bind path="user.phones[${s.index}].id">
            <input type="hidden" name="${status.expression}"
              value="${status.value}"/>
          </spring:bind>
          <tr>
            <td style="text-align: right">
              <spring:bind
                path="user.phones[${s.index}].type">
                <input type="text" name="${status.expression}"
                  value="${status.value}" size="5"/>
                <span class="fieldError">
                  ${status.errorMessage}</span>
              </spring:bind>
            </td>
            <td>
              <spring:bind
                path="user.phones[${s.index}].number">
                <input type="text" name="${status.expression}"
                  value="${status.value}" size="15"/>
                <span class="fieldError">
                  ${status.errorMessage}</span>
              </spring:bind>
            </td>
          </tr>
        </c:forEach>
      </table>
    </fieldset>
  </td>
</tr>
```

Listing 13.38

```
<spring:bind path="user.phones[${s.index}].id">
```

Listing 13.39

```
<sql driver="org.hsqldb.jdbcDriver" url="jdbc:hsqldb:db/appfuse"
    userid="sa" password="">
<classpath refid="classpath"/>

INSERT INTO app_user (id, first_name, last_name)
values (5, 'Julie', 'Raible');
INSERT INTO app_user (id, first_name, last_name)
    values (6, 'Abbie', 'Raible');

INSERT INTO phone (id, type, number, user_id)
    values (1, 'home', '303-555-1212', 5);

INSERT INTO phone (id, type, number, user_id)
    values (2, 'work', '720-555-1212', 5);
</sql>
```

Listing 13.40

```
<tr>
    <td style="text-align: right">
        Add New &raquo;
        <input type="text" name="new.phone.type" value="" size="5"/>
    </td>
    <td>
        <input type="text" name="new.phone.number"
            value="" size="15"/>
    </td>
</tr>
```

Listing 13.41

```
protected void onBind(HttpServletRequest request, Object o)
throws Exception {
    User command = (User) o;

    // check if user entered a "type" on the "Add New" row
    String type = request.getParameter("new.phone.type");
    if (!"".equals(type)) {
        if (command.getPhones() == null) {
            command.setPhones(new HashSet());
        }

        String number = request.getParameter("new.phone.number");
        command.getPhones().add(new Phone(type, number));
    }
}
```

```

    }

    super.onBind(request, command);
}

```

Listing 13.42

```

<table>
  <tr>
    <th style="text-align: left">Delete?</th>
    <th style="text-align: right">Type</th>
    <th style="text-align: left">Number</th>
  </tr>
  <c:forEach var="no" items="${user.phones}" varStatus="s">
    <spring:bind path="user.phones[${s.index}].id">
      <input type="hidden" name="${status.expression}"
        value="${status.value}"/>
    </spring:bind>
    <tr>
      <td>
        <spring:bind path="user.phones[${s.index}].id">
          <input type="checkbox" name="deletePhone"
            value="${status.value}"/>
        </spring:bind>
      </td>
      <td style="text-align: right">
        <spring:bind path="user.phones[${s.index}].type">
          <input type="text" name="${status.expression}"
            value="${status.value}" size="5"/>
          <span class="fieldError">${status.errorMessage}</span>
        </spring:bind>
      </td>
      <td>
        <spring:bind path="user.phones[${s.index}].number">
          <input type="text" name="${status.expression}"
            value="${status.value}" size="15"/>
          <span class="fieldError">${status.errorMessage}</span>
        </spring:bind>
      </td>
    </tr>
  </c:forEach>
  <tr>
    <td>Add New &raquo; </td>
    <td>
      <input type="text" name="new.phone.type" value=""
        size="5"/>
    </td>
    <td>
      <input type="text" name="new.phone.number" value=""

```

```

        size="15"/>
    </td>
</tr>
</table>

```

Listing 13.43

```

// check for deleted phone numbers
if (command.getPhones() != null) {
    String[] deletePhones = request.getParameterValues("deletePhone");
    Set phones = new HashSet(command.getPhones());
    for (int i = 0; deletePhones != null &&
        i < deletePhones.length; i++) {
        String phoneId = deletePhones[i];
        for (Iterator it = phones.iterator(); it.hasNext();) {
            Phone p = (Phone) it.next();
            if (p.getId().toString().equals(phoneId)) {
                command.getPhones().remove(p);
            }
        }
    }
}

super.onBind(request, command);

```

Listing 13.44

```

<property name="pages">
    <list>
        <value>wizard/name</value>
        <value>wizard/address</value>
        <value>wizard/phone</value>
    </list>
</property>

```

Listing 13.45

```

public WizardFormController() {
    setPages(new String[] { "wizard/name", "wizard/address",
        "wizard/phone" });
}

```

Listing 13.46

```
package org.appfuse.web;

// organize imports using your IDE

public class UserWizardController extends AbstractWizardFormController {
    private static Log log =
        LogFactory.getLog(UserWizardController.class);
    private UserManager userManager;

    public UserWizardController() {
        setCommandClass(User.class);
    }

    public void setUserManager(UserManager userManager) {
        this.userManager = userManager;
    }

    protected ModelAndView processFinish(HttpServletRequest request,
                                         HttpServletResponse response,
                                         Object command,
                                         BindException errors)
        throws Exception {
        if (log.isDebugEnabled()) {
            log.debug("executing 'processFinish'...");
        }

        User user = (User) command;
        userManager.saveUser(user);
        request.getSession()
            .setAttribute("message", "User added successfully.");

        return new ModelAndView(new RedirectView("users.html"));
    }
}
```

Listing 13.47

```
<bean id="userWizardController"
      class="org.appfuse.web.UserWizardController">
    <property name="commandName" value="user"/>
    <property name="userManager" ref="userManager"/>
    <property name="pages">
        <list>
            <value>wizard/name</value>
            <value>wizard/address</value>
        </list>
    </property>
</bean>
```

```

        <value>wizard/other</value>
    </list>
</property>
</bean>

```

Listing 13.48

```

<%@ include file="/taglibs.jsp"%>

<title>Wizard | Step 1</title>

<h3>Name</h3>

<form method="post" action="wizard.html">
<input type="hidden" name="_page" value="0"/>
<table>
<tr>
    <th><fmt:message key="user.firstName"/>:</th>
    <td>
        <spring:bind path="user.firstName">
            <input type="text" name="${status.expression}"
                value="${status.value}"/>
            <span class="fieldError">${status.errorMessage}</span>
        </spring:bind>
    </td>
</tr>
<tr>
    <th><fmt:message key="user.lastName"/>:</th>
    <td>
        <spring:bind path="user.lastName">
            <input type="text" name="${status.expression}"
                value="${status.value}"/>
            <span class="fieldError">${status.errorMessage}</span>
        </spring:bind>
    </td>
</tr>
<tr>
    <td></td>
    <td>
        <input type="submit" class="button" name="_target1"
            value="Next &raquo;"/>
        <input type="submit" class="button" name="_cancel"
            value="Cancel"/>
    </td>
</tr>
</table>
</form>

```


Listing 13.49

```
protected ModelAndView processCancel(HttpServletRequest request,
                                     HttpServletResponse response,
                                     Object command,
                                     BindException errors)

throws Exception {
    return new ModelAndView(new RedirectView("users.html"));
}
```

Listing 13.50

```
<%@ include file="/taglibs.jsp"%>

<title>Wizard | Step 2</title>

<h3>Address Information</h3>

<form method="post" action="wizard.html">
<input type="hidden" name="_page" value="1"/>
<table>
    <spring:nestedPath path="user.address">
    <tr>
        <th>City:</th>
        <td>
            <spring:bind path="city">
            <input type="text" name="${status.expression}"
                value="${status.value}" size="15"/>
            <span class="fieldError">${status.errorMessage}</span>
            </spring:bind>
        </td>
    </tr>
    <tr>
        <th>State:</th>
        <td>
            <spring:bind path="state">
            <input type="text" name="${status.expression}"
                value="${status.value}" size="2"/>
            <span class="fieldError">${status.errorMessage}</span>
            </spring:bind>
        </td>
    </tr>
    <tr>
        <th>Zip:</th>
        <td>
            <spring:bind path="zip">
            <input type="text" name="${status.expression}"
                value="${status.value}" size="10"/>
            <span class="fieldError">${status.errorMessage}</span>
            </spring:bind>
        </td>
    </tr>
</table>
</form>
```

```

        </spring:bind>
    </td>
</tr>
</spring:nestedPath>
<tr>
    <td></td>
    <td>
        <input type="submit" class="button" name="_target0"
            value="&laquo; Prev"/>
        <input type="submit" class="button" name="_target2"
            value="Next &raquo;"/>
        <input type="submit" class="button" name="_cancel"
            value="Cancel"/>
    </td>
</tr>
</table>
</form>

```

Listing 13.51

```

protected Object formBackingObject(HttpServletRequest request)
throws Exception {
    User command = new User();
    command.setAddress(new Address());

    return command;
}

```

Listing 13.52

```

<%@ include file="/taglibs.jsp"%>

<title>Wizard | Step 3</title>

<h3>Other Information</h3>

<form method="post" action="wizard.html">
<input type="hidden" name="_page" value="2"/>
<table>
<tr>
    <th><fmt:message key="user.birthDate"/>:</th>
    <td>
        <spring:bind path="user.birthDate">
            <input type="text" name="${status.expression}"
                value="${status.value}" size="11"/>
            <span class="fieldError">${status.errorMessage}</span>
        </spring:bind>
    </td>
</tr>
</table>

```

```

<tr>
  <td></td>
  <td>
    <input type="submit" class="button" name="_target1"
      value="&laquo; Prev"/>
    <input type="submit" class="button" name=" finish"
      value="Finish"/>
    <input type="submit" class="button" name="_cancel"
      value="Cancel"/>
  </td>
</tr>
</table>
</form>

```

Listing 13.53

```

protected void initBinder(HttpServletRequest request,
                          ServletRequestDataBinder binder) {
    // convert java.util.Date
    SimpleDateFormat dateFormat = new
        SimpleDateFormat("MM/dd/yyyy");
    dateFormat.setLenient(false);
    binder.registerCustomEditor(Date.class,
        new CustomDateEditor(dateFormat, true));
}

```

Listing 13.54

```

<field property="lastName" depends="required" page="1">

```

Listing 13.55

```

<input type="hidden" name="page" value="0"/>

```

Listing 13.56

```

<action-state id="search">
  <action bean="userListAction"/>
  <transition on="success" to="display.userList"/>
</action-state>

<view-state id="display.userList" view="pages/userList">
  <transition on="*" to="display.userForm">
    <action bean="userFormAction"/>
  </transition>
</view-state>

```

```

        <view-state id="display.user" view="pages/userForm">
            <transition on="submit" to="saveUser.action">
                <action bean="userFormAction"/>
            </transition>
            <transition on="cancel" to="showList.action"/>
        </view-state>

```

Listing 13.57

```

<input type="hidden" name="_flowExecutionId"
        value="${flowExecutionId}"/>
<input type="hidden" name="_eventId" value="submit"/>

```

Listing 13.58

```

package org.appfuse.web.flow;

// organize imports with your IDE

public class UserFormAction extends FormAction {
    private UserManager userManager;

    public void setUserManager(UserManager mgr) {
        this.userManager = mgr;
    }

    public UserFormAction() {
        setFormObjectName("user");
        setFormObjectClass(User.class);
        setFormObjectScope(ScopeType.FLOW);
    }

    public Object createFormObject(RequestContext context)
        throws InstantiationException, IllegalAccessException {
        User user = (User) super.createFormObject(context);
        user.setAddress(new Address());

        return user;
    }
}

```

Listing 13.59

```

<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE webflow PUBLIC "-//SPRING//DTD WEBFLOW//EN"
        "http://www.springframework.org/dtd/spring-webflow.dtd">

<webflow id="userFlow" start-state="setUpForm">

```

```

<action-state id="setupForm">
    <action bean="userFormAction"/>
    <transition on="success" to="display.nameForm"/>
</action-state>

<view-state id="display.nameForm" view="flow/name">
    <transition on="submit" to="display.addressForm">
        <action bean="userFormAction" method="bindAndValidate"/>
    </transition>
    <transition on="cancel" to="finish"/>
</view-state>

<view-state id="display.addressForm" view="flow/address">
    <transition on="previous" to="display.nameForm">
        <action bean="userFormAction" method="bindAndValidate"/>
    </transition>
    <transition on="submit" to="display.otherForm">
        <action bean="userFormAction" method="bindAndValidate"/>
    </transition>
    <transition on="cancel" to="finish"/>
</view-state>

<view-state id="display.otherForm" view="flow/other">
    <transition on="previous" to="display.addressForm"/>
    <transition on="submit" to="save">
        <action bean="userFormAction" method="bindAndValidate"/>
    </transition>
    <transition on="cancel" to="finish"/>
</view-state>

<action-state id="save">
    <action bean="userFormAction"/>
    <transition on="success" to="finish"/>
</action-state>

<end-state id="finish" view="redirect:/users.html"/>
</webflow>

```

Listing 13.60

```

<bean id="userFlow"
    class="org.springframework.web.flow.config.XmlFlowFactoryBean">
    <property name="location" value="/WEB-INF/flow-user.xml"/>
</bean>

```

Listing 13.61

```
<bean id="flowController"
      class="org.springframework.web.flow.mvc.FlowController">
  <property name="flowExecutionManager" ref="flowExecutionManager"/>
</bean>
```

Listing 13.62

```
<bean id="flowExecutionManager"
      class="org.springframework.web.flow.execution.servlet.ServletFlowExecutionManager">
  <property name="storage">
    <bean
      class="org.springframework.web.flow.execution.servlet.HttpSessionContinuationFlowExecutionStorage"/>
  </property>
</bean>
```

Listing 13.63

```
<prop key="/flow.html">flowController</prop>
```

Listing 13.64

```
<bean id="userFormAction"
      class="org.appfuse.web.flow.UserFormAction" autowire="byName"/>
```

Listing 13.65: *web/flow/name.jsp*

```
<%@ include file="/taglibs.jsp"%>

<title>Flow | Step 1</title>

<h3>Name</h3>

<form method="post" action="user.html">
  <input type="hidden" name="_flowExecutionId"
    value="${flowExecutionId}"/>
  <input type="hidden" name="_eventId" value="submit"/>
  <table>
  <tr>
    <th><fmt:message key="user.firstName"/>:</th>
    <td>
      <spring:bind path="user.firstName">
        <input type="text" name="${status.expression}"
```

```

        value="${status.value}"/>
        <span class="fieldError">${status.errorMessage}</span>
    </spring:bind>
</td>
</tr>
<tr>
    <th><fmt:message key="user.lastName"/>:</th>
    <td>
        <spring:bind path="user.lastName">
            <input type="text" name="${status.expression}"
                value="${status.value}"/>
            <span class="fieldError">${status.errorMessage}</span>
        </spring:bind>
    </td>
</tr>
</tr>
<tr>
    <td></td>
    <td>
        <input type="submit" class="button" value="Next &raquo;"/>
        <input type="submit" class="button" value="Cancel "
            onclick="this.form. eventId.value='cancel'"/>
    </td>
</tr>
</table>
</form>

```

Listing 13.66: *web/flow/address.jsp*

```

<%@ include file="/taglibs.jsp"%>

<title>Flow | Step 2</title>

<h3>Address Information</h3>

<form method="post" action="user.html">
    <input type="hidden" name="_flowExecutionId"
        value="${flowExecutionId}"/>
    <input type="hidden" name="_eventId" value="submit"/>
    <table>
        <spring:nestedPath path="user.address">
            <tr>
                <th>City:</th>
                <td>
                    <spring:bind path="city">
                        <input type="text" name="${status.expression}"
                            value="${status.value}" size="15"/>
                        <span class="fieldError">${status.errorMessage}</span>
                    </spring:bind>
                </td>
            </tr>
        </spring:nestedPath>
    </table>
</form>

```

```
|  |  |
| --- | --- |
| State: | <spring:bind path="state">         <input type="text" name="${status.expression}"             value="${status.value}" size="2"/>         <span class="fieldError">${status.errorMessage}</span>         </spring:bind>       </td> |
| Zip: | <spring:bind path="zip">         <input type="text" name="${status.expression}"             value="${status.value}" size="10"/>         <span class="fieldError">${status.errorMessage}</span>         </spring:bind>       </td> |
| </spring:nestedPath> | |
| <td></td> | <input type="submit" class="button" value="&laquo; Prev"             onclick="this.form. eventId.value='previous'"/>         <input type="submit" class="button" value="Next &raquo;" />         <input type="submit" class="button" value="Cancel"             onclick="this.form. eventId.value='cancel'"/>       </td> |
| </tr> | |

</table>
</form>

```

Listing 13.67: *web/flow/other.jsp*

```

<%@ include file="/taglibs.jsp"%>

<title>Flow | Step 3</title>

<h3>Other Information</h3>

<form method="post" action="user.html">
<input type="hidden" name="_flowExecutionId"
    value="${flowExecutionId}"/>
<input type="hidden" name="_eventId" value="submit"/>
<table>
<tr>
    <th><fmt:message key="user.birthDate"/>:</th>
    <td>

```



```

        <spring:bind path="user.birthDate">
        <input type="text" name="{status.expression}"
            value="{status.value}" size="11"/>
        <span class="fieldError">{status.errorMessage}</span>
        </spring:bind>
    </td>
</tr>
<tr>
    <td></td>
    <td>
        <input type="submit" class="button" value="⌂; Prev"
            onclick="this.form.eventId.value='previous'"/>
        <input type="submit" class="button" value="Finish"/>
        <input type="submit" class="button" value="Cancel"
            onclick="this.form.eventId.value='cancel'"/>
    </td>
</tr>
</table>
</form>

```

Listing 13.68

```

protected void initBinder(RequestContext context,
    DataBinder binder) {
    // convert java.util.Date
    SimpleDateFormat dateFormat =
        new SimpleDateFormat("MM/dd/yyyy");
    dateFormat.setLenient(false);
    binder.registerCustomEditor(Date.class,
        new CustomDateEditor(dateFormat, true));
}

```

Listing 13.69

```

public Event save(RequestContext context) throws Exception {
    User user = (User) context.getFlowScope().getAttribute("user");
    userManager.saveUser(user);
    return success();
}

```

Listing 13.70

```

<action-state id="save">
    <action bean="userFormAction"/>
    <transition on="success" to="finish"/>
</action-state>

```

Listing 13.71

```
<bean id="flowController"
      class="org.springframework.web.flow.mvc.FlowController"/>

<bean id="userFlow"
      class="org.springframework.web.flow.config.XmlFlowFactoryBean">
  <property name="location" value="/WEB-INF/flow-user.xml"/>
</bean>
```

Examples and References

Code

Listing A.1

```
<listener>
  <listener-class>net.sourceforge.myfaces.webapp.StartupServletContext
    Listener</listener-class>
</listener>
```

Listing A.2

```
<servlet>
  <servlet-name>faces</servlet-name>
  <servlet-class>javax.faces.webapp.FacesServlet</servlet-class>
  <load-on-startup>1</load-on-startup>
</servlet>
```

Listing A.3

```
<servlet-mapping>
  <servlet-name>faces</servlet-name>
  <url-pattern>*.html</url-pattern>
</servlet-mapping>
```

Listing A.4

```
<context-param>
  <param-name>
    javax.servlet.jsp.jstl.fmt.localizationContext
  </param-name>
  <param-value>messages</param-value>
</context-param>
```

Listing A.5

```
<%@ page language="java" errorPage="/error.jsp" %>
<%@ taglib uri="http://java.sun.com/jstl/core_rt" prefix="c" %>
<%@ taglib uri="http://java.sun.com/jstl/fmt_rt" prefix="fmt" %>
<%@ taglib uri="http://java.sun.com/jsf/html" prefix="h" %>
<%@ taglib uri="http://java.sun.com/jsf/core" prefix="f" %>
<%@ taglib uri="http://corejsf.com/validator" prefix="v" %>
<%@ taglib uri="http://displaytag.sf.net" prefix="display" %>
<%@ taglib uri="http://www.opensymphony.com/sitemesh/decorator"
    prefix="decorator" %>
<%@ taglib uri="http://myfaces.sourceforge.net/tld/myfaces_ext_0_9.tld"
    prefix="x"%>

<c:set var="datePattern"><fmt:message key="date.format"/></c:set>
```

Listing A.6

```
date.format=MM/dd/yyyy
```

Listing A.7

```
<f:view>
    <h1><h:outputText value="Hello JSF!"/></h1>
</f:view>
```

Listing A.8

```
package org.appfuse.web;

// organize imports with your IDE

public class BasePageTestCase extends TestCase {
    protected final Log log = LoggerFactory.getLog(getClass());
    protected static FacesContext facesContext;
    protected static MockServletConfig config;
    protected static MockServletContext servletContext;
    protected static WebApplicationContext ctx;

    // This static block ensures that Spring's BeanFactory and JSF's
    // FacesContext is only loaded once for all tests.
    static {
        servletContext = new MockServletContext("");
        servletContext.addInitParameter(
            ContextLoader.CONFIG_LOCATION_PARAM,
            "/WEB-INF/applicationContext*.xml");

        ServletContextListener contextListener =
```

```

        new ContextLoaderListener();
        ServletContextEvent event =
            new ServletContextEvent(servletContext);
        contextListener.contextInitialized(event);

        ctx = WebApplicationContextUtils
            .getRequiredWebApplicationContext(servletContext);
        config = new MockServletConfig(servletContext);
        facesContext = performFacesContextConfig();
    }

    protected static FacesContext performFacesContextConfig() {
        StartupServletContextListener facesListener =
            new StartupServletContextListener();
        ServletContextEvent event =
            new ServletContextEvent(servletContext);
        facesListener.contextInitialized(event);

        LifecycleFactory lifecycleFactory =
            (LifecycleFactory) FactoryFinder.getFactory(

                FactoryFinder.LIFECYCLE_FACTORY);

        Lifecycle lifecycle =
            lifecycleFactory.getLifecycle(getLifecycleId());

        FacesContextFactory facesCtxFactory =
            (FacesContextFactory) FactoryFinder.getFactory(
                FactoryFinder.FACES_CONTEXT_FACTORY);

        FacesContext ctx =
            facesCtxFactory.getFacesContext(servletContext,
                                            new MockHttpServletRequest(),
                                            new MockHttpServletResponse(),
                                            lifecycle);

        return ctx;
    }

    protected static String getLifecycleId() {
        String lifecycleId =
            servletContext.getInitParameter(
                FacesServlet.LIFECYCLE_ID_ATTR);

        return (lifecycleId != null) ?
            lifecycleId : LifecycleFactory.DEFAULT_LIFECYCLE;
    }

    /**
     * Get managed bean based on the bean name.
     */

```

```

    * @param beanName the bean name
    * @return the managed bean associated with the bean name
    */
    public Object getManagedBean(String beanName) {
        return getValueBinding(getJsfEl(beanName))
            .getValue(facesContext);
    }

    private Application getApplication() {
        ApplicationFactory appFactory =
            (ApplicationFactory) FactoryFinder.getFactory(
                FactoryFinder.APPLICATION_FACTORY);

        return appFactory.getApplication();
    }

    private ValueBinding getValueBinding(String el) {
        return getApplication().createValueBinding(el);
    }

    private String getJsfEl(String value) {
        return "#{ " + value + " }";
    }
}

```

Listing A.9

```

package org.appfuse.web;

// organize imports with your IDE

public class UserListTest extends BasePageTestCase {
    private UserList bean;

    protected void setUp() throws Exception {
        super.setUp();
        bean = (UserList) getManagedBean("userList");

        // create a new user
        User user = new User();
        user.setFirstName("Julie");
        user.setLastName("Raible");
        user.setBirthday(new Date());

        // persist to database
        UserManager userManager =
            (UserManager) ctx.getBean("userManager");
        userManager.saveUser(user);
    }
}

```

```

    public void testSearch() throws Exception {
        assertTrue(bean.getUsers().size() >= 1);
    }
}

```

Listing A.10

```

package org.appfuse.web;

// organize imports with your IDE

public class UserList {
    private UserManager mgr = null;

    public void setUserManager(UserManager userManager) {
        this.mgr = userManager;
    }

    public List getUsers() {
        return mgr.getUsers();
    }
}

```

Listing A.11

```

<%@ include file="/taglibs.jsp"%>

<f:view>
<f:loadBundle var="messages" basename="messages"/>
<title><fmt:message key="userList.title"/></title>

<h:form id="editUser">
<h:commandButton value="Add User" action="add" immediate="true"
styleClass=
    "button"/>

<h:dataTable value="#{userList.users}" var="user" styleClass="list"
    rowClasses="odd,even" id="userList">
    <h:column>
        <f:facet name="header">
            <h:outputText value="#{messages['user.id']}" />
        </f:facet>
        <h:commandLink action="#{userForm.edit}" value="#{user.id}">
            <f:param name="id" value="#{user.id}" />
        </h:commandLink>
    </h:column>
    <h:column>
        <f:facet name="header">
            <h:outputText value="#{messages['user.firstName']}" />

```

```

        </f:facet>
        <h:outputText value="#{user.firstName}"/>
    </h:column>
    <h:column>
        <f:facet name="header">
            <h:outputText value="#{messages['user.lastName']}"/>
        </f:facet>
        <h:outputText value="#{user.lastName}"/>
    </h:column>
    <h:column>
        <f:facet name="header">
            <h:outputText value="#{messages['user.birthday']}"/>
        </f:facet>
        <h:outputText value="#{user.birthday}">
            <f:convertDateTime pattern="#{messages['date.format']}"/>
        </h:outputText>
    </h:column>
</h:dataTable>

</h:form>
</f:view>

```

Listing A.12

```

</application> element.
<navigation-rule>
    <from-view-id>/userList.jsp</from-view-id>
    <navigation-case>
        <from-outcome>add</from-outcome>
        <to-view-id>/userForm.jsp</to-view-id>
    </navigation-case>
</navigation-rule>

```

Listing A.13

```

<h:commandLink action="#{userForm.edit}" value="#{user.id}">
    <f:param name="id" value="#{user.id}"/>
</h:commandLink>

```

Listing A.14

```

<!-- Step 1: Use a hidden dataTable to pull userList into request -->
<h:dataTable var="user" value="#{userList.users}" style="display:none"/>

<display:table name="userList.users" class="list" requestURI="/
userList.html"
    id="userList" export="true">
    <display:column sort="true" titleKey="user.id" media="html">

```



```

        <a href="javascript:viewUser('${userList.id}')">
            ${userList.id}</a>
    </display:column>
    <display:column property="id" media="csv excel xml pdf"
        titleKey="user.id"/>
    <display:column property="firstName" sort="true"
        titleKey="user.firstName"/>
    <display:column property="lastName" sort="true"
        titleKey="user.lastName"/>
    <display:column sort="true" titleKey="user.birthday" media="html">
        <!--<fmt:formatDate value="${userList.birthday}"
            pattern="yyyyMMdd"/>-->
        <fmt:formatDate value="${userList.birthday}"
            pattern="${datePattern}"/>
    </display:column>
    <display:column sort="true" titleKey="user.birthday"
        media="csv xml excel pdf">
        <fmt:formatDate value="${userList.birthday}"
            pattern="${datePattern}"/>
    </display:column>
</display:table>

<%-- JSF Hack for the Display Tag, from James Violette ~
    http://www.emeraldjb.com/roller/page/jviolett --%>

<%-- Step 2. Create a dummy actionLink, w/ no value --%>
<h:commandLink action="#{userForm.edit}" id="editUserLink">
    <f:param name="id"/>
</h:commandLink>

<%-- Step 3. Write a JavaScript function that's easy to call --%>
<script type="text/javascript">
function viewUser(id) {
    var f = document.forms['editUser'];
    f.elements['editUser:_link_hidden_'].value='editUser:editUserLink';
    f.elements['id'].value=id;
    f.submit();
}
highlightTableRows("userList");
</script>

```

Listing A.15

```

package org.appfuse.web;

// organize imports using your IDE

public class UserFormTest extends BasePageTestCase {
    private UserForm bean;
    private String userId;

```

```

protected void setUp() throws Exception {
    super.setUp();
    bean = (UserForm) getManagedBean("userForm");
    // verify that "userForm" has been defined as a managed-bean
    assertNotNull("define 'userForm' in faces-config.xml", bean);

    // create a new user
    User user = new User();
    user.setFirstName("Matt");
    user.setLastName("Raible");

    // persist to database
    UserManager userManager =
        (UserManager) ctx.getBean("userManager");
    userManager.saveUser(user);
    userId = user.getId().toString();
}

protected void tearDown() throws Exception {
    super.tearDown();
    bean = null;
}

public void testEdit() throws Exception {
    bean.setId(userId);
    assertEquals(bean.edit(), "success");
    assertNotNull(bean.getUser().getFirstName());
}

public void testSave() throws Exception {
    bean.setId(userId);
    bean.edit();
    assertNotNull(bean.getUser());
    bean.getUser().setFirstName("Jack");

    assertEquals(bean.save(), "success");
    assertNotNull(bean.getUser());
}

public void testRemove() throws Exception {
    User user = new User();
    user.setId(new Long(userId));
    bean.setUser(user);
    assertEquals(bean.delete(), "success");
}
}

```

Listing A.16

```
<managed-bean>
  <managed-bean-name>userForm</managed-bean-name>
  <managed-bean-class>org.appfuse.web.UserForm</managed-bean-class>
  <managed-bean-scope>request</managed-bean-scope>
  <managed-property>
    <property-name>id</property-name>
    <value>#{param.id}</value>
  </managed-property>
  <managed-property>
    <property-name>userManager</property-name>
    <value>#{userManager}</value>
  </managed-property>
</managed-bean>
```

Listing A.17

```
package org.appfuse.web;

// user your IDE to organize imports

public class UserForm {
    private final Log log = LogFactory.getLog(UserForm.class);
    private String id;
    public User user = new User();
    public UserManager mgr;

    public String getId() {
        return id;
    }

    public void setId(String id) {
        this.id = id;
    }

    public User getUser() {
        return user;
    }

    public void setUser(User user) {
        this.user = user;
    }

    public void setUserManager(UserManager userManager) {
        this.mgr = userManager;
    }

    public String edit() {
```

```

        if (id != null) {
            // assuming edit
            setUser(mgr.getUser(id));
        }

        return "success";
    }

    public String save() {
        mgr.saveUser(getUser());
        addMessage("user.saved", getUser().getFullName());

        return "success";
    }

    public String delete() {
        mgr.removeUser(getUser().getId().toString());
        addMessage("user.deleted", getUser().getFullName());

        return "success";
    }

    // Convenience methods =====
    public static String getRequestParam(String name) {
        return
            (String) FacesContext.getCurrentInstance()
                .getExternalContext()
                .getRequestParameterMap().get(name);
    }

    public void addMessage(String key, String arg) {
        // sure is a lot of work to get the named ResourceBundle
        // in JSF, eh?
        ApplicationFactory factory =
            (ApplicationFactory) FactoryFinder.
                getFactory(FactoryFinder.APPLICATION_FACTORY);
        String bundleName =
            factory.getApplication().getMessageBundle();
        ResourceBundle messages = ResourceBundle.getBundle(bundleName);

        // it's even more work to format a message with args
        MessageFormat form =
            new MessageFormat(messages.getString(key));

        String msg = form.format(new Object[] { arg });

        // add message to session so it can live past redirects
        // the MessageFilter class will take care of removing it
        HttpSession session =
            (HttpSession) FacesContext.getCurrentInstance()

```

```

        .getExternalContext()
        .getSession(true);
    session.setAttribute("message", msg);
}
}

```

Listing A.18

```

<%@ include file="/taglibs.jsp"%>

<f:view>
<f:loadBundle var="messages" basename="messages"/>

<head>
    <title><fmt:message key="userForm.title"/></title>
    <!-- JSCalendar -->
    <c:set var="ctx" scope="request"
        value="{pageContext.request.contextPath}"/>
    <script type="text/javascript"
        src="{ctx}/scripts/popcalendar.js"></script>
    <link rel="stylesheet" type="text/css"
        href="{ctx}/scripts/jscalendar-WH/theme.css"/>
</head>

<h:messages errorClass="error" layout="table" style="width: 100%"/>

<p>Please fill in user's information below:</p>

<h:form id="userForm">
<h:inputHidden value="#{userForm.user.id}"
    <f:convertNumber/>
</h:inputHidden>

<table class="detail">
<tr>
    <th>
        <h:outputLabel for="firstName"
            value="#{messages['user.firstName']}"/>:
    </th>
    <td>
        <h:inputText value="#{userForm.user.firstName}"
            id="firstName"/>
    </td>
</tr>
<tr>
    <th>
        <h:outputLabel for="lastName"
            value="#{messages['user.lastName']}"/>:
    </th>
    <td><h:inputText value="#{userForm.user.lastName}" id="lastName"/>

```

```

        <h:message for="lastName" styleClass="errorMessage"/>
    </td>
</tr>
<tr>
    <th>
        <h:outputLabel for="birthday"
            value="#{messages['user.birthday']}"/>:
    </th>
    <td><x:inputCalendar monthYearRowClass="yearMonthHeader"
        weekRowClass="weekHeader" id="birthday"
        currentDayCellClass="currentDayCell"
        value="#{userForm.user.birthday}"
        renderAsPopup="true"/>
        <h:message for="birthday" styleClass="errorMessage"/>
    </td>
</tr>
<tr>
    <td></td>
    <td>
        <h:commandButton value="Save" action="#{userForm.save}"
            id="save" styleClass="button"/>
        <c:if test="${not empty userForm.id}">
            <h:commandButton value="Delete" action="#{userForm.delete}"
                id="delete" styleClass="button"/>
        </c:if>
        <h:commandButton value="Cancel" action="cancel"
            immediate="true" id="cancel" styleClass="button"/>
    </td>
</tr>
</table>
</h:form>

<script type="text/javascript">
    document.forms["userForm"].elements["userForm:firstName"].focus();
</script>

</f:view>

```

Listing A.19

```

<navigation-rule>
    <from-view-id>/userList.jsp</from-view-id>
    <navigation-case>
        <from-outcome>add</from-outcome>
        <to-view-id>/userForm.jsp</to-view-id>
    </navigation-case>
    <navigation-case>
        <from-outcome>success</from-outcome>
        <to-view-id>/userForm.jsp</to-view-id>
    </navigation-case>
</navigation-rule>

```

```

<navigation-rule>
  <from-view-id>/userForm.jsp</from-view-id>
  <navigation-case>
    <from-outcome>cancel</from-outcome>
    <to-view-id>/userList.jsp</to-view-id>
  </navigation-case>
  <navigation-case>
    <from-outcome>success</from-outcome>
    <to-view-id>/userList.jsp</to-view-id>
  </navigation-case>
</navigation-rule>

```

Listing A.20

```

<%-- Success Messages --%>
<c:if test="${not empty message}">
  <div class="message">${message}</div>
</c:if>

```

Listing A.21

```

<h:inputText value="#{userForm.user.lastName}" id="lastName"
required="true"/>

```

Listing A.22

```

javax.faces.component.UIInput.REQUIRED=This is a required field.

```

Listing A.23

```

<h:form id="userForm" onsubmit="return validateUserForm(this)">

```

Listing A.24

```

<h:inputText value="#{userForm.user.lastName}" id="lastName"
  required="true">
  <v:commonsValidator type="required"
    arg="#{messages['user.lastName']}" />
</h:inputText>

```

Listing A.25

```

<h:commandButton value="Cancel" action="cancel" immediate="true"
id="cancel"
  styleClass="button" onclick="bCancel=true"/>

```

Listing A.26

```
<v:validatorScript functionName="validateUserForm"/>
```

Listing A.27

```
<config host="${host}" port="${port}"
  protocol="http" basepath="${webapp.name}"
  resultpath="${test.dir}/data" resultfile="web-tests-result.xml"
  summary="true" saveresponse="true"/>
```

Listing A.28

```
<!DOCTYPE project [
  <!ENTITY config SYSTEM "config.xml">
]>
<project basedir="." default="run-all-tests">
  <!-- Include messages.properties so we can test against
       keys, rather than values -->
  <property file="web/WEB-INF/classes/messages.properties"/>
  <property name="host" value="localhost"/>
  <property name="port" value="8080"/>

  <!-- runs all targets -->
  <target name="run-all-tests" depends="UserTests"
    description="Call and executes all test cases (targets)">
    <echo>All UserTests passed!</echo>
  </target>

  <!-- Verify adding a user, viewing and deleting a user works -->
  <target name="UserTests" description="Tests User CRUD">
    <canoo name="userTests">
      &config;
      <steps>
        <!-- View add screen -->
        <invoke url="/userList.html"/>
        <clickbutton label="Add User"/>

        <verifytitle text="${userForm.title}"/>

        <!-- Enter data and save -->
        <setinputfield name="userForm:firstName" value="Test"/>
        <setinputfield name="userForm:lastName" value="Name"/>
        <clickbutton label="Save"/>

        <!-- View user list -->
        <verifytitle text="${userList.title}"/>
      </steps>
    </canoo>
  </target>
</project>
```



```

        <!-- Delete first user in table -->
        <invoke url="/userList.html"/>

        <clicklink href="viewUser"/>
        <verifytitle text="\${userForm.title}"/>
        <clickbutton label="Delete"/>
        <verifytitle text="\${userList.title}"/>
    </steps>
</canoo>
</target>
</project>

```

Listing A.29

```

<filter>
    <filter-name>redirect</filter-name>
    <filter-class>org.apache.tapestry.RedirectFilter</filter-class>
</filter>

```

Listing A.30

```

<filter-mapping>
    <filter-name>redirect</filter-name>
    <url-pattern>/</url-pattern>
</filter-mapping>

```

Listing A.31

```

<servlet>
    <servlet-name>tapestry</servlet-name>
    <servlet-class>org.apache.tapestry.ApplicationServlet</servlet-
class>
    <load-on-startup>1</load-on-startup>
</servlet>

```

Listing A.32

```

<servlet-mapping>
    <servlet-name>tapestry</servlet-name>
    <url-pattern>/app</url-pattern>
</servlet-mapping>

```

Listing A.33

```
<application name="tapestry" engine-class="org.appfuse.web.PageEngine">
  <page name="Home" specification-path="/pages/home.page"/>
    <library id="contrib" specification-path="/org/apache/tapestry/
contrib
    /Contrib.library"/>
  </application>
```

Listing A.34

```
<context-param>
  <param-name>javax.servlet.jsp.jstl.fmt.localizationContext</param-
name>
  <param-value>messages</param-value>
</context-param>
```

Listing A.35

```
package org.appfuse.web;

// organize imports using your IDE

public class BasePageTestCase extends TapestryTestCase {
    protected final transient Log log = LogFactory.getLog(getClass());
    protected static WebApplicationContext ctx;
    protected MockEngine engine;
    protected MockHttpServletRequest request =
        new MockHttpServletRequest();
    protected MockHttpServletResponse response =
        new MockHttpServletResponse();

    // This static block ensures that Spring's BeanFactory is only
    // loaded once for all tests
    static {
        MockServletContext servletContext = new MockServletContext("");
        servletContext
            .addInitParameter(ContextLoader.CONFIG_LOCATION_PARAM,
                "/WEB-INF/applicationContext*.xml");

        ServletContextListener contextListener =
            new ContextLoaderListener();
        ServletContextEvent event =
            new ServletContextEvent(servletContext);
        contextListener.contextInitialized(event);

        ctx = WebApplicationContextUtils
            .getRequiredWebApplicationContext(servletContext);
    }
}
```

```

    }

    protected IPage getPage(Class clazz) throws Exception {
        AbstractInstantiator i = new AbstractInstantiator();
        BasePage result = (BasePage) i.getInstance(clazz);

        // used to indicate executing in a test, so messages lookup
        // will be friendly to tests
        result.setEngine(new MockEngine());
        result.setRequestCycle(getCycle(request, response));

        return result;
    }

    protected IRequestCycle getCycle(HttpServletRequest request,
                                     HttpServletResponse response)
    throws Exception {
        MockContext servletContext = new MockContext();
        MockServletConfig config =
            new MockServletConfig("servlet", servletContext);
        ApplicationServlet servlet = new ApplicationServlet();

        servlet.init(config);

        engine = new MockEngine();

        RequestContext context =
            new RequestContext(servlet, request, response);

        return new org.appfuse.web.MockRequestCycle(engine, context);
    }
}

```

Listing A.36

```

package org.appfuse.web;

// organize imports using your IDE

public class UserListTest extends BasePageTestCase {
    private UserList page;

    protected void setUp() throws Exception {
        super.setUp();
        page = (UserList) getPage(UserList.class);

        // this manager can be mocked if you want a "pure" unit test
        page.setUserManager((UserManager) ctx.getBean("userManager"));

        // default request cycle
    }
}

```

```

        page.setRequestCycle(getCycle(request, response));
    }

    protected void tearDown() throws Exception {
        super.tearDown();
        page = null;
    }

    public void testSearch() throws Exception {
        assertTrue(page.getUserManager().getUsers().size() >= 1);
    }
}

```

Listing A.37

```

package org.appfuse.web;

// organize imports using your IDE

public abstract class UserList extends BasePage {
    private final Log log = LoggerFactory.getLog(UserList.class);
    private String message;

    public abstract UserManager getUserManager();
    public abstract void setUserManager(UserManager manager);

    public void setMessage(String message) {
        this.message = message;
    }

    public String getMessage() {
        return this.message;
    }
}

```

Listing A.38

```

<title><span key="userList.title"/></title>

<span jwcid="@Conditional" condition="ognl:message != null">
    <div class="message">
        <span jwcid="@Insert" value="ognl:message" raw="true"/>
    </div>
</span>

<form jwcid="@Form">
    <input jwcid="@submit" value="Add User"/>
</form>

```

```

<table jwcid="table@contrib:Table" class="list" id="userList"
rowsClass="ognl:beans.evenOdd.next" row="ognl:row"
source="ognl:userManager.users"
    columns="id, firstName, lastName, birthday"
arrowUpAsset="ognl:assets.upArrow"
arrowDownAsset="ognl:assets.downArrow">
</table>

<script type="text/javascript">highlightTableRows("userList");</script>

```

Listing A.39

```

#-- messages for users.html page --
userList.title=Equinox ~ User List

id=Id
firstName=First Name
lastName=Last Name
birthday=Birthday
date.format=MM/dd/yyyy

```

Listing A.40

```

<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE page-specification
PUBLIC "-//Apache Software Foundation//Tapestry Specification 3.0/
/EN"
    "http://jakarta.apache.org/tapestry/dtd/Tapestry_3_0.dtd">

    <page-specification class="org.appfuse.web.UserList">
    <property-specification name="userManager"
        type="org.appfuse.service.UserManager">
        global.appContext.getBean("userManager")
    </property-specification>

    <property-specification name="message" type="java.lang.String"/>
    <property-specification name="row"/>

    <bean name="evenOdd" class="org.apache.tapestry.bean.EvenOdd"/>
    <context-asset name="upArrow" path="/images/arrow_up.png"/>
    <context-asset name="downArrow" path="/images/arrow_down.png"/>
</page-specification>

```

Listing A.41

```
<application name="tapestry" engine-class="org.appfuse.web.BaseEngine">
  <page name="Home" specification-path="/pages/home.page"/>
  <page name="users" specification-path="/pages/users.page"/>

  <library id="contrib" specification-
    path="/org/apache/tapestry/contrib/Contrib.library"/>
</application>
```

Listing A.42

```
-Dorg.apache.tapestry.disable-caching=true
```

Listing A.43

```
package org.appfuse.web;

// organize imports with your IDE

public class UserFormTest extends BasePageTestCase {
    private UserForm page;
    private Long userId;

    protected void setUp() throws Exception {
        super.setUp();

        page = (UserForm) getPage(UserForm.class);
        page.setUserManager((UserManager) ctx.getBean("userManager"));

        // create a new user
        User user = new User();
        user.setFirstName("Jack");
        user.setLastName("Raible");

        // persist to database
        UserManager userManager =
            (UserManager) ctx.getBean("userManager");
        userManager.saveUser(user);
        userId = user.getId();
    }

    protected void tearDown() throws Exception {
        super.tearDown();
        page = null;
    }
}
```

```

public void testEdit() throws Exception {
    MockRequestCycle cycle =
        (MockRequestCycle) page.getRequestCycle();
    cycle.addServiceParameter(userId);
    page.setRequestCycle(cycle);

    page.edit(cycle);

    assertNotNull(page.getUser());
}

public void testSave() throws Exception {
    MockRequestCycle cycle =
        (MockRequestCycle) page.getRequestCycle();
    cycle.addServiceParameter(userId);
    page.setRequestCycle(cycle);

    page.edit(cycle);
    assertNotNull(page.getUser());
    try {
        page.save(cycle);
        fail("Redirect didn't occur");
    } catch (PageRedirectException pe) {
        assertNotNull("redirect failed", pe);
    }
}

public void testRemove() throws Exception {
    User user = new User();
    user.setId(userId);
    page.setUser(user);

    try {
        page.delete(page.getRequestCycle());
        fail("Redirect didn't occur");
    } catch (PageRedirectException pe) {
        assertNotNull("redirect failed", pe);
    }
}
}

```

Listing A.44

```

package org.appfuse.web;

// organize imports using your IDE

public abstract class UserForm extends BasePage implements
    PageRenderListener {

```

```

public abstract UserManager getUserManager();
public abstract void setUserManager(UserManager manager);

public abstract void setUser(User user);
public abstract User getUser();

public void pageBeginRender(PageEvent event) {
    if ((getUser() == null) &&
        !event.getRequestCycle().isRewinding()) {
        setUser(new User());
    } else if (event.getRequestCycle().isRewinding()) {
        setUser(new User());
    }
}

public void cancel(IRequestCycle cycle) {
    if (log.isDebugEnabled()) {
        log.debug("entered 'cancel' method");
    }

    cycle.activate("users");
}

public void edit(IRequestCycle cycle) {
    Object[] parameters = cycle.getServiceParameters();
    Long id = (Long) parameters[0];

    if (log.isDebugEnabled()) {
        log.debug("fetching user with id: " + id);
    }

    setUser(getUserManager().getUser(id.toString()));
    cycle.activate(this);
}

public void save(IRequestCycle cycle) {
    if (log.isDebugEnabled()) {
        log.debug("entered 'save' method");
    }

    getUserManager().saveUser(getUser());

    UserList nextPage = (UserList) cycle.getPage("users");
    nextPage.setMessage(
        format("user.saved", getUser().getFullName()));
    throw new PageRedirectException(nextPage);
}

public void delete(IRequestCycle cycle) {
    if (log.isDebugEnabled()) {
        log.debug("entered 'delete 'method");
    }
}

```



```

    }

    getUserManager().removeUser(getUser().getId().toString());

    UserList nextPage = (UserList) cycle.getPage("users");
    nextPage.setMessage(
        format("user.deleted", getUser().getFullName()));
    throw new PageRedirectException(nextPage);
}
}

```

Listing A.45

```

<head>
    <title><span key="userForm.title"/></title>
    <link href="styles/calendar.css" type="text/css" rel="stylesheet"/>
</head>

<body jwcid="@Body">

<p>Please fill in user's information below:</p>

<form jwcid="@Form" name="userForm">
<input type="hidden" jwcid="@Hidden" value="ognl:user.id"/>
<table class="detail">
<tr>
    <th>
        <label for="firstName">
            <span key="firstName">First Name</span>:</label>
        </th>
<td><input jwcid="@TextField" type="text"
        value="ognl:user.firstName" id="firstName"/></td>
</tr>
<tr>
    <th>
        <label for="lastName">
            <span key="lastName">Last Name</span>:</label>
        </th>
<td><input jwcid="@TextField" type="text"
        value="ognl:user.lastName" id="lastName"/></td>
</tr>
<tr>
    <th>
        <label for="birthday"><span key="birthday">Birthday</span>:</
label>
        </th>
<td>
<input jwcid="@DatePicker" format="message:date.format"
        type="text" size="11"
        value="ognl:user.birthday" id="birthday"/>

```

```

        </td>
    </tr>
    <tr>
        <td></td>
        <td>
            <input type="submit" class="button" jwcid="@Submit"
                value="Save" id="save" listener="ognl:listeners.save"/>
            <span jwcid="@Conditional" condition="ognl:user.id != null">
                <input type="submit" class="button" jwcid="@Submit"
                    value="Delete" id="delete"
                    listener="ognl:listeners.delete"/>
            </span>
            <input type="submit" class="button" jwcid="@Submit"
                value="Cancel" id="cancel" listener="ognl:listeners.cancel"
                onclick="form.onsubmit = null"/>
        </td>
    </table>
</form>

<script type="text/javascript">
document.getElementById('firstName').focus();
</script>

</body>

```

Listing A.46

```

# Messages used for labels, errors and success messages
userForm.title=Equinox ~ User Details

user.saved=User <strong>{0}</strong> has been saved successfully.
user.missing=No user found with this id.
user.deleted=User <strong>{0}</strong> successfully deleted.

firstName=First Name
lastName=Last Name
birthday=Birthday
date.format=MM/dd/yyyy

```

Listing A.47

```

<?xml version="1.0"?>
<!DOCTYPE page-specification PUBLIC
    "-//Apache Software Foundation//Tapestry Specification 3.0//EN"
    "http://jakarta.apache.org/tapestry/dtd/Tapestry_3_0.dtd">

<page-specification class="org.appfuse.web.UserForm">
    <property-specification name="user" type="org.appfuse.model.User"/>
    <property-specification name="userManager"

```

```

        type="org.appfuse.service.UserManager">
        global.appContext.getBean("userManager")
    </property-specification>
</page-specification>

```

Listing A.48

```

<page name="userForm" specification-path="/pages/userForm.page"/>

```

Listing A.49

```

<a jwcid="@PageLink" page="userForm" id="formLink" style="display:none"/>
>
<button class="button" onclick="location.href=document.getElementById
('formLink').href">Add User</button>

```

Listing A.50

```

<tr jwcid="idColumnValue@Block">
    <a jwcid="@DirectLink" parameters="ognl:row.id"
        listener="ognl:requestCycle.getPage('userForm').listeners.edit">
        <span jwcid="@Insert" value="ognl:row.id"/>
    </a>
</tr>

```

Listing A.51

```

columns="id, firstName, lastName, birthday:new
java.text.SimpleDateFormat('MM/dd/yyyy').format(birthday) "

```

Listing A.52

```

<bean name="delegate"
class="org.apache.tapestry.valid.ValidationDelegate"
/>

```

Listing A.53

```

protected IValidationDelegate getValidationDelegate() {
    // be nice to unit tests
    if (getSpecification() != null) {
        return
            (IValidationDelegate) getBeans().getBean("delegate");
    }
}

```

```

    } else {
        return new ValidationDelegate();
    }
}

```

Listing A.54

```

<bean name="requiredValidator"
      class="org.apache.tapestry.valid.StringValidator">
  <set-property name="required" expression="true"/>
</bean>

```

Listing A.55

```

<component id="lastNameField" type="ValidField">
  <binding name="value" expression="user.lastName"/>
  <binding name="validator" expression="beans.requiredValidator"/>
  <message-binding name="displayName" key="lastName"/>
</component>

```

Listing A.56

```

<form jwcid="@Form" delegate="ognl:beans.delegate" name="userForm">

```

Listing A.57

```

<tr>
  <th>
    <label jwcid="@FieldLabel"
          field="ognl:components.lastNameField">Last Name</label>:
  </th>
  <td><input jwcid="lastNameField" type="text" id="lastName"/></td>
</tr>

```

Listing A.58

```

<span jwcid="@Conditional" condition="ognl:beans.delegate.hasErrors">
  <div class="error">
    <span jwcid="@Delegator"
          delegate="ognl:beans.delegate.firstError">
      Error Message
    </span>
  </div>
</span>

```

Listing A.59

```
public void save(IRequestCycle cycle) {
    if (log.isDebugEnabled()) {
        log.debug("entered 'save' method");
    }

    if (getValidationDelegate().getHasErrors()) {
        return;
    }

    getUserManager().saveUser(getUser());

    ...
}
```

Listing A.60

```
<bean name="requiredValidator"
    class="org.apache.tapestry.valid.StringValidator">
    <set-property name="required" expression="true"/>
    <set-property name="clientScriptingEnabled" expression="true"/>
</bean>
```

Listing A.61

```
package org.appfuse.web;

// organize imports using your IDE

public class UserWebTest extends WebTestCase {

    public UserWebTest(String name) {
        super(name);
        getTestContext()
            .setBaseUrl("http://localhost:8080/equinox-tapestry");
        getTestContext().setResourceBundleName("messages");
    }

    public void testWelcomePage() {
        beginAt("/");
        assertEqualsKey("index.title");
    }

    public void testAddUser() {
        beginAt("/app?service=page/userForm");
        assertEqualsKey("userForm.title");
        setFormElement("$TextField", "Spring");
    }
}
```

```

        setFormElement("lastNameField", "User");
        submit("$Submit");
        assertTextPresent("saved successfully");
    }

    public void testListUsers() {
        beginAt("/app?service=page/users");

        // check that table is present
        assertTablePresent("userList");

        // check that a set of strings are present somewhere in table
        assertTextInTable("userList",
            new String[] {"Spring", "User"});
    }

    public void testEditUser() {
        beginAt("/app?service=direct/0/users/$DirectLink&sp=1" +
            getInsertedUserId());
        assertFormElementEquals("$TextField", "Spring");
        submit("$Submit");
        assertTitleEqualsKey("userList.title");
    }

    /**
     * Convenience method to get the id of the inserted user
     * Assumes last inserted user is "Spring User"
     */
    public String getInsertedUserId() {
        String[] paths = {"WEB-INF/applicationContext*.xml"};
        ApplicationContext ctx = new
            ClassPathXmlApplicationContext(paths);
        List users = ((UserDAO) ctx.getBean("userDAO")).getUsers();
        // assumed that user inserted in testAddUser() is last user
        return "" + ((User) users.get(users.size() - 1)).getId();
    }
}

```

Listing A.62

```

<listener>
  <listener-class>
    com.opensymphony.xwork.spring.SpringObjectFactoryListener
  </listener-class>
</listener>

```

Listing A.63

```
<servlet>
  <servlet-name>action</servlet-name>
  <servlet-class>
    com.opensymphony.webwork.dispatcher.ServletDispatcher
  </servlet-class>
  <load-on-startup>1</load-on-startup>
</servlet>
```

Listing A.64

```
<servlet-mapping>
  <servlet-name>action</servlet-name>
  <url-pattern>*.html</url-pattern>
</servlet-mapping>
```

Listing A.65

```
<context-param>
  <param-name>
    javax.servlet.jsp.jstl.fmt.localizationContext
  </param-name>
  <param-value>messages</param-value>
</context-param>
```

Listing A.66

```
# Extension for actions. This uses by the webwork:form tag
webwork.action.extension=html
webwork.custom.i18n.resources=messages
```

Listing A.67

```
<%@ page language="java" errorPage="/error.jsp" %>
<%@ taglib uri="http://java.sun.com/jstl/core_rt" prefix="c" %>
<%@ taglib uri="http://java.sun.com/jstl/fmt_rt" prefix="fmt" %>
<%@ taglib uri="http://displaytag.sf.net" prefix="display" %>
<%@ taglib uri="http://www.opensymphony.com/sitemesh/decorator"
  prefix="decorator" %>
<%@ taglib uri="webwork" prefix="ww" %>

<c:set var="datePattern"><fmt:message key="date.format"/></c:set>
```

Listing A.68

```
date.format=MM/dd/yyyy
```

Listing A.69

```
package org.appfuse.web;

// use your IDE to organize imports

public class UserActionTest extends TestCase {
    private final Log log = LoggerFactory.getLog(UserActionTest.class);
    private static XmlWebApplicationContext ctx;
    private UserAction action;

    static {
        String[] paths = {"WEB-INF/applicationContext*.xml"};
        ctx = new XmlWebApplicationContext();
        ctx.setConfigLocations(paths);
        ctx.setServletContext(new MockServletContext(""));
        ctx.refresh();
    }

    protected void setUp() throws Exception {
        super.setUp();
        action = new UserAction();

        // grab the UserManager from the ApplicationContext or mock it
        UserManager mgr = (UserManager) ctx.getBean("userManager");
        action.setUserManager(mgr);

        // add a test user to the database
        User user = new User();
        user.setFirstName("Jack");
        user.setLastName("Raible");
        mgr.saveUser(user);

        ActionContext.getContext().setSession(new HashMap());
    }

    protected void tearDown() throws Exception {
        super.tearDown();
        action = null;
        ActionContext.getContext().setSession(null);
    }

    public void testSearch() throws Exception {
```



```

        assertEquals(action.list(), ActionSupport.SUCCESS);
        assertTrue(action.getUsers().size() >= 1);
    }
}

```

Listing A.70

```

package org.appfuse.web;

// use your IDE to organize imports

public class UserAction extends ActionSupport {
    private UserManager mgr = null;
    private List users;

    public void setUserManager(UserManager userManager) {
        this.mgr = userManager;
    }

    public List getUsers() {
        return users;
    }

    public String list() {
        users = mgr.getUsers();
        return SUCCESS;
    }
}

```

Listing A.71

```

<%@ include file="/taglibs.jsp"%>

<title><fmt:message key="userList.title"/></title>

<ww:set name="users" value="users" scope="request"/>
<button onclick="location.href='editUser.html'">Add User</button>

<display:table name="users" class="list" requestURI="" id="userList"
    export="true">
    <display:column property="id" sort="true" href="editUser.html"
        paramId="id" paramProperty="id" titleKey="user.id"/>
    <display:column property="firstName" sort="true"
        titleKey="user.firstName"/>
    <display:column property="lastName" sort="true"
        titleKey="user.lastName"/>
    <display:column sort="true" titleKey="user.birthday" media="html">
        <!--<fmt:formatDate value="{userList.birthday}"
            pattern="yyyyMMdd"/>-->

```

```

        <fmt:formatDate value="\${userList.birthday}"
pattern="\${datePattern}"/>
    </display:column>
    <display:column titleKey="user.birthday" media="csv xml excel pdf">
        <fmt:formatDate value="\${userList.birthday}"
            pattern="\${datePattern}"/>
    </display:column>
</display:table>

<script type="text/javascript">highlightTableRows("userList");</script>

```

Listing A.72

```

<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE beans PUBLIC "-//SPRING//DTD BEAN//EN"
    "http://www.springframework.org/dtd/spring-beans.dtd">

<beans>
    <bean id="userAction" class="org.appfuse.web.UserAction"
        singleton="false">
        <property name="userManager">
            <ref bean="userManager"/>
        </property>
    </bean>
</beans>

```

Listing A.73

```

<context-param>
    <param-name>contextConfigLocation</param-name>
    <param-value>
        /WEB-INF/applicationContext*.xml
        /WEB-INF/action-servlet.xml
    </param-value>
</context-param>

```

Listing A.74

```

<!DOCTYPE xwork PUBLIC "-//OpenSymphony Group//XWork 1.0//EN"
    "http://www.opensymphony.com/xwork/xwork-1.0.dtd">

<xwork>
    <!-- Include webwork defaults (from WebWork-2.1 JAR). -->
    <include file="webwork-default.xml"/>

    <package name="default" extends="webwork-default">

        <action name="users" class="userAction" method="list">

```

```

        <result name="success"
            type="dispatcher">userList.jsp</result>
        </action>
    </package>
</xwork>

```

Listing A.75

```

public class UserActionTest extends TestCase {
    private final Log log = LoggerFactory.getLog(UserActionTest.class);
    private static XmlWebApplicationContext ctx;
    private UserAction action;
    private String userId;

    static {
        String[] paths = {"/WEB-INF/applicationContext*.xml"};
        ctx = new XmlWebApplicationContext();
        ctx.setConfigLocations(paths);
        ctx.setServletContext(new MockServletContext(""));
        ctx.refresh();
    }

    protected void setUp() throws Exception {
        super.setUp();
        action = new UserAction();

        // grab the UserManager from the ApplicationContext or mock it
        UserManager mgr = (UserManager) ctx.getBean("userManager");
        action.setUserManager(mgr);

        // add a test user to the database
        User user = new User();
        user.setFirstName("Jack");
        user.setLastName("Raible");
        mgr.saveUser(user);
        userId = user.getId().toString();

        ActionContext.getContext().setSession(new HashMap());
    }

    protected void tearDown() throws Exception {
        super.tearDown();
        action = null;
        ActionContext.getContext().setSession(null);
    }

    public void testSearch() throws Exception {
        assertEquals(action.list(), ActionSupport.SUCCESS);
        assertTrue(action.getUsers().size() >= 1);
    }
}

```

```

public void testEdit() throws Exception {
    action.setId(userId);
    assertEquals(action.edit(), "success");
    assertNotNull(action.getUser().getFirstName());
}

public void testSave() throws Exception {
    action.setId(userId);
    action.edit();
    assertNotNull(action.getUser());
    action.getUser().setFirstName("Jack");

    assertEquals(action.save(), "success");
    assertNotNull(action.getUser());
}

public void testRemove() throws Exception {
    User user = new User();
    user.setId(new Long(userId));
    action.setUser(user);
    assertEquals(action.delete(), "deleteSuccess");
}
}

```

Listing A.76

```

public class UserAction extends ActionSupport {
    private UserManager mgr;
    private List users;
    private User user;
    private String id;
    private String cancel;
    private String delete;

    public void setCancel(String cancel) {
        this.cancel = cancel;
    }

    public void setDelete(String delete) {
        this.delete = delete;
    }

    public void setUserManager(UserManager userManager) {
        this.mgr = userManager;
    }

    public List getUsers() {
        return users;
    }
}

```

```

public void setId(String id) {
    this.id = id;
}

public User getUser() {
    return user;
}

public void setUser(User user) {
    this.user = user;
}

public String cancel() {
    return SUCCESS;
}

public String delete() {

    mgr.removeUser(user.getId().toString());

    List args = new ArrayList();
    args.add(user.getFullName());
    ActionContext.getContext().getSession().put("message",
        getText("user.deleted", args));

    return "deleteSuccess";
}

public String edit() {
    // check for an add
    if (id != null) {
        user = mgr.getUser(id);
    } else {
        user = new User();
    }
    return SUCCESS;
}

public String execute() {
    if (cancel != null) {
        return "cancel";
    }

    if (delete != null) {
        return delete();
    }

    return save();
}

```

```

public String save() {
    mgr.saveUser(user);

    List args = new ArrayList();
    args.add(user.getFullName());

    ActionContext.getContext().getSession().put("message",
        getText("user.saved", args));

    return SUCCESS;
}

public String list() {
    users = mgr getUsers();
    return SUCCESS;
}
}

```

Listing A.77

```

<%@ include file="/taglibs.jsp"%>

<title><fmt:message key="userForm.title"/></title>

<p>Please fill in user's information below:</p>

<ww:form name="'userForm'" action="'saveUser'" method="'post'">
<ww:hidden name="'user.id'" value="user.id"/>
    <ww:textfield label="getText('user.firstName')"
        name="'user.firstName'"
        value="user.firstName" id="user.firstName"/>
    <ww:textfield label="getText('user.lastName')"
        name="'user.lastName'"
        value="user.lastName" required="'true'"/>
</tr>
    <th>
        <label for="user.birthday">
            <fmt:message key="user.birthday"/>:
        </label>
    </th>
    <td>
        <ww:set name="birthday" scope="request"
            value="(user.birthday instanceof java.util.Date) ?
user.birthday
            : ''"/>
        <input type="text" size="11" name="user.birthday"
            id="user.birthday"
            value="<fmt:formatDate value='${birthday}'"
pattern="${datePattern}"
            /></td>
    </tr>
</ww:form>

```

```

        [${datePattern}]
    </td>
</tr>
<tr>
    <td></td>
    <td>
        <input type="submit" class="button" name="save" value="Save"/>
        <c:if test="${not empty param.id}">
            <input type="submit" class="button" name="delete"
                value="Delete"/>
        </c:if>
        <input type="submit" class="button" name="cancel"
            value="Cancel"/>
    </td>
</ww:form>
<script type="text/javascript">
    document.forms["userForm"].elements["user.firstName"].focus();
</script>

```

Listing A.78

```

<default-interceptor-ref name="defaultStack"/>

<!-- List of Users -->
<action name="users" class="userAction" method="list">
    <result name="success" type="dispatcher">userList.jsp</result>
</action>

<!-- Edit User -->
<action name="editUser" class="userAction" method="edit">
    <result name="success" type="dispatcher">userForm.jsp</result>
    <result name="error" type="dispatcher">userList.jsp</result>
</action>

<!-- Save User -->
<action name="saveUser" class="userAction">
    <result name="cancel" type="redirect">users.html</result>
    <result name="deleteSuccess" type="redirect">users.html</result>
    <result name="success" type="redirect">users.html</result>
</action>

```

Listing A.79

```

<%-- Error Messages --%>
<ww:if test="hasFieldErrors()">
    <div class="error">
        <ww:iterator value="fieldErrors">
            <ww:iterator value="value">
                <ww:property/><br/>
            </ww:iterator>
        </ww:iterator>
    </div>
</ww:if>

```

```

        </ww:iterator>
    </ww:iterator>
</div>
</ww:if>

<%-- Success Messages --%>
<c:if test="${not empty message}">
    <div class="message">${message}</div>
</c:if>

```

Listing A.80

```
<default-interceptor-ref name="defaultStack"/>
```

Listing A.81

```

<!DOCTYPE validators PUBLIC
    "-//OpenSymphony Group//XWork Validator 1.0//EN"
    "http://www.opensymphony.com/xwork/xwork-validator-1.0.dtd">

<validators>
    <validator name="requiredstring"
class="com.opensymphony.xwork.validator.
        validators.RequiredStringValidator"/>
</validators>

```

Listing A.82

```

<!DOCTYPE validators PUBLIC
    "-//OpenSymphony Group//XWork Validator 1.0//EN"
    "http://www.opensymphony.com/xwork/xwork-validator-1.0.dtd">
<validators>
    <field name="user.lastName">
        <field-validator type="requiredstring">
            <message key="errors.required">
                Last Name is required.
            </message>
        </field-validator>
    </field>
</validators>

```

Listing A.83

```

<!-- Save User (Validation Enabled) -->
<action name="saveUserWithValidation" class="userAction"
    method="save">
    <interceptor-ref name="validationWorkflowStack"/>

```



```

    <result name="input" type="dispatcher">userForm.jsp</result>
    <result name="success" type="redirect">users.html</result>
    <result name="error" type="dispatcher">userForm.jsp</result>
</action>

```

Listing A.84

```

public String execute() {
    if (cancel != null) {
        return "cancel";
    }

    if (delete != null) {
        return delete();
    }

    return SUCCESS;
}

```

Listing A.85

```

<action name="saveUser" class="userAction">
    <result name="cancel" type="redirect">users.html</result>
    <result name="deleteSuccess" type="redirect">users.html</result>
    <result name="success" type="chain">saveUserWithValidation</result>
</action>

```

Listing A.86

```

<validator name="requiredstring"
class="com.opensymphony.webwork.validators.
JavaScriptRequiredStringValidator"/>

```

Listing A.87

```

<ww:form name="'userForm'" action="'saveUser'" method="'post'"
validate="true">

```

Listing A.88

```

<input type="submit" class="button" name="cancel" value="Cancel"
onclick="form.onsubmit=null"/>

```

Listing A.89

```
package org.appfuse.web;

// organize imports using your IDE

public class UserWebTest extends WebTestCase {

    public UserWebTest(String name) {
        super(name);
        getTestContext()
            .setBaseUrl("http://localhost:8080/equinox-webwork");
        getTestContext().setResourceBundleName("messages");
    }

    public void testWelcomePage() {
        beginAt("/");
        assertEqualsKey("index.title");
    }

    public void testAddUser() {
        beginAt("/editUser.html");
        assertEqualsKey("userForm.title");
        setFormElement("user.firstName", "Spring");
        setFormElement("user.lastName", "User");
        submit("save");
        assertEqualsKey("userList.title");
    }

    public void testListUsers() {
        beginAt("/users.html");

        // check that table is present
        assertTablePresent("userList");

        //check that a set of strings are present somewhere in table
        assertTextInTable("userList",
            new String[]{"Spring", "User"});
    }

    public void testEditUser() {
        beginAt("/editUser.html?id=" + getInsertedUserId());
        assertFormElementEquals("user.firstName", "Spring");
        submit("save");
        assertEqualsKey("userList.title");
    }

    public void testDeleteUser() {
        beginAt("/editUser.html?id=" + getInsertedUserId());
        assertEqualsKey("userForm.title");
    }
}
```

```

        submit("delete");
        assertEqualsKey("userList.title");
    }

    /**
     * Convenience method to get the id of the inserted user
     * Assumes last inserted user is "Spring User"
     */
    public String getInsertedUserId() {
        String[] paths = {"WEB-INF/applicationContext*.xml"};
        ApplicationContext ctx = new
            ClassPathXmlApplicationContext(paths);
        List users = ((UserDAO) ctx.getBean("userDAO")).getUsers();
        // assumed that user inserted in testAddUser() is last user
        return "" + ((User) users.get(users.size() - 1)).getId();
    }
}

```

